TVS MOTOR COMPANY	Document Name: R&D/SWCoC/ TVSM_DIAGNOSTIC_SPECIFICATION	Type: Draft Issue: v11.4.1 Date: 29-04-2026

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TVSM Diagnostic Specification

Vehicle information: EV-platform [2W and 3W]
Applicable components: VCU, BMS, CLU, ABS, MCU, i VCU and LCM
Protocol (CAN): UDS-ISO 14229

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1. Introduction

Serial diagnosis provides the possibility of quickly accessing information and functions of the control unit without disconnecting the cable harness or disconnecting from Vehicle.
Serial diagnosis provides:

1) Readout of the ECU's information [Software, Hardware, Programing date, etc.]
2) Readout of the ECU's input signals.
3) Readout of the ECU's internal information, e.g., the content of the fault memory.
4) Control of connected actuators.
5) Reprogramming the ECUs via CAN.

Scope:

This document contains specifications of Diagnostic Software which is used to diagnose the ECU in the vehicle network.

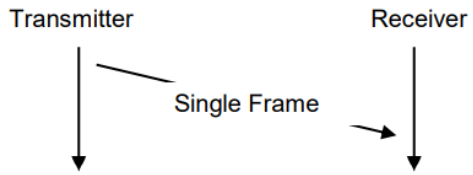
1.1 CAN Communication

1.1.1 Data stream

The ECU supports CAN diagnostic messages using the Unacknowledged Segmented Data Transfer (USDT) mechanism as specified in ISO 15765-2. Accordingly, the following two types of CAN data streams are supported:

a) Non-segmented data stream:

Consists of a Single Frame (SF) of non-segmented CAN message. This can be used for messages with maximum data length of 7 data bytes.



b) Segmented data stream:

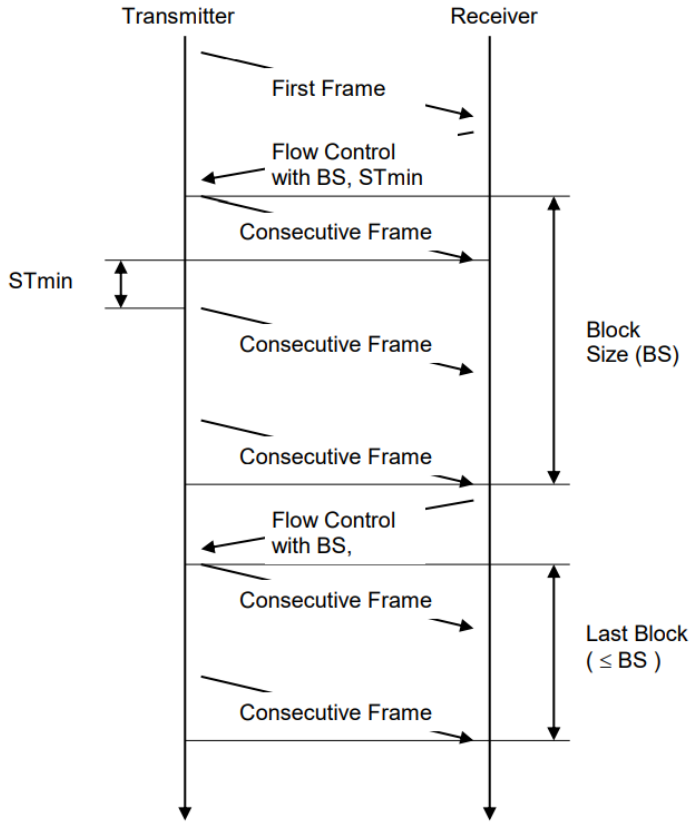
The communication consists of multiple frames of segmented CAN messages. This can be used when the maximum data length exceeds 7 data bytes.

The data stream comprises of a First Frame (FF) and a few Consecutive Frames (CF) containing the subsequent data bytes, following the First Frame. A Handshake in the form of a Flow Control (FC) must be sent by the receiver node after reception of the First Frame.

The Flow Control should contain information regarding the maximum number of Consecutive Frames the receiver can allow (Block Size – BS) after which the sender must wait for an authorization

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from the receiver to continue, and the minimum time the sender must wait between transmission of two consecutive frames (Separation Time – STmin).



1.1.2 Message Types

Each one of the above CAN message types of Single Frame (SF), First Frame (FF), Consecutive Frame (CF) or Flow Control (FC), shall have an 8-byte long Data Field consisting of the following two pieces of information:

- ♦ Protocol Control Information (PCI): Indicates the frame type, number of frames, data length etc.
- ♦ Transmission data (Data): Contains the data to be sent.

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Data Field Structure					
Frame Type	Byte 1		Byte 2	Byte 3	Bytes 4-8
	Bits 7-4	Bits 3-0			
Single Frame (SF)	0000	SF_DL	DB 1	DB 2	DB 3 – 7
First Frame (FF)	0001	FF_DL		DB 1	DB 2 – 6
Consecutive Frame (CF)	0010	SN	DB x_1	DB x_2	DB x_3-x_7
Flow Control (FC)	0011	FS	BS	STmin	N/A

SF_DL : 4 Bit Single Frame Data length
FF_DL : 12 Bit First Frame Data length
SN : Sequence Number
FS : Status of data stream
BS : Block Size
STmin : Minimum Separation Time
DB : Data byte

1.1.2.1 Single Frame

Single Frame message type is used for data blocks with a maximum data length of 7 Bytes and is identified by the upper nibble of PCI set to 0000B. Single Frame Data length SF_DL is computed as:

Single Frame data length (SF_DL):

Hex Value	Description
0-7	Single Frame Data length (SF_DL) contains number of data Bytes

1.1.2.2 First Frame

First Frame message type is used for data blocks more than 7 Bytes long, in combination with one or more Consecutive Frames and is identified by the upper nibble of PCI set to 0001B. First Frame Data length FF_DL is computed as:

First Frame Data length (FF_DL):

Hex Value	Description
000 – 007	Reserved
008 – FFF	First Frame Data length (FF_DL) contains the number of data bytes. 12 Bit FF_DL theoretically supports sending of up to 4096 data bytes.

1.1.2.3 Consecutive Frame

Consecutive Frames follow the First Frame and are identified by the upper nibble of PCI set to 0010B. The lower nibble of the PCI has a Sequence Number (SN) which is determined as follows:

- The first Frame is regarded as containing Sequence Number 0.
- Hence the first Consecutive Frame always holds Sequence Number 1.
- Sequence Number is incremented by one for every additional Consecutive Frame.
- A Flow Control frame sent ‘in between’ does not affect the Sequence Number counting.

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- When the Sequence Number reaches the value of 15, the following Consecutive Frame will be sent with Sequence Number set to 0.

Definition of Sequence Number (SN):

Protocol Data name	FF	CF	CF	CF	CF	CF	CF	CF	CF
SN (hex)	0	1	2	...	E	F	0	1	...

Hex Value	Description
0 – F	Sequence Number (SN) The Sequence Number has a range of 0 to 15. SN-Value is stored to the lower 4 bits of the Byte 1

1.1.2.4 Flow Control Frame

After the First Frame or the last Consecutive Frame of a block of frames, when further CFs are required to complete the data transfer, a Flow Control message must be sent by the receiver node and is identified by the upper nibble of PCI set to 0011B. With the Flow Control, the receiving node notifies the transmitting node how many messages it can receive between two Flow Control frames (maximum Block Size, BSmax) and at what rate between Consecutive Frames (minimum Separation Time, STmin). The lower nibble of the Flow Control PCI shall have a Flow Control Status (FS) set to "Continue to Send (CTS)".

Stream Status (FS)

Hex Value	Description
0	ContinueToSend (CTS) The Receiver is ready and awaits the BS announced maximal no of Consecutive Frames.
1	Wait (WT) The Flow Control Status Wait is supported.
2	Overflow (OVFLW) This status is sent to indicate that the data length indicated by the first frame is too large for the receiver buffer. Not supported as of now.

Block Size (BSmax)

Hex Value	Description
00	Block Size (BSmax) No more Flow Control frames will be sent during the transmission of the segmented Message. All Consecutive Frames will be sent without stopping for additional Flow Control frames
01 – FF	Block Size (BSmax) Defines the maximal number of Consecutive Frames till next Flow Control.

Notes: This value is Server dependent, based on its capabilities.

Separation Time (STmin)

Hex Value	Description
00 – 7F	Separation Time (STmin)

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	The Separation Time defines the minimal allowed time (in ms) between 2 Consecutive Frames. The range is from 0 to 127ms as per the ISO standards.

1.1.3 Addressing modes.

The ECU supports two types of addressing modes:

1) **Physical addressing Mode:** All physically addressed messages will be received by the ECU on ECU_PHY_ADDR (see "Communication Parameters" section below for used CAN ID). It is possible to execute any of the services supported in the active diagnostic session of the ECU, when the request is made using this mode. Response will be sent on TESTER_ADDR (see "Communication Addresses "section below for used CAN ID), if not disallowed by the request.

2) **Functional addressing Mode**: Broadcasting messages to all ECUs in the network.
Functional Address: **0x7DF**

Note: *Functional addressing mode can be used for the following services only:*

1. *Diagnostic Session control [$10]*
2. *Control DTC settings [$85]*
3. *Communication Control [$28]*
4. *Tester Present [$3E]*

1.2 Diagnostic Sessions

The ECU supports the following Diagnostic Sessions:

1) Normal / Default Session

At ignition ON, the ECU enables the Normal diagnostic session by default. In this session both the controller functions and diagnostic services are active. But this session does not permit the execution of all supported diagnostic services of the ECU mode (see "Availability of Services" section below for services supported during this session).

2) Extended Diagnostic Session

During this session all the major controller functions will be set to passive and only the diagnostic services will be active. This session allows the execution of all diagnostic services supported by the ECU.

3) Reprogramming Session

This session is only available on ECUs installed with customer reprogramming bootloader software. The ECU starts execution from the bootloader software for this session.

Available function in each-mode:

	Normal Diag session	Extended/Test Diag session	Re-programming session
Vehicle Functions	Available	Not Available	Not Available
Failure monitoring	Available	Available	Not Available

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CAN TX/RX	Available	Available	Not Available- Except Diag ID's
Precondition	none	Vehicle Speed=0kmph	Vehicle Speed=0kmph
Lamps if any	OFF	ON	ON
Diag indication on CAN Bus	DiagMode=0 [Normal Mode]	DiagMode=1 [Extended Mode]	No CAN ID/No Signal Since it is reprograming mode

1.3 Communication Parameters

Parameter	Name	Value		
Communication Speed (Baud rate)	BR-CAN	500 kBd ± 2kBd		
Tester address (CAN-IDs):	TESTER_ADDR	Refer Below Table		
Tester address (CAN-IDs):	SG_ADDR	Refer Below Table		
		Min	**Max**	**Recommended**
Possible Separation Time by send (*)	STmin	0ms	127ms	10ms
Maximum time the client must wait or respond to the UDS request (*)	P2	0ms	150ms	50ms
The maximum allowable additional time the ECU requires to generate the response after sending pending response NRC (0x78).	P2*	0ms	5000ms	5000ms
S3 is the timeout in the diagnostic server for leaving a non-default session. Normal/Default Diagnostic Session ($01) Reprogramming Session ($02) Extended Diagnostic Session ($03)	 S3 NormDiag S3 ExtdDiag S3 Reprog	 Infinite 5000ms 5000ms	 Infinite 5000ms 5000ms	 Infinite 5000ms 5000ms

Note: The pending response NRC (0x78) handling should be limited to 10 times.

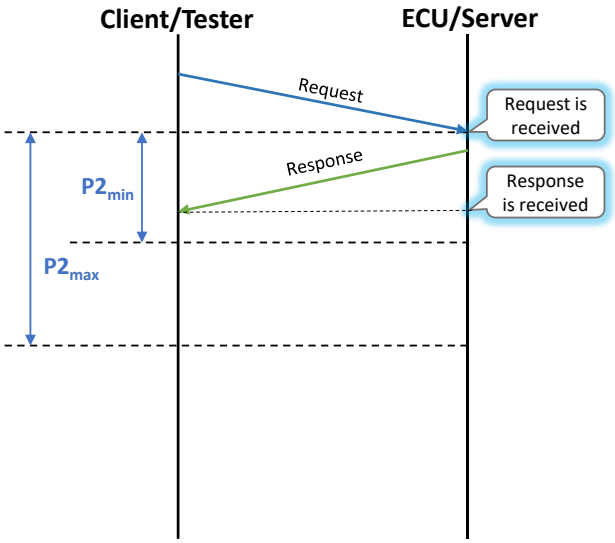
Network Layer Target Timings	Name	ValueThe
Time for transmission of the CAN frame (any N_PDU) on the receiver side	N_Ar	1000ms
Frame transmission timeout	N_As	1000ms
The max time from sending an FF to the reception of the FC	N_Bs	1000ms
Time from reception of an FF to transmission of the FC	N_Br	0ms*
Time from reception of an CF to transmission of the FC	N_Cs	0ms*
The max time from sending an FC to the next CF	N_Cr	1000ms

*As fast as possible

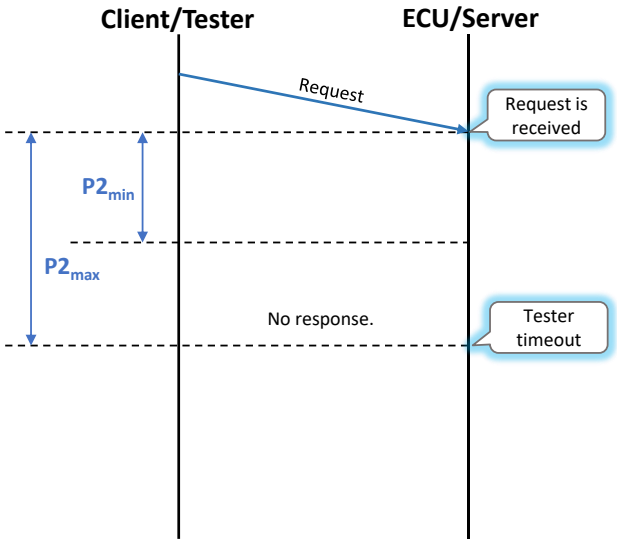
1.3.1 Timing Diagram and Timeout handling

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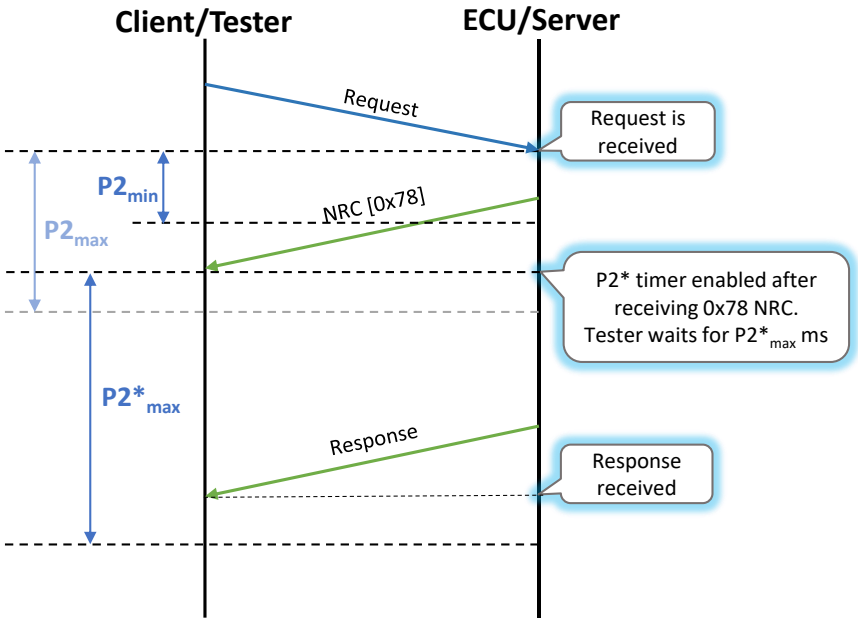
Case 1: Positive response received within $P2_{max}$ time.



Case 2: No response received within $P2_{max}$ time. Tester timeout happens.

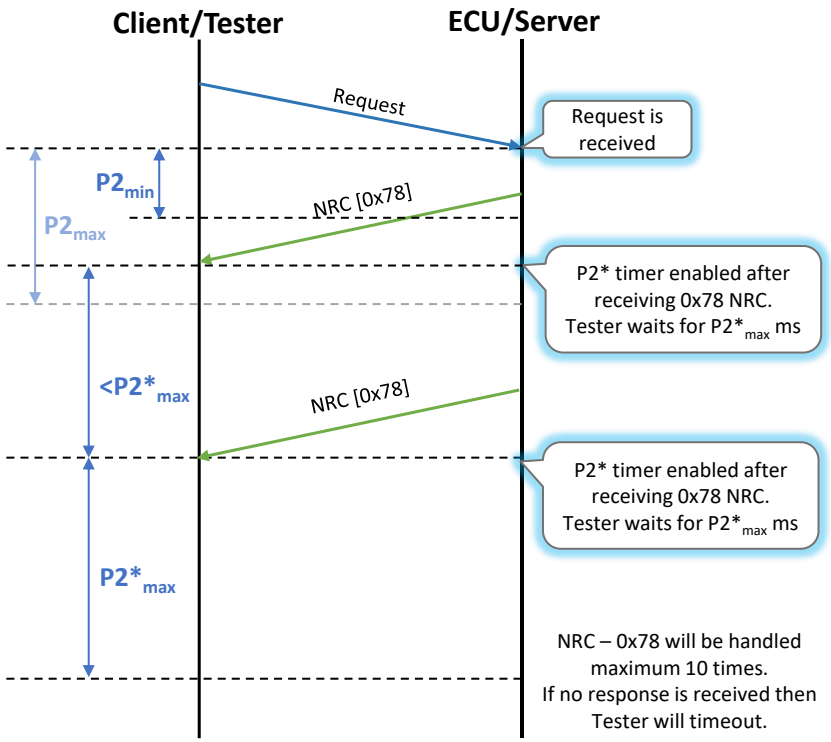


Case 3: Pending response handling.
Tester needs to wait $P2+P2_{max}$ ms to receive the response.



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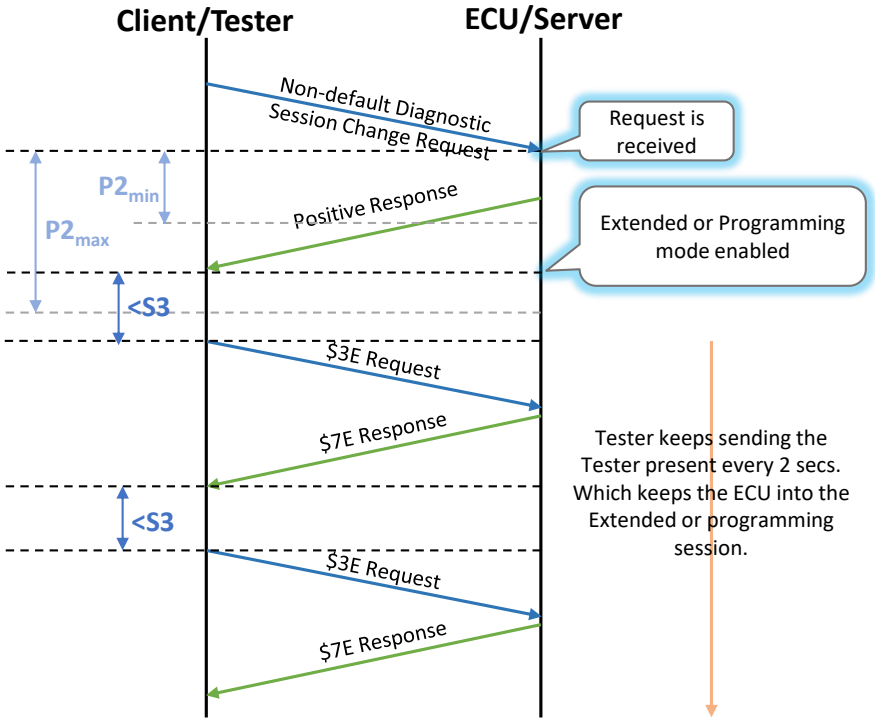
Case 4: Multiple Pending responses handling.
Max 10 times.



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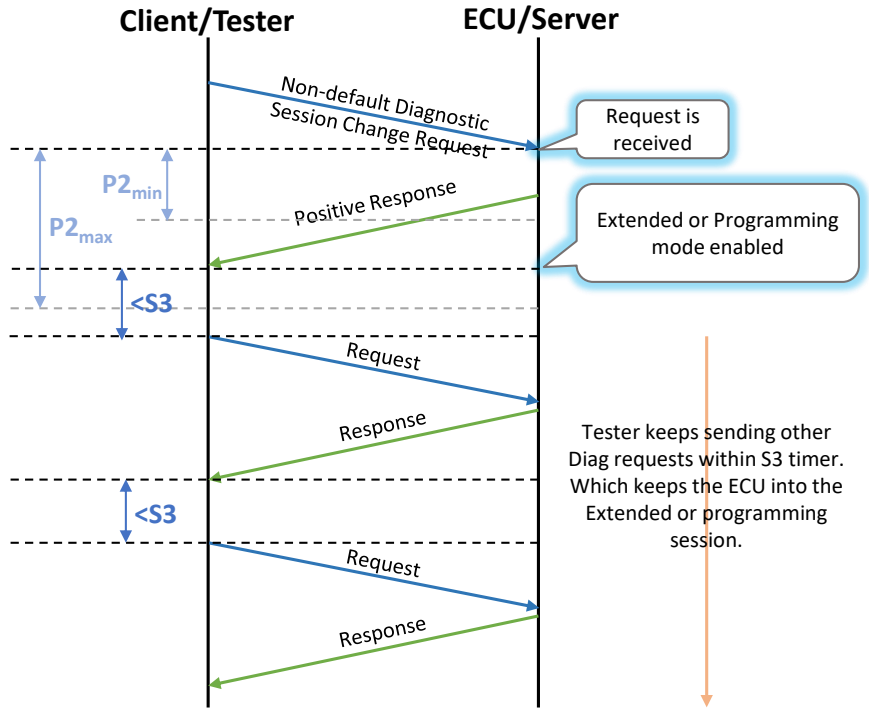
1.3.2 S3 Timeout handling

Case 1: Keep sending Tester Present request within 2 seconds.



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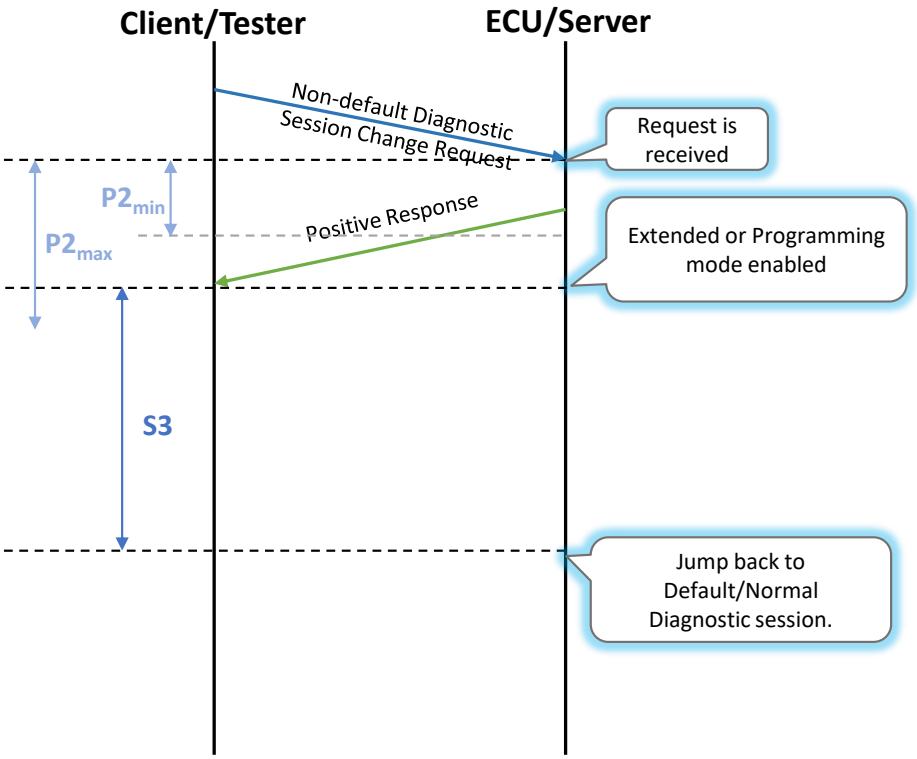
Case 2: Other diagnostic requests are being made within S3 timeout.



i

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Case 3: Neither Tester Present nor any other diagnostic service request has been made.



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1.4 Deleted

1.5 Communication Failures

Unexpected Messages

As a rule, the arrival of an unexpected message or a message with illegal format from any node will be ignored.

Missing Messages

If the ECU senses a communication failure (e.g. timing violations of the Network Layer) during a tester request, the ECU discards the request, and no response will be sent. The tester must monitor that no ECU-response has been sent within P2 NormRsp, and hence the tester can repeat his previous request. During Extended Diagnostic Session, if no request is received from the tester for P3 ExtdDiagMax (criteria for broken communication wire) ECU will exit to the Default Session.

1.6 Overview of project specific supported services / functions

SID	Description	Secured	Session Type						
			Default 01H	Extended 03H	Reprogram 02H	Default 01H	Extended 03H	Reprogram 02H	Client 40H
			Application Software			Bootloader			
10H	Diagnostic Session Control		✓	✓	✓	✓	✓	✓	✓
11H	ECU Reset		X	✓	✓	X	✓	✓	✓
14H	Clear Diagnostic Information		✓	✓	X	X	X	X	X
19H	Read DTC Information		✓	✓	X	X	X	X	X
22H	Read Data By Identifier		✓	✓	✓	✓	✓	✓	✓
23H	Read Memory By Address	✓	X	✓	X	X	✓	X	X
27H	Security Access		X	✓	✓	X	✓	✓	✓
28H	Communication Control		X	✓	✓	X	✓	✓	✓
2EH	Write Data By Identifier	✓	X	✓	X	X	✓	✓	✓
31H	Routine Control		X	✓	✓	X	✓	✓	✓
34H	Request Download	✓	X	X	X	X	X	✓	✓
36H	Transfer Data	✓	X	X	X	X	X	✓	✓
37H	Request Transfer Exit	✓	X	X	X	X	X	✓	✓
3EH	Tester Present		✓	✓	✓	✓	✓	✓	✓
83H	Access Timing Parameter	✓	X	✓	X	X	X	X	X
85H	Control DTC Setting		X	✓	✓	X	X	X	X
86H	Response On Event		✓	✓	X	X	X	X	X
87H	Link Control	✓	X	X	X	X	✓	✓	X

✓ – Applicable ✘ – Not applicable

Note:

1. The Memory, Erase and the EOL related Routine Control need not be implemented on the Function Software.

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2. Pre-condition check routine should be supported in both Bootloader (Positive Response) and Function Software.
3. Implementation of the Communication control service in Bootloader Software is optional, shall be implemented if any messages other than Diagnostics are being transmitted.
4. Implementation of services is on a needed basis, based on the specific ECU requirements to be fulfilled.

1.7 Deleted

1.8 Overview Negative Response Table

Server Negative Response (Hex)	Response Description
10H	**General Reject** This NRC indicates that the requested action has been rejected by the server.
11H	**Service not supported** This NRC shall be sent if the requested SID not supported
12H	**Subfunction not supported** This NRC shall be sent if the requested sub-function is either generally not supported
13H	**Incorrect message length or Invalid format** This NRC shall be sent if the length of the message is wrong.
22H	**Conditions not correct** This NRC indicates that the requested action will not be taken because the server prerequisite conditions are not met.
24H	**Request Sequence Error** This NRC indicates that the requested action will not be taken because the server expects a different sequence of request messages or message as sent by the client. This may occur when sequence sensitive requests are issued in the wrong order.
31H	**Request out of Range** This NRC indicates that the requested action will not be taken because the server has detected that the request message contains a parameter which attempts to substitute a value beyond its range of authority
33H	**Security Access Denied** This NRC indicates that the requested action will not be taken because the server's security strategy has not been satisfied by the client.
35H	**Invalid Key** This NRC indicates that the server has not given security access because the key sent by the client did not match with the key in the server's memory.
36H	**Exceed Number of Attempts**

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Server Negative Response (Hex)	Response Description
	This NRC indicates that the requested action will not be taken because the client has unsuccessfully attempted to gain security access more times than the server's security strategy will allow.
71H	**Transfer Data Suspended** This NRC indicates that a data transfer operation was halted due to some fault. The active transfer Data sequence shall be aborted.
72H	**General Programming Failure** This NRC indicates that the server detected an error when erasing or programming a memory location in the permanent memory device (e.g. Flash Memory).
73H	**Wrong Block Sequence Counter** This NRC indicates that a data transfer operation was halted due to some fault. The active transfer Data sequence shall be aborted.
78H	**Response pending** This NRC indicates that the request message was received correctly, and that all parameters in the request message were valid, but the action to be performed is not yet completed and the server is not yet ready to receive another request. As soon as the requested service has been completed, the server shall send a positive response message or negative response message with a response code different from this. NOTE: If NRC 78 is repeating for 10 times, then send NRC10(General reject) as response.
7EH	**Subfunction not supported in active diagnostic session** This NRC indicates that the requested action will not be taken because the server does not support the requested sub-function in the session currently active.
7FH	**Service not supported in active diagnostic session** This NRC is in general supported by each diagnostic service, as not otherwise stated in the data link specific implementation document, therefore it is not listed in the list of applicable response codes of the diagnostic services.

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2. Flashing Procedure

2.1 Flashing Sequence

This section has been moved to a separate document. Please refer to Annexure[f].

3. Functional Requirements

3.1 Diagnostic Session Control: $10

The service Start Diagnostic Session is used to select a specific ECU Diagnostic mode (session). Depending on the session selected, actions are authorized, prohibited, or limited. The following Session Types are supported.

Sub Function	Session	Support
$01	Default Diagnostic Session	Yes
$02	Programming Session	Yes
$03	Extended Diagnostic Session	Yes
$40	Client Session	Yes
$50	EOL Session	Yes

3.1.1 Default Diagnostic Session: $10 $01

This diagnostic session enables the default diagnostic session in the server(s) and does not support any diagnostic application timeout handling provisions (e.g. no Tester Present service is necessary to keep the session active). If any other session than the default Session has been active in the server and the default Session is once again started, then the following implementation rules shall be followed: The server shall stop the current diagnostic session when it has sent the Diagnostic Session Control positive response message and shall start the newly requested diagnostic session afterwards.

If the server has sent a Diagnostic Session Control positive response message it shall have re-locked the server if the client unlocked it during the diagnostic session.

Request Message		**Message Length: 2 Bytes**
Byte number	Value (Hex)	Parameter Name
1	10	Service ID: Diagnostic Session Control
2	01	Sub function: Default Diagnostic Session

Positive Response Message	**Message Length: 6 Bytes**

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Byte number	Value (Hex)	Parameter Name
1	50	Response ID: Diagnostic Session Control
2	01	Default Diagnostic Session
3	xx	P2ServerMax (MSB)
4	xx	P2ServerMax (LSB)
5	yy	P2*ServerMax (MSB)
6	yy	P2*ServerMax (LSB)

Note: Resolution of P2 is 1ms and P2* is 10ms.

3.1.2 Programming Session: $10 $02

This diagnostic Session enables all diagnostic services required to support the memory programming of a server. In case the server runs the programming Session in the boot software, the programming Session shall only be left via an ECU Reset (0x11) service initiated by the client, a Diagnostic Session Control (0x10) service with session Type equal to default Session, or a session layer timeout in the server. In case the server runs in the boot software when it receives the Diagnostic Session Control (0x10) service with session Type equal to default Session or a session layer timeout occurs, and a valid application software is present for both cases then the server shall restart the application software.

Request Message		**Message Length: 2 Bytes**
Byte number	Value (Hex)	Parameter Name
1	10	Service ID: Diagnostic Session Control
2	02	Sub function: Programming Session

Positive Response Message		**Message Length: 6 Bytes**
Byte number	Value (Hex)	Parameter Name
1	50	Response ID: Diagnostic Session Control
2	02	Programming Session
3	xx	P2ServerMax (MSB)
4	xx	P2ServerMax (LSB) (in ms)
5	xx	P2*ServerMax (MSB)
6	xx	P2*ServerMax (LSB) (in ms)

Note1: Resolution of P2 is 1ms and P2* is 10ms.

Note2: Inform to Rider or Display

Precondition:

1) Vehicle speed = 0KMPH

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3.1.3 Extended Diagnostic Session: $10 $03

This diagnostic Session can be used to enable all diagnostic services required to support the adjustment of functions like "Idle Speed, CO Value, etc." in the server's memory. It can also be used to enable diagnostic services, which are not specifically tied to the adjustment of functions.

Request Message		**Message Length: 2 Bytes**
Byte number	Value (Hex)	Parameter Name
1	10	Service ID: Diagnostic Session Control
2	03	Sub function: Extended Diagnostic Session

Positive Response Message		**Message Length: 6 Bytes**
Byte number	Value (Hex)	Parameter Name
1	50	Response ID: Diagnostic Session Control
2	03	Extended Diagnostic Session
3	XX	P2ServerMax (MSB)
4	XX	P2ServerMax (LSB) (in ms)
5	XX	P2*ServerMax (MSB)
6	XX	P2*ServerMax (LSB) (in ms)

Note: Resolution of P2 is 1ms and P2* is 10ms.

Note1: Inform on CAN bus that it is **Test mode** / ECU is not functional

Note2: Inform to Rider or Display

Precondition:

Vehicle speed = 0KMPH

3.1.4 Client Session: $10 $40

This diagnostic Session can be used to enable all diagnostic services required to support the adjustment of functions like "programing client services. It can also be used to enable diagnostic services, which are used to reprogram the server.

Request Message		**Message Length: 2 Bytes**
Byte number	Value (Hex)	Parameter Name
1	10	Service ID: Diagnostic Session Control
2	40	Sub function: Client Session

Positive Response Message		**Message Length: 6 Bytes**
Byte number	Value (Hex)	Parameter Name
1	50	Response ID: Diagnostic Session Control
2	40	Client Session

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3	XX	P2ServerMax (MSB)
4	XX	P2ServerMax (LSB) (in ms)
5	XX	P2*ServerMax (MSB)
6	XX	P2*ServerMax (LSB) (in ms)

Note1: Inform on CAN bus that it is **Test mode** / ECU Feature Application is not running

Note2: Inform to Rider or Display

Note3: This session is applicable to the ECU which is supposed to reprogram other ECUs on the network.

Precondition:

Vehicle speed 0 /1KMPH

3.1.5 EOL Session: $10 $50

This diagnostic Session shall be used to enable all diagnostic services required to support the EOL services.

Request Message		**Message Length: 2 Bytes**
Byte number	Value (Hex)	Parameter Name
1	10	Service ID: Diagnostic Session Control
2	50	Sub function: EOL Session

Positive Response Message		**Message Length: 6 Bytes**
Byte number	Value (Hex)	Parameter Name
1	50	Response ID: Diagnostic Session Control
2	50	EOL Session
3	XX	P2ServerMax (MSB)
4	XX	P2ServerMax (LSB) (in ms)
5	XX	P2*ServerMax (MSB)
6	XX	P2*ServerMax (LSB) (in ms)

Precondition:

Vehicle speed 0 /1KMPH

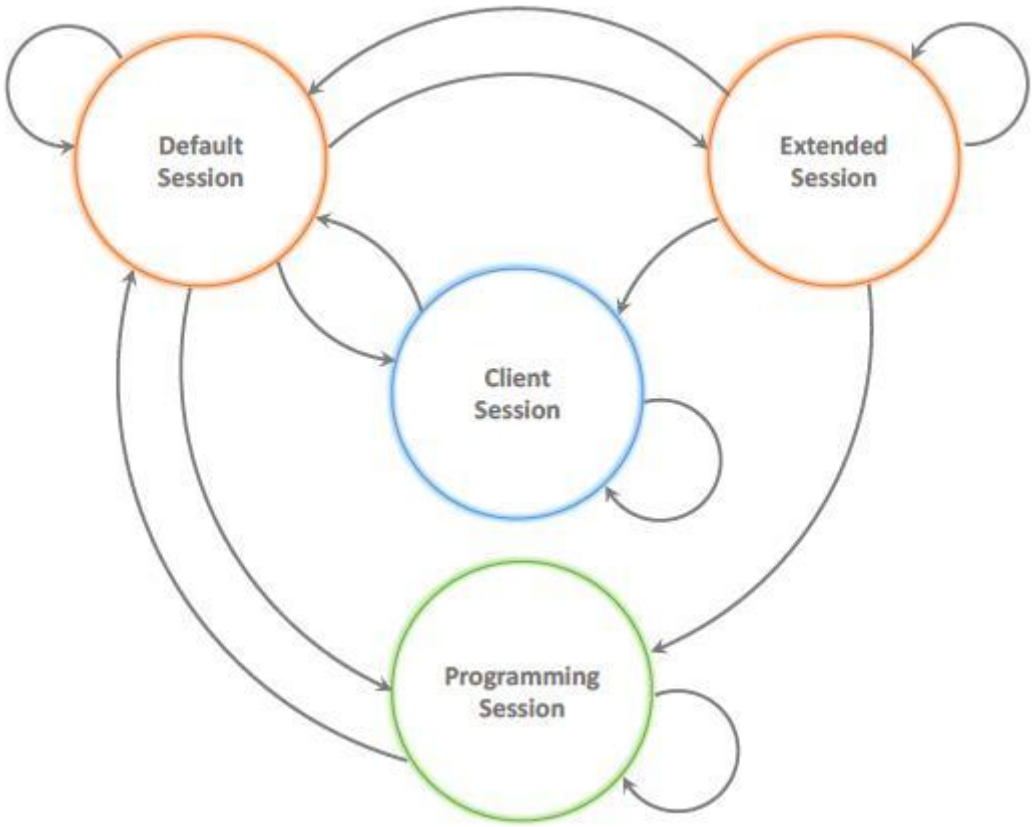
3.1.6 Supported NRCs

Negative Response	
Server Response **(Hex)**	Response Description
12	Subfunction not supported.
13	Incorrect message length or Invalid format.

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22	Pre-conditions are not met to enter the requested diagnostic session.
78	Response pending
7E	Subfunction not supported in active diagnostic session

3.1.7 Service $10 Flow Diagram



Note: Transition from Default to Programming session is only allowed in bootloader but not in application.

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3.2 Diagnostic Trouble Codes

Format of DTC configuration is:
DTC_CONFIG (dtc_name, dtc_number, obd_relevant, priority, mil_relevant, test_cycle, healing_cycle, conf_threshold, heal_threshold, flt_absent_inc_rate, flt_absent_dec_rate, flt_threshold, flt_present_inc_rate, flt_present_dec_rate, fail_func_link, description_string).

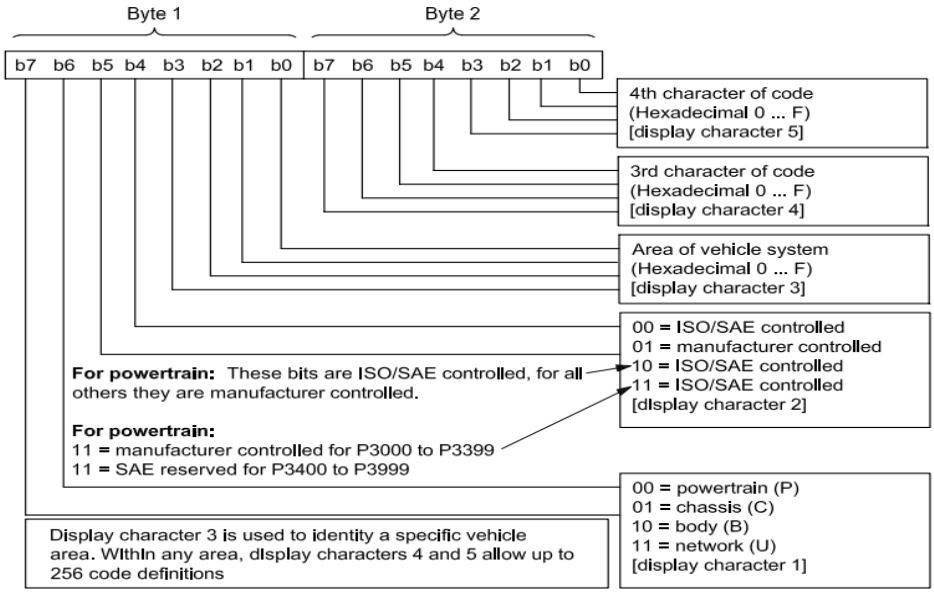
- conf_threshold: Specifies the DTC confirmation threshold i.e. the number of test cycles a fault must be detected for the DTC to transition from 'PENDING' to 'CONFIRMED & PENDING'. The confirmation threshold must not be greater than ***10***.

- heal_threshold: Specifies the DTC healing threshold i.e. the number of healing cycles a fault must be absent for it to be cleared from the fault store. Must be set to ***40*** for DTCs.

Note:
The list of the DTCs can be found in the annexure “TVSM_Diagnostics_Info_Ver7”.

statusOfDTC: bit field name	Bit #	Bit state	Description
testFailed	0	1	DTC is failed at the time of the request
testFailedThisOperationCycle	1	1	DTC failed on the current operation cycle
pendingDTC	2	1	DTC failed on the current or previous operation cycle
confirmedDTC	3	1	DTC is confirmed at the time of the request
testNotCompletedSinceLastClear	4	0	DTC test were completed since the last code clear
testFailedSinceLastClear	5	1	DTC test failed at least once since last code clear
testNotCompletedThisOperationCycle	6	0	DTC test completed this operation cycle
warningIndicatorRequested	7	0	Server is not requesting warningIndicator to be active

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3.3 Read DTC Information: $19

This service reads the faults information.

Sub-Function supported information		
Byte number	Value (Hex)	Name
1	02	Report DTC by Status Mask
2	04	Report DTC Snapshot Record by DTC Number
3	0A	Report Supported DTC

3.3.1 Report Number Of DTCs By Status Mask: $19 $01

This service allows a diagnostic client to request the **number of Diagnostic Trouble Codes (DTCs)** that match a specified **status mask**. The ECU filters its stored DTCs based on the mask and responds with:

- **DTCStatusAvailabilityMask** – indicates which status bits are supported for filtering.
- **DTCFormatIdentifier** – specifies the format used for DTC encoding.
- **DTCCount** – a 2-byte value representing the number of matching DTCs.

Request Message	Message Length: 3 Bytes	
Byte number	Value (Hex)	Parameter Name
1	19	Service ID: Read DTC Information
2	01	Sub function: Report Number of DTC By Status Mask
3	XX	DTC Status Mask

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Positive Response Message		Message Length: 6 Bytes
Byte number	Value (Hex)	Parameter Name
1	59	Response ID: Read DTC Information
2	01	Sub function: Report Number of DTC By Status Mask
3	FF	DTC Status Availability Mask
4	01	DTC Format Identifier = ISO_14229-1_DTCFormat
5-6	XX	DTC Count

Negative Response	
Server Response **(Hex)**	Response Description
11	Service is not supported
12	Sub-function not supported
13	Incorrect message length or Invalid format
31	The client specified a DTCMaskRecord that was not recognized by the server
78	Response pending
7F	Service not supported in active diagnostic session.

3.3.2 Read DTC by Status Mask: $19 $02

The purpose of this sub-function is to retrieve a list of DTCs which satisfies a tool-defined mask. For each DTC, the ECU performs a bitwise AND with the status mask and the DTC status. Whenever the result of this operation is non-zero, the corresponding DTC is returned.

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	19	Service ID: Read DTC Information
2	02	Sub function: Report DTC by Status Mask
3	XX	DTC Status Mask

Positive Response Message		Message Length: 3 + (n * 4) Bytes
Byte number	Value (Hex)	Parameter Name
1	59	Response ID: Read DTC Information
2	02	Sub function: Report DTC by Status Mask
3	FF	DTC Availability Mask
4	XX	DTC 1 [High byte]
5	XX	DTC 1 [Middle byte]
6	XX	DTC 1 [Low byte]
7	XX	Status of DTC 1
…..	…….	…….

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(n-1)*4 +3	XX	Status of DTC (n-1)
(n*4)	XX	DTC n [High byte]
(n*4) +1	XX	DTC n [(Middle byte]
(n*4) +2	XX	DTC n [(Low byte]
(n*4) +3	XX	Status of DTC n

Negative Response	
Server Response **(Hex)**	Response Description
11	Service is not supported
12	Sub-function not supported
13	Incorrect message length or Invalid format
31	DTC mask record not recognized by the server.
78	Response pending
7F	Service not supported in active diagnostic session.

3.3.3 Report DTC Snapshot Record: $19 $04

Request Message		**Message Length: 6 Bytes**
Byte number	Value (Hex)	Parameter Name
1	19	Service ID: Read DTC Information
2	04	Sub function: Report DTC Snapshot Record by DTC Number
3	XX	DTC (High byte)
4	XX	DTC (Middle byte)
5	XX	DTC (Low byte)
6	01 – FF	DTC Snapshot Record Number

Positive Response Message		**Message Length: Variable length**
Byte number	Value (Hex)	Parameter Name
1	59	Response ID: Read DTC Information
2	04	Sub function: Report DTC Snapshot Record by DTC Number
3	0x00 – 0xFF	DTC (High byte)
4	0x00 – 0xFF	DTC (Middle byte)
5	0x00 – 0xFF	DTC (Low byte)
6	0x00 – 0xFF	Status of DTC
7	0x00 – 0xFF	DTC Snapshot Record Number#1
8	0x00 – 0xFF	DTC Snapshot Record Number of Identifiers#1
9	0x00 – 0xFF	dataIdentifier#1 byte#1 (MSB)
10	0x00 – 0xFF	dataIdentifier#1 byte#2 (LSB)
11	0x00 – 0xFF	snapshotData#1 byte#1
..	..	..

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11 + (p-1)	0x00 – 0xFF	snapshotData#1 byte#p
..	..	..
r-(m-1)-2	0x00 – 0xFF	dataIdentifier#w byte#1 (MSB)
r-(m-1)-1	0x00 – 0xFF	dataIdentifier#w byte#2 (LSB)
r-(m-1)	0x00 – 0xFF	snapshotData#w byte#1
..	..	...
r	0x00 – 0xFF	snapshotData#w byte#m
...	...	... (Next DTC Snapshot Record Number, IF ANY)

Negative Response	
Server Response **(Hex)**	Response Description
11	Service is not supported
12	Sub-function not supported
13	Incorrect message length or Invalid format
31	Client specified an invalid DTC Snapshot Record Number.
78	Response pending
7F	Service not supported in active diagnostic session.

3.3.4 ReportDTCExtDataRecordByDTCNumber : $19 $06

This service allows a diagnostic client to request a specific **extended data record** for a given DTC. The client provides:

- A **DTC number** (high, middle, low byte),
- A **DTCExtDataRecordNumber** indicating which extended data record to retrieve.

The ECU searches for the exact DTC and returns:

- The **DTC number and its status**,
- The requested **extended data record**.

If the client sets the record number to 0xFE or 0xFF, the ECU returns **all available extended data records** for that DTC.

Request Message		**Message Length: 6 Bytes**
Byte number	Value (Hex)	Parameter Name
1	19	Service ID: Read DTC Information
2	06	Sub function: reportDTCExtDataRecordByDTCNumber
3-5	XX	DTC Number
6	XX	DTC Extended Data Record Number

Positive Response Message		**Message Length: Variable length**
Byte number	Value (Hex)	Parameter Name
1	59	Response ID: Read DTC Information
2	06	Sub function: reportDTCExtDataRecordByDTCNumber

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3-5	XX	DTC Number (High Byte , Middle Byte , Low Byte)
6	0x00 – 0xFF	Status of DTC
7	0x00 – 0xFF	DTC Extended Data Record Number#1
8	0x00 – 0xFF	DTC Extended Data bytes for Record #1
8+(p-1)	0x00 – 0xFF	Extended Data#1 byte #p
:	:	:
#t	0x00 – 0xFF	Extended Data #1 byte#x
..	..	..
#t + 1	0x00 – 0xFF	Extended Data#1 byte #1
..	..	..
#t+1+(q-1)	0x00 – 0xFF	Extended Data #x byte #q
…	…	… (Next DTC Extended Record Number, IF ANY)

Negative Response	
Server Response **(Hex)**	Response Description
11	Service is not supported
12	Sub-function not supported
13	Incorrect message length or Invalid format
31	The client specified an invalid DTCExtDataRecordNumber
78	Response pending
7F	Service not supported in active diagnostic session.

3.3.5 Report supported DTCs: $19 $0A

To retrieve the status of all DTCs supported by the server.

Request Message		**Message Length: 2 Bytes**
Byte number	Value (Hex)	Parameter Name
1	19	Service ID: Read DTC Information
2	0A	Sub function: Report Supported DTCs [Current and Past DTCs]

Positive Response Message		**Message Length: 3 + (n*4) Bytes**
Byte number	Value (Hex)	Parameter Name
1	59	Response ID: Read DTC Information
2	0A	Sub function: All DTC's
3	FF	DTC Availability Mask
4	XX	DTC 1 [High byte]
5	XX	DTC 1 [Middle byte]
6	XX	DTC 1 [Low byte]
7	XX	Status of DTC 1

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8	XX	DTC 2 [High byte]
…..	…….	…….
(n-1)*4 +3	XX	Status of DTC (n-1)
(n*4)	XX	DTC n [High byte]
(n*4) +1	XX	DTC n [Middle byte]
(n*4) +2	XX	DTC n [(Low byte]
(n*4) +3	XX	Status of DTC n

Negative Response	
Server Response **(Hex)**	Response Description
11	Service is not supported
12	Sub-function not supported
13	Incorrect message length or Invalid format
31	Functional Group Identifier/DTC Mask Record was not recognized by the server.
78	Response pending
7F	Service not supported in active diagnostic session.

3.4 Clear Diagnostic Information: $14 $FF FF FF

Request Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	14	Service ID: Clear Diagnostic Information
2	FF	Group of DTC High byte
3	FF	Group of DTC Middle byte
4	FF	Group of DTC Low byte

Positive Response Message		**Message Length: 1 Byte**
Byte number	Value (Hex)	Parameter Name
1	54	Response ID: Clear Diagnostic Information

Negative Response	
Server Response **(Hex)**	Response Description
11	Service is not supported
13	Incorrect message length or Invalid format
22	Internal conditions within the server prevent the clearing of DTC related information stored in the server.
31	The specified group of DTC parameter is not supported.

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72	Server detects an error while writing to a memory location.
78	Response pending
7F	Service not supported in active diagnostic session.

3.4.1 Clear Diagnostic Information Specific DTC: $14 $XX XX XX

Request Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	14	Service ID: Clear Diagnostic Information
2	XX	DTC High byte
3	XX	DTC Middle byte
4	XX	DTC Low byte

Positive Response Message		**Message Length: 1 Byte**
Byte number	Value (Hex)	Parameter Name
1	54	Response ID: Clear Diagnostic Information (Clear specific single DTC)

Negative Response	
Server Response **(Hex)**	Response Description
11	Service is not supported
13	Incorrect message length or Invalid format
22	Internal conditions within the server prevent the clearing of DTC related information stored in the server.
31	The specified DTC parameter is not supported.
72	Server detects an error while writing to a memory location.
78	Response pending
7F	Service not supported in active diagnostic session.

Note: Clearing of an individual DTC is optional.

3.5 ECU Reset: $11

The ECU Reset service is used by the client to request a server reset. This service requests the server to effectively perform a server reset based on the content of the reset Type parameter value embedded in the ECU Reset request message. The ECU Reset positive response message (if required) shall be sent before the reset is executed in the server(s). After a successful server reset the server shall activate the default session.

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Request Message		**Message Length: 2 Bytes**
Byte number	Value (Hex)	Parameter Name
1	11	Service ID: ECU Reset
2	XX	Sub function: 01 -> Hard Reset, 02 -> Key ON-OFF, 03 -> Soft Reset

Positive Response Message		**Message Length: 2 Bytes**
Byte number	Value (Hex)	Parameter Name
1	51	Response ID: ECU Reset
2	XX	Sub function: 01 -> Hard Reset, 02 -> Key ON-OFF, 03 -> Soft Reset

Negative Response	
Server Response **(Hex)**	Response Description
12	Subfunction not supported in any diagnostic session.
13	Incorrect message length or Invalid format.
22	Criteria for the ECU Reset request is not met.
33	Server is not in unlocked state.
78	Response pending.
7E	Sub-function not supported in active session.
7F	Service not supported in active session.

01 - Hard Reset:
This simulates the power-on /start-up sequence typically performed after a server has been previously disconnected from its power supply (i.e. battery).

02 - Key ON-OFF Reset:
This is like the driver turning the ignition key off and back on. This reset condition should simulate a key-off-on sequence (i.e. interrupting the switched power supply).

03 - Soft Reset:
This causes the server to immediately restart the application program if applicable. (Can be a watchdog reset.)

Function to Reset:
1. All Global and static variable re-initiate
2. Enable DTC transmission
3. Enable Tx normal message transmission
4. Jump back to Application if Application valid
5. Keep default settings in Vehicle

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3.6 Read Data by Identifier: $22

The Read Data by Identifier service allows the client to request data record values from the server identified by one or more Data Identifiers. The client request message contains one or more two-byte Data Identifier values that identify data record(s) maintained by the server.
The format and definition of the Data Record shall be vehicle manufacturer or system supplier specific, and may include analog input and output signals, digital input and output signals, internal data, and system status information if supported by the server.

Supported NRCs

Negative Response	
Server Response **(Hex)**	Response Description
13	Incorrect message length or Invalid Format.
14	Response length exceeds the limit of the underlying transport protocol.
22	Operating conditions of the server are not met to perform the required action.
31	The requested DID is either is not supported or has not been assigned.
33	If the DID is secured and the server is not in unlocked state.
7F	Service not supported in active session.

3.6.1 Test Stamp: $22 $10 FF

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	10	ID High byte: Test Stamp High Byte
3	FF	ID Low byte: Test Stamp High Byte

Positive Response Message		**Message Length: 6 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	10	ID High byte: Session status
3	FF	ID Low byte: Session status
4	XX	Test-stamp Data byte 1 (MSB)
5	XX	Test-stamp Data byte 2
6	XX	Test-stamp Data byte 3 (LSB)

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The Test-stamp Data bytes are to be mutually decided between Diagnostic Tool provider and TVSM.
Note: It is used for both Read $22 and write $2E services.

3.6.2 Active Diagnostic Session: $22 $F1 86

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Session status
3	86	ID Low byte: Session status

Positive Response Message		**Message Length: 5 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Session status
3	86	ID Low byte: Session status
4	XX	Active Session 0x01 => Default session 0x02 => Programming session 0x03 => Extended session
5	XX	Software Identifier 0x01 => Bootloader Software 0x02 => Application Software

3.6.3 VIN (Vehicle Identification Number): $22 $F1 90

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: VIN Request High byte
3	90	ID Low byte: VIN Request Low byte

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Positive Response Message		Message Length: 20 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: VIN Request High byte
3	90	ID Low byte: VIN Request Low byte
4	XX	VIN byte 1 [ASCII] (MSB)
5	XX	VIN byte 2 [ASCII]
6	XX	VIN byte 3 [ASCII]
7	XX	VIN byte 4 [ASCII]
8	XX	VIN byte 5 [ASCII]
9	XX	VIN byte 6 [ASCII]
10	XX	VIN byte 7 [ASCII]
11	XX	VIN byte 8 [ASCII]
12	XX	VIN byte 9 [ASCII]
13	XX	VIN byte 10 [ASCII]
14	XX	VIN byte 11 [ASCII]
15	XX	VIN byte 12 [ASCII]
16	XX	VIN byte 13 [ASCII]
17	XX	VIN byte 14 [ASCII]
18	XX	VIN byte 15 [ASCII]
19	XX	VIN byte 16 [ASCII]
20	XX	VIN byte 17 [ASCII] (LSB)

Note: It is used for both Read $22 and write $2E services.

3.6.4 SKU (Stock keeping Unit): $22 $ F1 91

This DID is used to read the vehicle variant information.

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: SKU High byte
3	91	ID Low byte: SKU Low byte

Positive Response Message		Message Length: 19 Bytes
Byte number	Value (Hex)	Parameter Name

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1	62	Response ID: Read data by Identifier
2	F1	ID High byte: SKU High byte
3	91	ID Low byte: SKU Low byte
4-19	XX	SKU (ASCII)

Note: Unused bytes of the SKU shall be filled with 0x2D (at the end).

3.6.5 Application Software Version: $22 $ F1 81

This service is used to read the Application Software details.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Software Details Request High byte
3	81	ID Low byte: Software Details Request Low byte

Positive Response Message		**Message Length: 27 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Software Details Request High byte
3	81	ID Low byte: Software Details Request Low byte
4	XX	Vehicle Category and Engine type
5-15	XX	SAP SW part Number [ASCII]
16-18	XX	Vehicle Identification [ASCII]
19	XX	Software type [HEX]
20	XX	ECU Identification [HEX]
21	XX	Major Version [HEX]
22	XX	Minor Version [HEX]
23	XX	Micro Version [HEX]
24	XX	Software Release Category [Hex]
25	XX	Year
26	XX	Month
27	XX	Date

Note:
1. Refer to the Reference document (c) for the value table and Reference document (b) for the example.
2. Unused data bytes shall be set to 0x2D (at the end).

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3.6.6 Hardware Version: $22 $ F1 93

Hardware details

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Hardware Details Request High byte
3	93	ID Low byte: Hardware Details Request Low byte

Positive Response Message		Message Length: 21 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Hardware Details Request High byte
3	93	ID Low byte: Hardware Details Request Low byte
4-14	XX	SAP HW part Number [ASCII]
15	XX	ECU Identification [HEX]
16	XX	Major Version [HEX]
17	XX	Minor Version [HEX]
18	XX	Micro Version [HEX]
19	XX	BOM [HEX]
20	XX	Reserved Byte
21	XX	Reserved Byte

Note:

1. Refer to the Reference document (c) for the value table and Reference document (b) for the example.
2. Unused data bytes shall be set to 0x2D(at the end).

3.6.7 Checksum: $22 $ F1 F0

Read the application Checksum.

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	ServiceID: Read Data by Identifier
2	F1	ID High byte: Checksum Details Request High byte
3	F0	ID Low byte: Checksum Details Request Low byte

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Positive Response Message		Message Length: 5 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Programming Date Request High byte
3	F0	ID Low byte: Programming Date Request Low byte
4	XX	Checksum Byte1 [Hex] (MSB)
5	XX	Checksum Byte2 [Hex] (LSB)

3.6.8 Pack Variant: $22 $ F1 01

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: pack variant ID high byte
3	01	ID Low byte: pack variant ID low byte

Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: pack variant ID high byte
3	01	ID Low byte: pack variant ID low byte
4	XX	Variant

3.6.9 ODO Read: $22 $ F1 0A

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: ODO read ID high byte
3	0A	ID Low byte: ODO read ID low byte

Positive Response Message		Message Length: 7 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: ODO read ID high byte

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3	0A	ID Low byte: ODO read ID low byte
4 - 7	XX	Odometer

3.6.10 IGN Counter Read: $22 $ F1 10

The DID serves as a counter for ignition activations.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: IGN counter read high byte
3	10	ID Low byte: IGN counter read low byte

Positive Response Message		**Message Length: 6 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: IGN counter read high byte
3	10	ID Low byte: IGN counter read low byte
4	XX	Hex Value (MSB)
5	XX	Hex Value
6	XX	Hex Value (LSB)

3.6.11 Reprograming times Read: $22 $ F1 02

The DID is used to read the no.of times the ECU is programmed (Re-flashed via UDS).

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Reprogramming times read ID high byte
3	02	ID Low byte: Reprogramming times read ID low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Reprogramming times read ID high byte

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3	02	ID Low byte: Reprogramming times read ID low byte
4	XX	Hex Value

3.6.12 Crash Counter Read: $22 $ F1 20

When the Watchdog reset occurs, this counter value can be incremented and stored in the EEPROM.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Crash counter read ID high byte
3	20	ID Low byte: Crash counter read ID low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Crash counter read ID high byte
3	20	ID Low byte: Crash counter read ID low byte
4	XX	Hex Value

3.6.13 ECU Hardware Serial Number: $22 $ F1 8A

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: ECU serial number ID high byte
3	8A	ID Low byte: ECU serial number ID low byte

Positive Response Message		**Message Length: 9 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: ECU serial number ID high byte
3	8A	ID Low byte: ECU serial number ID low byte
4	XX	Hex Value (MSB)

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5	XX	Hex Value
6	XX	Hex Value
7	XX	Hex Value
8	XX	Hex Value
9	XX	Hex Value (LSB)

3.6.14 Vehicle Software Version: $22 $ F1 89

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Vehicle software version ID high byte
3	89	ID Low byte: Vehicle software version ID low byte

Positive Response Message		**Message Length: 12 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Vehicle software version ID high byte
3	89	ID Low byte: Vehicle software version ID low byte
4	XX	ASCII Value (MSB)
5	XX	ASCII Value
6	XX	ASCII Value
7	XX	ASCII Value
8	XX	ASCII Value
9	XX	ASCII Value
10	XX	ASCII Value
11	XX	ASCII Value
12	XX	ASCII Value (LSB)

3.6.15 Vehicle Level Battery Pack Counter: $22 $ F1 A0

This service is used to read the number of battery packs supported in the vehicle.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name

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1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: vehicle level battery pack counter high byte
3	A0	ID Low byte: vehicle level battery pack counter low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: vehicle level battery pack counter high byte
3	A0	ID Low byte: vehicle level battery pack counter low byte
4	XX	Hex Value

Negative Response	
Server Response **(Hex)**	Response Description
13	Incorrect message length or Invalid format
14	Diagnostic response too long
22	Conditions Not Correct
31	Request out of range
33	Security Access Denied
78	Response pending

3.6.16 Bootloader Software Version: $22 $ F1 85

This service is used to read the bootloader software version.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: BL Software Details Request High byte
3	85	ID Low byte: BL Software Details Request Low byte

Positive Response Message		**Message Length: 27 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: BL Software Details Request High byte
3	85	ID Low byte: BL Software Details Request Low byte
4	XX	Vehicle Category and Engine type

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Positive Response Message		Message Length: 27 Bytes
Byte number	Value (Hex)	Parameter Name
5-15	XX	SAP SW part Number [ASCII]
16-18	XX	Vehicle Identification [ASCII]
19	XX	Software type [HEX]
20	XX	ECU Identification [HEX]
21	XX	BL Major Version [HEX]
22	XX	BL Minor Version [HEX]
23	XX	BL Micro Version [Hex]
24	XX	Software Category [Hex]
25	XX	Year
26	XX	Month
27	XX	Date

Note: Refer to the Reference document (c) for the value table.

3.6.17 GA Number/Software Bom ID: $22 $ F1 88

This service is used to read the General Assembly/Software assembly number (8 Bytes).

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: GA number Request High byte
3	88	ID Low byte: GA number Request Low byte

Positive Response Message		Message Length: 11 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: GA number Request High byte
3	88	ID Low byte: GA number Request Low byte
4	XX	GA number [ASCII] (MSB)
5	XX	GA number [ASCII]
6	XX	GA number [HEX]
7	XX	GA number [HEX]
8	XX	GA number [HEX]
9	XX	GA number [HEX]
10	XX	GA number [HEX]

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11	XX	GA number [HEX] (LSB)

3.6.18 Application Presence Pattern: $22 $ F1 82

This service is used to read the Application Presence Pattern (4 Bytes).

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Application Presence Pattern High byte
3	82	ID Low byte: Application Presence Pattern Low byte

Positive Response Message		**Message Length: 7 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Application Presence Pattern High byte
3	82	ID Low byte: Application Presence Pattern Low byte
4	XX	Application Presence Pattern: Byte 1 (MSB)
5	XX	Application Presence Pattern: Byte 2
6	XX	Application Presence Pattern: Byte 3
7	XX	Application Presence Pattern: Byte 4 (LSB)

3.6.19 Fingerprint/Tester serial Number: $22 $ F1 84

This service is used to read Fingerprint (2 Bytes).

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Fingerprint request High byte
3	84	ID Low byte: Fingerprint request Low byte

Positive Response Message		**Message Length: 5 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Fingerprint request High byte

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3	84	ID Low byte: Fingerprint request Low byte
4	XX	Index (MSB)
5	XX	Index (LSB)

Value Table		
Sl. No.	Index (Hex)	Client/Tester Name
1	00 01	Infotainment System
2	00 02	Telematics Control unit
3	00 03	MAHLE tool
4	00 04	CAPL script
5	00 05	R&D SW CoC UDS Programmer
6	00 06 – 00 FF	Reserved for future

3.6.20 Vehicle's Current Mobility Status: $22 $ F1 0C

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Vehicle's Current Mobility Status ID high byte
3	0C	ID Low byte: Vehicle's Current Mobility Status ID low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Vehicle's Current Mobility Status ID high byte
3	0C	ID Low byte: Vehicle's Current Mobility Status ID low byte
4	XX	01: Vehicle is Immobilized 00: Vehicle is Mobilized

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3.6.21 Partition Verify: $22 $ F1 1B

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: partition verify ID high byte
3	1B	ID Low byte: partition verify ID low byte

Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: partition verify ID high byte
3	1B	ID Low byte: partition verify ID low byte
4	XX	01: Partition Present 00: Partition not present

Note: This is microcontroller dependent. This is valid where the flash memory partitioning is required.

3.6.22 Key Verify: $22 $ F1 1C

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Key verify high byte
3	1C	ID Low byte: Key verify low byte

Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Key verify high byte
3	1C	ID Low byte: Key verify low byte
4	XX	01: Key Present 00: Key not present

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3.6.23 EMS Unique Code: $22 $ F1 21

The purpose of this service is to read the Vendor's Unique code to identify which manufacturer has manufactured the hardware.
At present, the following vendor codes exist:

1. 21669 for the SGS
2. 20430 for the Flextronics

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte
3	21	ID Low byte

Positive Response Message		**Message Length: 8 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte
3	21	ID Low byte
4	XX	EMS Unique Code [ASCII] (MSB)
5	XX	EMS Unique Code [ASCII]
6	XX	EMS Unique Code [ASCII]
7	XX	EMS Unique Code [ASCII]
8	XX	EMS Unique Code [ASCII] (LSB)

Note: Writing of EMS unique code should be done while performing FCT at the supplier EOL.

3.6.24 Date of Manufacturing: $22 $ F1 22

The purpose of this service is to read the Date of Manufacturing of the ECU board.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte:
3	22	ID Low byte:

Positive Response Message		**Message Length: 6 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier

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2	F1	ID High byte:
3	22	ID Low byte:
4	XX	Date (MSB)
5	XX	Month
6	XX	Year (LSB)

Note: Writing of Date of manufacturing should be done while performing FCT at the supplier EOL.

3.6.25 Battery UID or BIN: $22 $ F1 1E

The purpose of this service is to read the Battery unique Identification Number [BIN] (12 Bytes)

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte
3	1E	ID Low byte

Positive Response Message		**Message Length: 15 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte
3	1E	ID Low byte
4	XX	Data Byte 1 (ASCII): Make (TVS marked as “T”)
5	XX	Data Byte 2 (ASCII): Battery Type (MSB)
6	XX	Data Byte 3 (ASCII): Battery Type (LSB)
7	XX	Data Byte 4 (ASCII): Month Code
8	XX	Data Byte 5 (ASCII): Year Code
9	XX	Data Byte 6 (ASCII): Plant Code
10	XX	Data Byte 7 (ASCII): Check Digit
11	XX	Data Byte 8 (HEX): Running Serial Number (MSB)
12	XX	Data Byte 9 (HEX): Running Serial Number
13	XX	Data Byte 10 (HEX): Running Serial Number
14	XX	Data Byte 11 (HEX): Running Serial Number
15	XX	Data Byte 12 (HEX): Running Serial Number (LSB)

Note: Writing of Battery unique code in the Battery pack should be done at the TVSM EOL only which will be limited to one-time. For more Byte-wise details, please refer to the reference document “KE210130_A3”.

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3.6.26 Theta: $22 $ F1 30

The purpose of this service is to read the motor's Initial angle of sync (Theta).

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte
3	30	ID Low byte

Positive Response Message		**Message Length: 13 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte
3	30	ID Low byte
4	XX	Theta Init (MSB)
5	XX	Theta Init (LSB) [0 – 4095; range of 2 bytes]
6	XX	Sine Offset [0 – 255]
7	XX	Cosine Offset [0 – 255]
8	XX	Correction gain (MSB)
9	XX	Correction gain (LSB)
10	XX	Phase lag [0 – 255]
11	XX	Theta Calib status [0 – 3]
12	XX	Theta Init Offset (MSB)
13	XX	Theta Init Offset (LSB)

3.6.27 UDS Spec Version: $22 $ 10 01

The purpose of this service is to read the TVSM Diagnostic Specification document version. This is used by the UDS client to check the compliance of the Software with TVSM Diagnostic specification.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	10	ID High byte
3	01	ID Low byte

Positive Response Message		**Message Length: 5 Bytes**
Byte number	Value (Hex)	Parameter Name

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1	62	Response ID: Read data by Identifier
2	10	ID High byte
3	01	ID Low byte
4	XX	Major Version of the UDS Specification Document
5	XX	Minor Version of the UDS Specification Document

3.6.28 TPMS Pairing Data: $22 $ F1 31

The purpose of this service is to read the MAC ID of all the Tire Pressure Measurement Systems in the vehicle.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte
3	31	ID Low byte

Positive Response Message		**Message Length: Variable length**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte
3	31	ID Low byte
4	X'X''	Position of the tire-1 Upper Nibble (X'): 1 – Front, 2 – Rear Lower Nibble (X''): 0 – Centre (Single), 1 – Left, 2 – Right
5	XX	Data Byte 1 (HEX): MAC ID of tire-1 TPMS (MSB)
6	XX	Data Byte 2 (HEX): MAC ID of tire-1 TPMS
7	XX	Data Byte 3 (HEX): MAC ID of tire-1 TPMS
8	XX	Data Byte 4 (HEX): MAC ID of tire-1 TPMS
9	XX	Data Byte 5 (HEX): MAC ID of tire-1 TPMS
10	XX	Data Byte 6 (HEX): MAC ID of tire-1 TPMS (LSB)
11	Y'Y''	Position of the tire-2 Upper Nibble (X'): 1 – Front, 2 – Rear Lower Nibble (X''): 0 – Centre (Single), 1 – Left, 2 – Right
12	YY	Data Byte 1 (HEX): MAC ID of tire-2 TPMS (MSB)
13	YY	Data Byte 2 (HEX): MAC ID of tire-2 TPMS
14	. . .	. . .

Note:

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1. The length of the response is variable and depends on the Number of TPMS devices present in the vehicle.
2. For example, the response for a vehicle with 1 TPMS will have response length of 10 Bytes.
the response for a vehicle with 2 TPMS will have response length of 17 Bytes and so on.

3.6.29 Tire Pressure Threshold: $22 $ F1 33

This shall be used when all the wheels of the vehicle have the same pressure threshold values.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier: Live Parameter
2	F1	ID High byte: Read Data by Identifier High byte
3	33	ID Low byte: Read Data by Identifier Low byte

Positive Response Message		**Message Length: 7 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Read Data by Identifier High byte
3	33	ID Low byte: Read Data by Identifier Low byte
4	XX	Tire Pressure Threshold: Very Low Pressure
5	XX	Tire Pressure Threshold: Low Pressure
6	XX	Tire Pressure Threshold: High Pressure
7	XX	Tire Pressure Threshold: Very High Pressure

3.6.30 Deleted

3.6.31 Deleted

3.6.32 Immobilization feature Status: $22 $ F1 36

The DID is used to read the status of immobilization feature in the vehicle.
The immobilization status shall be interpreted as mentioned below.
01: Enable
00: Disable

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name

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1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Immobilizer High byte
3	36	ID Low byte: Immobilizer Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Immobilizer High byte
3	36	ID Low byte: Immobilizer Low byte
4	XX	Immobilization Enable Status

3.6.33 Hill Hold Feature: $22 $ F1 37

The purpose of this DID is to read the pitch angle of the vehicle.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Hill Hold High byte
3	37	ID Low byte: Hill Hold Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Hill Hold High byte
3	37	ID Low byte: Hill Hold Low byte
4	XX	HEX: Pitch angle (degrees)

3.6.34 MAC_ID of Bluetooth chip: $22 $ B1 D5

The purpose of this DID is to read the MAC_ID of Bluetooth chip.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	B1	ID High byte: MAC_ID High byte

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3	D5	ID Low byte: MAC_ID Low byte

Positive Response Message		**Message Length: 9 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	B1	ID High byte: MAC_ID High byte
3	D5	ID Low byte: MAC_ID Low byte
4	XX	Byte1[Hex][MSB]
5	XX	Byte2[Hex]
6	XX	Byte3[Hex]
7	XX	Byte4[Hex]
8	XX	Byte5[Hex]
9	XX	Byte6[Hex][LSB]

3.6.35 IMEI Number: $22 $ F1 43

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: IMEI number High byte
3	43	ID Low byte: IMEI number Low byte

Positive Response Message		**Message Length: 18 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: IMEI number High byte
3	43	ID Low byte: IMEI number Low byte
4	XX	Byte1[ASCII][MSB]
5	XX	Byte2[ASCII]
6	XX	Byte3[ASCII]
7	XX	Byte4[ASCII]
8	XX	Byte5[ASCII]
9	XX	Byte6[ASCII]
10	XX	Byte7[ASCII]
11	XX	Byte8[ASCII]

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12	XX	Byte9[ASCII]
13	XX	Byte10[ASCII]
14	XX	Byte11[ASCII]
15	XX	Byte12[ASCII]
16	XX	Byte13[ASCII]
17	XX	Byte14[ASCII]
18	XX	Byte15[ASCII] [LSB]

3.6.36 Battery Voltage: $22 $ E1 42

Voltage shall be provided in the hex format with resolution of 0.1V

Example:

12.8V shall be represented as 0x80.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	E1	ID High byte: Battery voltage High byte
3	42	ID Low byte: Battery voltage Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	E1	ID High byte: Battery voltage High byte
3	42	ID Low byte: Battery voltage Low byte
4	XX	Battery Voltage [Hex]

3.6.37 Photosensor: $22 $ E1 1B

The luminance percentage is returned as a single byte value with a resolution of 1%

Example:

If the illumination is 58%, then the response byte value will be 0x3A

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	E1	ID High byte: Photosensor High byte

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3	1B	ID Low byte: Photosensor Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	E1	ID High byte: Photosensor High byte
3	1B	ID Low byte: Photosensor Low byte
4	XX	Luminance Percentage [Hex]

3.6.38 Ignition Status: $22 $ E1 1C

The ignition status is returned as a Boolean value in a 1 byte

Example:

If the ignition is ON, then the response byte will be 0x1

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	E1	ID High byte: IGN status High byte
3	1C	ID Low byte: IGN status Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	E1	ID High byte: IGN status High byte
3	1C	ID Low byte: IGN status Low byte
4	XX	Ignition Status [Boolean]

3.6.39 Current Fuel Volume: $22 $ E0 64

This is the calculated Fuel volume remaining in the fuel tank. This shall be provided in hex format with a resolution of 0.1 Liter.

Example:

If the value is 6.5l, then the response byte will be 0x41

If the value is 7.1l, then the response byte will be 0x47 & 0x04 is default I/O control mode

Request Message	**Message Length: 3 Bytes**

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Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	E0	ID High byte: Fuel volume High byte
3	64	ID Low byte: Fuel volume Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	E0	ID High byte: Fuel volume High byteh
3	64	ID Low byte: Fuel volume Low byte
4	XX	Fuel Volume [Hex]

3.6.40 Fuel level sensor resistance: $22 $ E0 65

Fuel level sensor value shall be provided in hex format with 1ohm resolution

Example:

If the resistance value is 300 ohms, the response will be 0x012C

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	E0	ID High byte: Fuel level sensor resistance High byte
3	65	ID Low byte: Fuel level sensor resistance Low byte

Positive Response Message		**Message Length: 5 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	E0	ID High byte: Fuel level sensor resistance High byte
3	65	ID Low byte: Fuel level sensor resistance Low byte
4	XX	Fuel level sensor resistance [Hex]
5	XX	Fuel level sensor resistance [Hex]

3.6.41 Gear position: $22 $ E1 27

Gear position value shall be provided in ASCII format in 1 byte

Gear-ASCII Value	Gear-Hex Value	Gear Description
0	0X30	Reserved
1	0X31	Gear 1
2	0X32	Gear 2
3	0X33	Gear 3

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4	0X34	Gear 4
5	0X35	Gear 5
6	0X36	Gear 6
7	0X37	Gear 7
8	0X38	Gear 8
9	0X39	Gear 9
F	0X46	Forward
N	0X4E	Neutral
P	0X50	Parking
I	0X49	Invalid Signal
R	0X52	Reverse
G	0X47	Reserved

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	E1	ID High byte: Gear position High byte
3	27	ID Low byte: Gear position Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	E1	ID High byte: Gear position High byte
3	27	ID Low byte: Gear position Low byte
4	XX	Gear position [ASCII]

3.6.42 High beam Input switch: $22 $ E1 C8

High beam input switch status will be provided as Boolean value in 1 byte.

Example:

If the High beam input status is active, then the response byte value will be 0x1.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	E1	ID High byte: High beam input switch High byte
3	C8	ID Low byte: High beam input switch Low byte

Positive Response Message	**Message Length: 4 Bytes**

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Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	E1	ID High byte: High beam input switch High byte
3	C8	ID Low byte: High beam input switch Low byte
4	XX	Byte1[Boolean]

3.6.43 Indicator switch Activation Status: $22 $ E1 C9

This DID represent the activation status of indicator switch. Each switch is assigned a bit.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	E1	ID High byte: Indicator input switch High byte
3	C9	ID Low byte: Indicator input switch Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	E1	ID High byte: Indicator input switch High byte
3	C9	ID Low byte: Indicator input switch Low byte
4	XX	Bit 7-2: unused (shall be set to 0) Bit 1: Left Indicator Bit 0: Right Indicator

3.6.44 Distance Unit: $22 $ E1 17

This DID represent the unit of the distance.
If the byte value is 0, unit is in kilometers (default).
If the byte value is 1, unit is in miles.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	E1	ID High byte: Distance unit High byte
3	17	ID Low byte: Distance unit Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name

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1	62	Response ID: Read data by Identifier
2	E1	ID High byte: Distance unit High byte
3	17	ID Low byte: Distance unit Low byte
4	XX	Byte1[Boolean]

3.6.45 Indicator Activation Status: $22 $ E1 B3

The indicator activation status shall be provided in Boolean format

- Bit_3 -> Rear Right Indicator
- Bit_2 -> Front Right Indicator
- Bit_1 -> Rear Left Indicator
- Bit_0 -> Front Left Indicator

Example:

Rear Right & Front Right Indicators are ON. Then response byte value will be 0x0C

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	E1	ID High byte: Indicator status High byte
3	B3	ID Low byte: Indicator status Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	E1	ID High byte: Indicator status High byte
3	B3	ID Low byte: Indicator status Low byte
4	XX	Byte1[Hex]

3.6.46 Bluetooth Connection status: $22 $ B1 D0

BLE connection status shall be provided in 1 byte.

Example:

If BLE is in the connected state, then the response byte value shall be 0x1.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	B1	ID High byte: Bluetooth connection status High byte
3	D0	ID Low byte: Bluetooth connection status Low byte

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Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	B1	ID High byte: Bluetooth connection status High byte
3	D0	ID Low byte: Bluetooth connection status Low byte
4	XX	Byte1[Boolean]

3.6.47 Mode connection Status: $22 $ E1 D0

The control switch status shall be provided in Boolean format

- Bit_3 -> Control switch 4(DOWN)
- Bit_2 -> Control switch 3(MENU)
- Bit 1 -> Control switch 2(BACK)
- Bit_0 -> Control switch 1(UP)

Example: If Menu is entered, 0x04 is the value to be written

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	E1	ID High byte: Mode connection status High byte
3	D0	ID Low byte: Mode connection status Low byte

Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	E1	ID High byte: Mode connection status High byte
3	D0	ID Low byte: Mode connection status Low byte
4	XX	Byte1[Boolean]

3.6.48 Hazard Switch Status: $22 $ E1 D3

Hazard input switch status shall be provided as Boolean value in 1 byte.

Example: If the Hazard input switch status is active then the response value will be 0x1.

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	E1	ID High byte: Hazard switch status High byte
3	D3	ID Low byte: Hazard switch status Low byte

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Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	E1	ID High byte: Hazard switch status High byte
3	D3	ID Low byte: Hazard switch status Low byte
4	XX	Byte1[Boolean]

3.6.49 Speedometer SW version number: $22 $ F1 95

Software version number format shall be MM:MM:mm:ii (in BCD format).

- MM-Major number
- mm - minor number
- li-iteration

Example: If the version is 00.00.12.3, then the frame shall be 0x00 0x00 0x0C 0x03.

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Speedometer SW version number High byte
3	95	ID Low byte: Speedometer SW version number Low byte

Positive Response Message		Message Length: 7 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Speedometer SW version number High byte
3	95	ID Low byte: Speedometer SW version number Low byte
4	XX	Byte1[Hex]
5	XX	Byte2[Hex]
6	XX	Byte3[Hex]
7	XX	Byte4[Hex]

3.6.50 Bluetooth firmware number: $22 $ B1 D1

Firmware version number format shall be MM: mm (in BCD format)

- MM-Major number
- mm-minor number

Example: If the version is 00:02 then the received frame shall be 0x00 0x02

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Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	B1	ID High byte: Bluetooth firmware version number High byte
3	D1	ID Low byte: Bluetooth firmware version number Low byte

Positive Response Message		Message Length: 6 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	B1	ID High byte: Bluetooth firmware version number High byte
3	D1	ID Low byte: Bluetooth firmware version number Low byte
4	XX	Byte1[Hex]
5	XX	Byte2[Hex]
6	XX	Byte3[Hex]

3.6.51 ABS variant: $22 $ F1 97

2-byte NVM is allocated for ABS Variant.

- 0XE1 -> U489C-flared
- 0xE2 -> U489c-naked
- 0xE3 -> N597

Example: If the variant is U489c Flared then the value is 0x00E1

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: ABS variant High byte
3	97	ID Low byte ABS variant Low byte

Positive Response Message		Message Length: 5 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: ABS variant High byte
3	97	ID Low byte: ABS variant Low byte
4	XX	ABS variant [Hex]
5	XX	ABS variant [Hex]

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3.6.52 Cruise functionality status: $22 $ F1 9E

This DID is used to read whether the Cruise functionality is available or not in the vehicle variant. Cruise Functionality status shall be considered as mentioned below.

Value description

0-Not available

1-Available

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Cruise functionality status High byte
3	9E	ID Low byte: Cruise functionality status Low byte

Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Cruise functionality status High byte
3	9E	ID Low byte: Cruise functionality status Low byte
4	XX	Byte1[Boolean]

3.6.53 Deleted

3.6.54 Deleted

3.6.55 Battery level for front tire: $22 $ B1 D7

Battery level for front tires shall be provided in 1 byte with 0.1V resolution.

Example: 3.8V shall be represented as 0x26

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	B1	ID High byte: Battery level for front tire High byte
3	D7	ID Low byte: Battery level for front tire Low byte

Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	B1	ID High byte: Battery level for rear tire High byte
3	D7	ID Low byte: Battery level for rear tire Low byte

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4	XX	Byte1[Hex]

3.6.56 Battery level for rear tire: $22 $ B1 D8

Battery connection for rear tire shall be provided in 1 byte with 0.1V.
Example: 3.8V shall be represented as 0x26

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	B1	ID High byte: Battery level for rear tire High byte
3	D8	ID Low byte: Battery level for rear tire Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	B1	ID High byte: Battery level for rear tire High byte
3	D8	ID Low byte: Battery level for rear tire Low byte
4	XX	Battery level [Hex]

3.6.57 Front tire pressure: $22 $ B1 D9

Read pressure value for front tire shall be provided in 2 bytes with 0.1psi resolution. Example:35.6 psi in front tire shall be read 0x01 and 0x64

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	B1	ID High byte: Front Tire Pressure High byte
3	D9	ID Low byte: Front Tire Pressure Low byte

Positive Response Message		**Message Length: 5 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	B1	ID High byte: Front Tire Pressure High byte
3	D9	ID Low byte: Front Tire Pressure Low byte
4	XX	Front pressure value [Hex]

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5	XX	Front pressure value [Hex]

3.6.58 Rear tire pressure: $22 $ B1 DA

Read pressure value of rear tire shall be provided in 2 bytes with 0.1 resolution

Example: 35.6 psi in front tire shall be read as 0x01 and 0x64

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	B1	ID High byte: Rear Tire Pressure High byte
3	DA	ID Low byte: Rear Tire Pressure Low byte

Positive Response Message		**Message Length: 5 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	B1	ID High byte: Rear Tire Pressure High byte
3	DA	ID Low byte: Rear Tire Pressure Low byte
4	XX	Rear pressure value [Hex]
5	XX	Rear pressure value [Hex]

3.6.59 Tire Pressure threshold for front and rear: $22 $ F1 9A

This DID shall be used when the tire pressure thresholds for front and rear tires are different.
Threshold for Front and Rear tire for low or high pressure shall be provided in Hex format

- 0th position -> Front very Low tire pressure
- 1st position -> Front Low tire pressure
- 2nd byte position -> Front high tire pressure
- 3rd byte position -> Front very high tire pressure
- 4th position -> Rear very low tire pressure
- 5th position -> Rear low tire pressure
- 6th position -> Rear high tire pressure
- 7th position -> Rear very high tire pressure

01 indicates both rear and front tire low pressure value and 02 indicates both rear and front tire high pressure value.

Example:

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Front very low tire pressure = 13 psi (0x0D)
Front low tire pressure = 19 psi(0x13)
Front high tire pressure = 40 psi (0x28)

Front very high tire pressure = 44(0x2C)
Rear very low tire pressure = 15(0x0F)
Rear low tire pressure = 21 psi(0x15)
Rear high tire pressure = 42 psi (0x2A)
Rear very high tire pressure = 46(0x2E)

Then the value to be written is:
$10 $0B $2E $F1 $9A $0D $13 $28
$21 $2C $15 $21 $2A $2E

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Tire Pressure threshold for front and rear High byte
3	9A	ID Low byte: Tire Pressure threshold for front and rear Low byte

Positive Response Message		**Message Length: 11 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Tire Pressure threshold for front and rear High byte
3	9A	ID Low byte: Tire Pressure threshold for front and rear Low byte
4	XX	Byte1[Hex][MSB]
5	XX	Byte2[Hex]
6	XX	Byte3[Hex]
7	XX	Byte4[Hex]
8	XX	Byte5[Hex]
9	XX	Byte6[Hex]
10	XX	Byte7[Hex]
11	XX	Byte8[Hex][LSB]

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3.6.60 Cruising range: $22 $ C1 11

The purpose of this DID is to provide the cruising range values. It shall be provided in 3 bytes.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	C1	ID High byte: Cruising range High byte
3	11	ID Low byte: Cruising range Low byte

Positive Response Message		**Message Length: 6 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	C1	ID High byte: Cruising range High byte
3	11	ID Low byte: Cruising range Low byte
4	XX	Byte1[Hex][MSB]
5	XX	Byte2[Hex]
6	XX	Byte3[Hex] [LSB]

3.6.61 Headlamp control: $22 $ FF 55

The purpose of this DID is to read the Head lamp control enable/disable status.

Value description:

0 – disabled

1 – enabled

The reserved bytes shall be kept as 0.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	FF	ID High byte: Head lamp control High byte
3	55	ID Low byte: Head lamp control Low byte

Positive Response Message		**Message Length: 10 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	FF	ID High byte: Head lamp control High byte
3	55	ID Low byte: Head lamp control Low byte
4	XX	Byte1[Hex][MSB]: Head lamp control enable/disable status
5	XX	Byte2[Hex] Reserved
6	XX	Byte3[Hex]: Reserved

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7	XX	Byte4[Hex]: Reserved
8	XX	Byte5[Hex]: Reserved
9	XX	Byte6[Hex]: Reserved
10	XX	Byte7[Hex] [LSB]: Reserved

3.6.62 MAC ID for Go pro pairing: $22 $ F2 00

GO PRO ID shall be provided in Hex format (6 bytes).

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F2	ID High byte: Go pro High byte
3	00	ID Low byte: Go pro Low byte

Positive Response Message		**Message Length: 9 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F2	ID High byte: Go pro High byte
3	00	ID Low byte: Go pro Low byte
4	XX	Byte1[Hex][MSB]
5	XX	Byte2[Hex]
6	XX	Byte3[Hex]
7	XX	Byte4[Hex]
8	XX	Byte5[Hex]
9	XX	Byte6[Hex][LSB]

3.6.63 Cruise set status: $22 $ F1 9B

The DID is used to read the status of the cruise set status.
Cruise Set status shall be considered as mentioned below.

Value description:

0-Inactive
1-Active

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier

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2	F1	ID High byte: Cruise set status High byte
3	9B	ID Low byte: Cruise set status Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Cruise set status High byte
3	9B	ID Low byte: Cruise set status Low byte
4	XX	Byte1[Boolean]

3.6.64 Climate Controlled seat: $22 $ F2 05

Climate controlled seat feature enable status shall be considered as mentioned below.
1 – enabled
0 - disabled

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F2	ID High byte: Climate controlled seat High byte
3	05	ID Low byte: Climate controlled range Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F2	ID High byte: Climate controlled seat High byte
3	05	ID Low byte: Climate controlled range Low byte
4	XX	Byte1[Boolean]

3.6.65 Cruise reset status: $22 $ F1 9C

The DID is used to read the reset status of the cruise.
Cruise Reset status shall be considered as mentioned below.
Value description:
0-Inactive
1-Active

Request Message	**Message Length: 3 Bytes**

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Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Cruise reset status High byte
3	9C	ID Low byte: Cruise reset status Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Cruise reset status High byte
3	9C	ID Low byte: Cruise reset status Low byte
4	XX	Byte1[Boolean]

3.6.66 Cruise ON or OFF status: $22 $ F1 9D

This DID is used to read the ON or OFF status of cruise feature.
Cruise On/Off status shall be considered as mentioned below.

Value description:

0-Inactive
1-Active

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Cruise on or off status High byte
3	9D	ID Low byte: Cruise on or off status Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Cruise on or off status High byte
3	9D	ID Low byte: Cruise on or off status Low byte
4	XX	Byte1[Boolean]

3.6.67 Quick Shifter (QS) functionality status: $22 $ F1 9F

QS Functionality status shall be considered as mentioned below.

Value description:

0-Inactive

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1-Active

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: QS functionality status High byte
3	9F	ID Low byte: QS functionality status Low byte

Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: QS functionality status High byte
3	9F	ID Low byte: QS functionality status Low byte
4	XX	Byte1[Boolean]

3.6.68 Brake switch status: $22 $ F1 A3

Brake Switch status shall be considered as mentioned below.

Value description:

0-Inactive
1-Front Active
2-Rear Active
3-Both Active

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Brake switch status High byte
3	A3	ID Low byte: Brake switch status Low byte

Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Brake switch status High byte
3	A3	ID Low byte: Brake switch status Low byte
4	XX	Byte1[Hex]

3.6.69 Brake Lamp Control Status: $22 $ F1 A2

Brake Lamp Control status shall be considered as mentioned below.

Value description:

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0-Inactive
1-Active

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Brake lamp Control status High byte
3	A2	ID Low byte: Brake lamp Control status Low byte

Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Brake lamp Control status High byte
3	A2	ID Low byte: Brake lamp Control status Low byte
4	XX	Byte1[Boolean]

3.6.70 BTO animation enable: $22 $ F2 06

BTO animation enables status shall be considered as mentioned below.

Value description:

Enabled-1
Disabled-0

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F2	ID High byte: BTO animation enable High byte
3	06	ID Low byte: BTO animation enable Low byte

Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F2	ID High byte: BTO animation enable High byte
3	06	ID Low byte: BTO animation enables Low byte
4	XX	Byte1[Boolean]

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3.6.71 Tacho Swipe: $22 $ F2 07

Tacho Swipe value shall be considered as mentioned below

Value description:

Enable – 1(default)

Disable – 0

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F2	ID High byte: Tacho swipe High byte
3	07	ID Low byte: Tacho swipe Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F2	ID High byte: Tacho swipe High byte
3	07	ID Low byte: Tacho swipe Low byte
4	XX	Byte1[Boolean]

3.6.72 Celebration Alerts: $22 $ F2 08

The purpose of this DID is to read the celebration alerts. It shall be provided in 2 bytes

Value mapped to Byte1:

- Enable – 1(default)
- Disable – 0

Value mapped to Byte2:

- 00 -> Default Alerts (Birthday, etc.)
- 01 -> Days of National Importance (only for IN)
- 02 -> Vehicle related Alerts (First Month, Anniversary, etc.)
- FF -> Enabled for all Celebration Alerts

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F2	ID High byte: Celebration alerts High byte
3	08	ID Low byte: Celebration alerts Low byte

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Positive Response Message		Message Length: 5 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F2	ID High byte: Celebration alerts High byte
3	08	ID Low byte: Celebration alerts Low byte
4	XX	Byte1[Hex]
5	XX	Byte2[hex]

3.6.73 Navigation for TPMS/Low fuel: $22 $ F1 A6

To enable/disable rerouting when fuel level is Low or the tire pressure is low/critical.

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Navigation for TPMS/Low fuel High byte
3	A6	ID Low byte: Navigation for TPMS/Low fuel Low byte

Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Navigation for TPMS/Low fuel High byte
3	A6	ID Low byte: Navigation for TPMS/Low fuel Low byte
4	XX	Value Description: Enable: 1 Disable: 0

3.6.74 Display Language: $22 $ F1 A5

Display language shall be interpreted as mentioned below

Value description:

EN-0(default)
FR-1
DE-2
ES-3
JP-4

Request Message	Message Length: 3 Bytes

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Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Language High byte
3	A5	ID Low byte: Language Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Language High byte
3	A5	ID Low byte: Language Low byte
4	XX	Byte1[Hex]

3.6.75 Celebration setting: $22 $ F2 09

Value description:

Celebration dates can be read through tester. It shall be provided in 3 bytes
Happy Birthday Celebration, Vehicle Anniversary Celebration shall be written in this DID
Celebration End Date.

Value Mapped to Byte1:

- 00 - do not consider
- 01 - Vehicle Anniversary
- 02 - Happy Birthday

Value Mapped to Byte2:

- 00 - Default

Value Mapped to Byte3:

- 00 - Default

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F2	ID High byte: celebration setting High byte
3	09	ID Low byte: celebration setting Low byte

Positive Response Message		**Message Length:6 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F2	ID High byte: celebration setting High byte

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3	09	ID Low byte: celebration setting Low byte
4	XX	Byte1[Hex][MSB]
5	XX	Byte2[Hex]
6	XX	Byte3[Hex][LSB]

3.6.76 Enable/Disable TPMS: $22 $ B1 DB

Status of TPMS shall be as mentioned below.

Value Description:

0-Not Configured
1 – both configured
2- Front Configured
3- Rear Configured

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	B1	ID High byte: Enable/Disable TPMS High byte
3	DB	ID Low byte: Enable/Disable TPMS Low byte

Positive Response Message		**Message Length:4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	B1	ID High byte: Enable/Disable TPMS High byte
3	DB	ID Low byte: Enable/Disable TPMS Low byte
4	XX	Byte1[Hex][MSB]

3.6.77 Country DID: $22 $ F1 A7

Vehicle Country code can be read through this DID.

Value Description:

IN-0(default)
EU-1
LATAM-2
AU-3
JP-4
AF - 5

Request Message	**Message Length: 3 Bytes**

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Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Country DID High byte
3	A7	ID Low byte: Country DID Low byte

Positive Response Message		**Message Length:4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Country DID High byte
3	A7	ID Low byte: Country DID Low byte
4	XX	Byte1[Hex]

3.6.78 Fog Lamp: $22 $ FF 56

FOG Lamp enable status shall be considered as mentioned below

Value Description:

Enable - 1 (default)/Disable - 0

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	FF	ID High byte: Fog lamp High byte
3	56	ID Low byte: Fog lamp Low byte

Positive Response Message		**Message Length:4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	FF	ID High byte: Fog lamp High byte
3	56	ID Low byte: Fog lamp Low byte
4	XX	Byte1[Boolean]

3.6.79 Cluster housing temperature: $22 $ E0 67

Housing temperature of the cluster shall be read through this DID.

Temperature value from 0 - 140 deg C

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	E0	ID High byte: Cluster housing temperature High byte
3	67	ID Low byte: Cluster housing temperature Low byte

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Positive Response Message		Message Length:4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	E0	ID High byte: Cluster housing temperature High byte
3	67	ID Low byte: Cluster housing temperature Low byte
4	XX	Byte1[Hex]

3.6.80 Ambient light sensor value: $22 $ E0 66

Ambient light sensor value shall be read though this DID.
Temperature value from 0 - 1000 lux

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	E0	ID High byte: Ambient light sensor value High byte
3	66	ID Low byte: Ambient light sensor value Low byte

Positive Response Message		Message Length:4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	E0	ID High byte: Ambient light sensor value High byte
3	66	ID Low byte: Ambient light sensor value Low byte
4	XX	Byte1[Hex]

3.6.81 Time and Date: $22 $ E1 2B

Time and date are read in this DID
The time and date format will be HHmmDDMMYYYY (24Hr Hex Format)
Example:
If the date is April 3rd, 2019, and time is 06:45Pm then the response byte value shall be 0x12 0x2D 0x03 0x04 0x14 0x13

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	E1	ID High byte: Time and Date High byte
3	2B	ID Low byte: Time and Date Low byte

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Positive Response Message		Message Length:9 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	E1	ID High byte: Time and Date High byte
3	2B	ID Low byte: Time and Date Low byte
4	XX	Hour [Hex]
5	XX	Minute [Hex]
6	XX	Day [Hex]
7	XX	Month [Hex]
8	XX	Year [MSB][Hex]
9	XX	Year [LSB][Hex]

3.6.82 Quick shifter cluster switch test: $22 $ 11 25

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	11	ID High byte: Quick shifter cluster switch test High byte
3	25	ID Low byte: Quick shifter cluster switch test Low byte

Positive Response Message		Message Length:4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	11	ID High byte: Quick shifter cluster switch test High byte
3	25	ID Low byte: Quick shifter cluster switch test Low byte
4	XX	Byte1[Hex]

3.6.83 Traction Control (TC) functionality status: $22 $ 09 09

TC Functionality status can be read through DID

Value description:

0-Deactive

1-Active.

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name

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1	22	Service ID: Read Data by Identifier
2	09	ID High byte: Cruise TC functionality status High byte
3	09	ID Low byte: Cruise TC functionality status Low byte

Positive Response Message		**Message Length:4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte: Cruise TC functionality status High byte
3	09	ID Low byte: Cruise TC functionality status Low byte
4	XX	Cruise TC functionality status [Hex]

3.6.84 Variant control: $22 $ 09 07

This DID is used to read the cluster variant.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte: Variant control High byte
3	07	ID Low byte: Variant control Low byte

Positive Response Message		**Message Length:4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte: Variant control High byte
3	07	ID Low byte: Variant control Low byte
4	XX	0- base variant 1-keyless variant

3.6.85 Dealer ID: $22 $ F1 00

The purpose of this DID is to read the ID of the dealer that sold the vehicle.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Dealer ID High byte

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3	00	ID Low byte: Dealer ID Low byte

Positive Response Message		**Message Length: 8 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Dealer ID High byte
3	00	ID Low byte: Dealer ID Low byte
4	XX	Dealer ID [Hex][MSB]
5	XX	Dealer ID [Hex]
6	XX	Dealer ID [Hex]
7	XX	Dealer ID [Hex]
8	XX	Dealer ID [Hex][LSB]

3.6.86 Project code: $22 $ F1 03

The purpose of this DID is to read the project code assigned to the product inside TVSM.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Project code High byte
3	03	ID Low byte: Project code Low byte

Positive Response Message		**Message Length:6 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Project code High byte
3	03	ID Low byte: Project code Low byte
4	XX	Project code [ASCII] (MSB)
5	XX	Project code [ASCII]
6	XX	Project code [ASCI] (LSB)

3.6.87 Date of last programming: $22 $ F1 04

The purpose of this DID is to read the date of the last time when the ECU was programmed/reprogrammed.

Request Message	**Message Length: 3 Bytes**

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Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Date of last programming High byte
3	04	ID Low byte: Date of last programming Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Date of last programming High byte
3	04	ID Low byte: Date of last programming Low byte
4	XX	Month[Hex][MSB]
5	XX	Date[Hex]
6	XX	Year[Hex][LSB]

3.6.88 MFR/Projector lamp ID - Head lamp: $22 $ 09 11

The DID can be used to read the Headlamp's Current Beam Status.

Value description:

b0-Low beam on/off status

b1-High beam on/off status

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte: MFR/Projector lamp ID - Head lamp High byte
3	11	ID Low byte: MFR/Projector lamp ID - Head lamp Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte: MFR/Projector lamp ID - Head lamp High byte
3	11	ID Low byte: MFR/Projector lamp ID - Head lamp Low byte
4	XX	Head lamp on/off status [Hex] 00 – Head lamp is off. 01 – low beam is on. 02 – High beam is on.

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3.6.89 MFR/Projector lamp ID - DRL: $22 $ 09 12

The DID can be used to read the DRL's Current ON/OFF Status.

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte: MFR/Projector lamp ID - DRL High byte
3	12	ID Low byte: MFR/Projector lamp ID - DRL Low byte

Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte: MFR/Projector lamp ID - DRL High byte
3	12	ID Low byte: MFR/Projector lamp ID - DRL Low byte
4	XX	DRL on/off status [Hex] 0-DRL is off 1-DRL is on

3.6.90 Head lamp Intensity Level: $22 $ 09 13

The DID can be used to read the Headlamp's Current Intensity Level.

Value description:

0-Level 1
1-Level 2
2-Level 3

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte: Dynamic Head lamp ID - High byte
3	13	ID Low byte: Dynamic Head Lamp ID - Low byte

Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte: Dynamic Head lamp ID - High byte
3	13	ID Low byte: Dynamic Head lamp ID - Low byte
4	XX	Dynamic Head Lamp Intensity level [Hex]

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3.6.91 Cornering/Blend Lamp: $22 $ 09 14

The DID can be used to read the Cornering/Bend Lamp's Current ON/OFF Status.

Value description:

Lamp OFF - 0

Lamp ON - 1

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte: Cornering/Blend lamp ID - High byte
3	14	ID Low byte: Cornering/Blend lamp ID - Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte: Cornering/Blend lamp ID - High byte
3	14	ID Low byte: Cornering/Blend lamp ID - Low byte
4	XX	Cornering/Blend lamp ON/OFF Status [Hex]

3.6.92 Turn Signal Lamp (TSL): $22 $ 09 15

The DID can be used to read the TSL's Current ON/OFF Status.

Value description:

b0-TSL Left Front

b1-TSL Left Rear

b2-TSL Right Front

b3-TSL Right Rear

b4-TSL auto turn off

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte: TSL ID - High byte
3	15	ID Low byte: TSL ID - Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte: TSL- High byte

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3	15	ID Low byte: TSL- Low byte
4	XX	Possible values: 0x00-TSL is off 0x03-TSL left is on. 0x0C-TSL right is on. 0x0F-Both TSL are on. 0x10- TSL is off automatically

3.6.93 Parking Lamp: $22 $ 09 16

The DID can be used to read the Parking lamp's ON/OFF Status.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte: Parking Lamp ID - High byte
3	16	ID Low byte: Parking Lamp ID - Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte: Parking Lamp - High byte
3	16	ID Low byte: Parking Lamp- Low byte
4	XX	Parking Lamp ON/OFF Status [Hex] 0-OFF 1-ON

3.6.94 Live Fuel Economy Dashboard: $22 $ 09 17

The DID can be used to read the Current Fuel Economy in km/l.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte: Live Fuel Economy Dashboard ID - High byte
3	17	ID Low byte: Live Fuel Economy Dashboard - Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier

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2	09	ID High byte: Live Fuel Economy Dashboard - High byte
3	17	ID Low byte: Live Fuel Economy Dashboard Low byte
4	XX	Fuel Economy [Hex]

3.6.95 Telematics Status: $22 $ 09 18

This DID provides status of various stages of TCU for PKI registration and enrollment process

Value description:

- b0 - Registration Status (0 -> Not-Executed/Failed, 1 -> Pass)
- b1 - Birth Certificate Installation Status (0 -> Not-Executed/Failed, 1 -> Pass)
- b2 - Enrollment Initiated Status (0 -> Not-Executed/Failed, 1 -> Pass)
- b3 - Device Certificate Installation Status (0 -> Not-Executed/Failed, 1 -> Pass)
- b4 - Enrollment Status (0 -> Not-Executed/Failed, 1 -> Pass)
- b5 - Renewal Status (0 -> Not-Executed/Failed, 1 -> Pass)
- b6 - Health Ok Status (0 -> Not-Executed/Failed, 1 -> Pass)
- b7 - VIN Mapping Status (0 -> Not-Executed/Failed, 1 -> Pass)
- b8 - EOL Baseline Download Status (0 -> Not-Executed/Failed, 1 -> Pass)

Note: b5 will be set to 1(Pass) only when Certificate Renewal is triggered explicitly, else it will be 0 means Not-Executed
which is expected and should not be considered as FAIL.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte: Telematics ID - High byte
3	18	ID Low byte: Telematics ID - Low byte

Positive Response Message		**Message Length: 5 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte: Telematics Status- High byte
3	18	ID Low byte: Telematics Status- Low byte
4	XX	Telematics Status (MSB) [Hex]
5	XX	Telematics Status (LSB)[Hex]

3.6.96 Drive modes/ Ride modes: $22 $ 09 01

Drive modes / Ride modes value description is dependent to project.
Values are dependent on the project.

Request Message	**Message Length: 3 Bytes**

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Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte: Drive mode/Ride mode - High byte
3	01	ID Low byte: Drive mode/Ride mode - Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte: Drive mode/Ride mode - High byte
3	01	ID Low byte: Drive mode/Ride mode - Low byte
4	XX	Drive mode/Ride mode status [Hex]

3.6.97 Primary Key Fob: $22 $ F1 38

The purpose of this DID is to read the MAC_ID of Key fob(s)

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: MAC_ID High byte
3	38	ID Low byte: MAC_ID Low byte

Positive Response Message		**Message Length: minimum 10 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: MAC_ID High byte
3	38	ID Low byte: MAC_ID Low byte
4-9	XX	MAC Id of Key fob[Hex]

3.6.98 Take Me Bike: $22 $ 09 02

The DID indicates the activation status of the Take Me Bike feature.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte: Take Me Bike High byte
3	02	ID Low byte: Take Me Bike Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name

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1	62	Response ID: Read data by Identifier
2	09	ID High byte: Take Me Bike - High byte
3	02	ID Low byte: Take Me Bike - Low byte
4	XX	Take Me Bike status [Hex] 0 – inactive 1 – active

3.6.99 Dynamic Brake Light Status: $22 $09 03

This DID read the dynamic brake light enable status

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte: Dynamic brake light High byte
3	03	ID Low byte: Dynamic brake light Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte: Dynamic brake light High byte
3	03	ID Low byte: Dynamic brake light Low byte
4	XX	Value Description : Disabled – 0 Enabled - 1

3.6.100 Fuel Filler Flap: $22 $09 04

This DID is used to read the status of fuel filler flap.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte: Fuel Filler Flap High byte
3	04	ID Low byte: Fuel Filler Flap Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier

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2	09	ID High byte: Fuel Filler Flap High byte
3	04	ID Low byte: Fuel Filler Flap Low byte
4	XX	Fuel Filler Flap status: Open -1 / Close - 0

3.6.101 Voice Assist Feature : $22 $A1 2B

This DID indicates the Voice Assist feature enable status

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	A1	ID High byte: Voice Assist Feature High byte
3	2B	ID Low byte: Voice Assist Feature Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	A1	ID High byte: Voice Assist Feature High byte
3	2B	ID Low byte: Voice Assist Feature Low byte
4	XX	Feature Disabled – 0 Feature Enabled - 1

3.6.102 No. of Key fobs Paired: $22 $44 3C

This DID is used to read the total number of key fobs learnt with keyless Ecu

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	44	ID High byte: NO. of key fobs Paired High byte
3	3C	ID Low byte: NO. of key fobs Paired Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	44	ID High byte: NO. of key fobs Paired High byte
3	3C	ID Low byte: NO. of key fobs Paired Low byte

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4	XX	Number of key fob Paired [Hex]

3.6.103 Service Due Date : $22 $E1 2C

This DID contains the next service due date.
The service date format shall be (DDMMYYYY 24Hr BCD format)

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	E1	ID High byte: Service Date High byte
3	2C	ID Low byte: Service Date Low byte

Positive Response Message		**Message Length: 7 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	E1	ID High byte: Service Date High byte
3	2C	ID Low byte: Service Date Low byte
4	XX	DAY
5	XX	MONTH
6,7	XX	YEAR

3.6.104 Service Distance: $22 $E1 2D

This DID records the distance value for the next scheduled service.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	E1	ID High byte: Service Distance High byte
3	2D	ID Low byte: Service Distance Low byte

Positive Response Message		**Message Length: 5 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	E1	ID High byte: Service Distance High byte
3	2D	ID Low byte: Service Distance Low byte
4	XX	MSB Byte1 [Hex]
5	XX	LSB Byte[Hex]

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3.6.105 Keyfob Authentication Status: $22 $09 05

This DID is used to read the authentication status of Key fob .

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte: Keyfob Authentication Status High byte
3	05	ID Low byte: Keyfob Authentication Status Low byte

Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte: Key fob Authentication Status High byte
3	05	ID Low byte: Keyfob Authentication Status Low byte
4	XX	Authentication Successful - 1 Authentication failed - 0

3.6.106 ECU Authentication Status: $22 $09 06

This DID is used to read the authentication status of ECU.
This is for Keyless entry (Based on requirement of Keyless entry, this DID should be configured).

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte: ECU Authentication Status High byte
3	06	ID Low byte: ECU Authentication Status Low byte

Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte: ECU Authentication Status High byte
3	06	ID Low byte: ECU Authentication Status Low byte
4	XX	Authentication Successful - 1 Authentication failed - 0

3.6.107 Map Mirroring Enable Status: $22 $09 1B

This DID is used to read the Map Mirroring enable/disable status.

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Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte: Map Mirroring Status High byte
3	1B	ID Low byte: Map Mirroring Status Low byte

Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte: Map Mirroring Status High byte
3	1B	ID Low byte: Map Mirroring Status Low byte
4	XX	Map Mirroring Enabled - 1 Map Mirroring disabled - 0

3.6.108 Traffic Screen Enable Status: $22 $09 1C

This DID is used to read the Traffic Screen enable/disable status.

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte: Traffic Screen Enable Status High byte
3	1C	ID Low byte: Traffic Screen Enable Status Low byte

Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte: Traffic Screen Enable Status High byte
3	1C	ID Low byte: Traffic Screen Enable Status Low byte
4	XX	Traffic Screen Enabled - 1 Traffic Screen disabled - 0

3.6.109 Music Enable Status: $22 $09 1D

This DID is used to read the Music enable/disable status.

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte: Music Enable Status High byte

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3	1D	ID Low byte: Music Enable Status Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte: Music Enable Status High byte
3	1D	ID Low byte: Music Enable Status Low byte
4	XX	Music Enabled - 1 Music disabled - 0

3.6.110 SMS Notification Enable Status: $22 $09 1E

This DID is used to read the SMS notification enable/disable status.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte: SMS Notification Enable Status High byte
3	1E	ID Low byte: SMS Notification Enable Status Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte: SMS Notification Enable Status High byte
3	1E	ID Low byte SMS Notification Enable Status Low byte
4	XX	SMS Notification Enabled - 1 SMS Notification disabled - 0

3.6.111 Product Variant ID: $22 $F1 B1

It consists of a lookup table used to identify a separate project Variant from a product platform. Ex: Such as iQube 09 from iQube platform etc.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Product Variant ID High byte
3	B1	ID Low byte: Product Variant ID Low byte

Positive Response Message	**Message Length: 4 Bytes**

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Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Product Variant ID High byte
3	B1	ID Low byte: Product Variant ID Low byte
4	XX	Product Variant ID [HEX]

3.6.112 Battery Type_0: $22 $F1 B2

It contains information of each BMS pack type that the vehicle is initially configured to.

Request Message	**Message Length: 3 Bytes**

Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F1	ID High byte: Battery Type_0 ID High byte
3	B2	ID Low byte: Battery Type_0 ID Low byte

Positive Response Message	**Message Length: 6 Bytes**

Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F1	ID High byte: Battery Type_0 ID High byte
3	B2	ID Low byte: Battery Type_0 ID Low byte
4-6	XX	Battery_Type_0[HEX]

3.6.113 RPM Threshold- a for Gear Shifting: $22 $09 00

When the engine RPM goes above this Threshold value, the system shall suggest shifting to a higher gear.

Request Message	**Message Length: 3 Bytes**

Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	00	ID Low byte

Positive Response Message	**Message Length: 5 Bytes**

Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	00	ID Low byte
4,5	XX	RPM Threshold-a

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3.6.114 RPM Threshold- b for Gear shifting: $22 $09 08

When the engine RPM goes above this Threshold value, the system shall suggest shifting to a higher gear

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	08	ID Low byte

Positive Response Message		**Message Length: 5 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	08	ID Low byte
4,5	XX	RPM Threshold-b

3.6.115 RPM Threshold- c for Gear shifting: $22 $09 0A

When the engine RPM goes above this Threshold value, the system shall suggest shifting to a higher gear.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	0A	ID Low byte

Positive Response Message		**Message Length: 5 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	0A	ID Low byte
4,5	XX	RPM Threshold-c

3.6.116 RPM Threshold- d for Gear shifting -: $22 $09 0B

When the engine RPM goes above this Threshold value, the system shall suggest shifting to a higher gear.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name

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1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	0B	ID Low byte

Positive Response Message		**Message Length: 5 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	0B	ID Low byte
4,5	XX	RPM Threshold-d

3.6.117 RPM Threshold- e for Gear shifting: $22 $09 0C

When the engine RPM goes below this Threshold value, the system shall suggest shifting to a lower gear.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	0C	ID Low byte

Positive Response Message		**Message Length: 5 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte:
3	0C	ID Low byte
4,5	XX	RPM Threshold-e

3.6.118 RPM Threshold- f for Gear shifting: $22 $09 0D

When the engine RPM goes below this Threshold value, the system shall suggest shifting to a lower gear.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	0D	ID Low byte

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Positive Response Message		Message Length: 5 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	0D	ID Low byte
4,5	XX	RPM Threshold-f

3.6.119 RPM Threshold- g for Gear shifting: $22 $09 0E

When the engine RPM goes below this Threshold value, the system shall suggest shifting to a lower gear.

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	0E	ID Low byte

Positive Response Message		Message Length: 5 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	0E	ID Low byte
4,5	XX	RPM Threshold-g

3.6.120 RPM Threshold- h for Gear shifting: $22 $09 0F

When the engine RPM goes below this Threshold value, the system shall suggest shifting to a lower gear.

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	0F	ID Low byte

Positive Response Message		Message Length: 5 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte

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3	0F	ID Low byte
4,5	XX	RPM Threshold-h

3.6.121 ODO Threshold- x for Gear Shifting: $22 $09 10

The Odometer Threshold value for optimal gear shift suggestion

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	10	ID Low byte

Positive Response Message		**Message Length: 5 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	10	ID Low byte
4	XX	ODO Threshold

3.6.122 Coolant Temp Threshold- y for Gear Shifting: $22 $09 1F

The Coolant temperature threshold value for optimal gear shift suggestion

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	1F	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	1F	ID Low byte
4	XX	Coolant Temperature Threshold

3.6.123 Indicator Blinking Period: $22 $F2 01

The Turn Signal lamps shall blink with this configured value.
Resolution: 10

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Example: A value of 10 indicates 100ms.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F2	ID High byte
3	01	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F2	ID High byte
3	01	ID Low byte
4	XX	Indicator blinking Period [HEX]

3.6.124 Engine High Temp Threshold: $22 $F2 02

The DID is used to read the Engine High Temperature Threshold.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F2	ID High byte
3	02	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F2	ID High byte
3	02	ID Low byte
4	XX	High Threshold value [HEX]

3.6.125 Fuel Thermistor Threshold: $22 $F2 03

The DID is used to read the Fuel Thermistor Threshold.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	F2	ID High byte
3	03	ID Low byte

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Positive Response Message		Message Length: 5 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	F2	ID High byte
3	03	ID Low byte
4	XX	Fuel Thermistor Threshold [HEX] MSB
5	XX	Fuel Thermistor Threshold [HEX] LSB

3.6.126 Suspension Setting Active Status: $22 $09 2A

The DID is used to check Enable /Disable Suspension Setting.

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	2A	ID Low byte

Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	2A	ID Low byte
4	XX	Suspension Setting [HEX] Value Description: 00-Disabled, 01-Enabled

3.6.127 Battery Type Erase Status: $22 $09 2B

This DID is used to check whether the Battery type value has been erased from NvM.

Request Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Response ID: Read data by Identifier
2	09	ID High byte
3	2B	ID Low byte

Positive Response Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Service ID: Read Data by Identifier
2	09	ID High byte

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3	2B	ID Low byte
4	XX	Value Description: 0-Not erased, 01-Erased

3.6.128 Battery Type Feature Check Enable: $22 $09 2C

The DID is used to Enable /Disable Battery Type Check Feature.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	2C	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	2C	ID Low byte
4	XX	Battery Type Check Enable [HEX] 01- Enable,00- Disable

3.6.129 Enable or Disable Emergency Brake Lamp:$22 $09 2D

The DID is used To Enable or Disable Emergency Brake Lamp functionality

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	2D	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	2D	ID Low byte
4	XX	Emergency Brake Lamp Enable [HEX] 01- Enable,00- Disable

3.6.130 Vehicle variant and Region Parameters: $22 $09 2E

The DID is used to configure the vehicle settings before delivering the vehicle to the customer

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Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	2E	ID Low byte

Positive Response Message		Message Length:8 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	2E	ID Low byte
4	00-0F	Vehicle Type [HEX]
5	00-0F	Vehicle Trim Level[HEX]
6	00-0F	Model Year[HEX]
7	00-0F	Region of Sale[HEX]
8	00-FF	Language [HEX]

3.6.131 Vehicle Steering Status: $22 $09 31

The DID is used to monitor the real-time state of the steering lock system. Assist in vehicle security and anti-theft mechanisms. It support diagnostics and fault detection in the steering lock/unlock process and enable service technicians to identify issues related to steering lock functionality.

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	31	ID Low byte

Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	31	ID Low byte
4	XX	0- steering locked 1 - steering unlocked 2 - Error During steering Lock

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		3 - Error During steering unlock

3.6.132 Seat And Fuel Cap Solenoid Status: $22 $09 32

This DID is used to Indicate whether the seat and fuel cap locking mechanisms are active Control and monitor the enable/disable state of seat and fuel cap release buttons. Support safety and anti-theft features by restricting unauthorized access. Assist in diagnostics of the solenoid relay and associated control circuits.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	32	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	32	ID Low byte
4	XX	0- solenoid disengaged 1 - solenoid engaged

3.6.133 VCU Sleep Status: $22 $09 33

This DID is used to monitor the power state of the VCU in real time. Support energy-saving strategies by confirming sleep mode entry. Assist in diagnostics of wakeup and sleep transitions and Ensure proper coordination with other ECUs in the vehicle network.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	33	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	33	ID Low byte
4	XX	0- VCU in sleep mode 1 - VCU in wakeup mode

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3.6.134 Ignition Load Relay Status: $22 $09 34

This DID is used to Monitor the ignition state of the vehicle. Determine whether electrical loads are being powered. Support diagnostics of ignition relay and power distribution circuits. Enable coordination between ECUs based on ignition status

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	34	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	34	ID Low byte
4	XX	0- Vehicle Ignition OFF 1-Vehicle Ignition ON

3.6.135 Find Me TSL Status: $22 $09 35

This DID is used to read the status of the Turn Signal Lamp (TSL) system. It indicates whether the turn signal lamps are in their default state, actively ON (blinking/indicating), or turned OFF. This DID is used to monitor the operational state of the vehicle's turn indicators for signaling and diagnostic purposes.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	35	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	35	ID Low byte
4	XX	0- TSL_Default 1 - TSL_ON 2. TSL OFF

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3.6.136 PWM Signal Overrun: $22 $09 36

This DID is used to Detect timing violations in PWM signal generation. Monitor real-time performance of control loops. Assist in diagnosing ECU overload or scheduling issues and Ensure proper operation of PWM-controlled components.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	36	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	36	ID Low byte
4	XX	0– No PWM Overrun 1-PWM Overrun

3.6.137 Watchdog Reset Count: $22 $09 37

This DID is used to read the number of times a watchdog reset has been triggered in the ECU. Watchdog resets occur when the ECU fails to respond within predefined time, indicating potential software lockups or timing issues.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	37	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	37	ID Low byte
4	XX	Watchdog reset count

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3.6.138 HSD diagnostics pin status: $22 $09 38

This DID is used to read the diagnostic status of the High-Side Driver (HSD) channels in the ECU. It reflects whether any fault condition has been detected via the diagnostic feedback pin(s) associated with the HSD circuitry.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	38	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	38	ID Low byte
4	XX	0-Diagnostic Fault Active 1 - No diagnostic fault

3.6.139 HSD overcurrent value: $22 $09 39

This DID is used to monitor the electrical load on high-side driver channels. Detect overcurrent conditions to protect actuators and power electronics. Diagnose wiring faults, short circuits, or component failures in HSD-controlled systems.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	39	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	39	ID Low byte
4	XX	HSD overcurrent value

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3.6.140 Authentication Fail Count: $22 $09 3A

This DID is used to Monitor system security and access control. Diagnosing repeated unauthorized access attempts. Assist in identifying potential software or network security issues.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	3A	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	3A	ID Low byte
4	XX	Authentication fail count

3.6.141 EEPROM Operation Fail Count:$22 $09 3B

This DID is used to Monitor the reliability of EEPROM memory operations. Diagnose potential hardware or memory corruption issues. Support maintenance and preventive actions for ECU memory health.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	3B	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	3B	ID Low byte
4	XX	EEPROM operation fail count

3.6.142 Hardware Security Module (CSEC) Init Status: $22 $09 3C

This Data Identifier DID indicates the initialization status of the Hardware Security Module (HSM), specifically CSEC (Cryptographic Services Engine). The HSM/CSEC is responsible for secure operations such as key management, encryption/decryption, and authentication. This

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DID allow monitoring whether the module is uninitialized, currently initializing, fully initialized, or has failed to initialize.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	3C	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	3C	ID Low byte
4	XX	0- CSEC_Not init 1- CSEC_Init in progress 2- CSEC_Init done 3 - CSEC_init Failed

3.6.143 Hot Seat Feature : $22 $09 2F

The DID is used to Enable or Disable Hot Seat Feature

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	2F	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	2F	ID Low byte
4	XX	0-Disable 1-Enable

3.6.144 Cool Seat Feature : $22 $09 30

The DID is used to Enable or Disable Hot Seat Feature

Request Message	**Message Length: 3 Bytes**

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Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	30	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	30	ID Low byte
4	XX	0-Disable 1-Enable

3.6.145 Variant Configuration Method: $22 $09 43

The DID is used to read the method of variant configuration in cluster. If Value is set as 00, Cluster shall allow variant configuration via CAN bus message. If Value is set as 01, Cluster shall allow variant configuration via DID.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier
2	09	ID High byte
3	43	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	09	ID High byte
3	43	ID Low byte
4	XX	00-CAN 01-DID

3.6.146 A/A2 License configuration: $22 $09 44

Cluster shall support both A and A2 license configurations within the same software package, Cluster shall support switching b/w A and A2 license configurations via UDS using DID. Based on A/A2 license configuration enabled, the following CAN signal shall be updated.

Request Message	**Message Length: 3 Bytes**

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Byte number	Value (Hex)	Parameter Name
1	22	Response ID: Read data by Identifier
2	09	ID High byte
3	44	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Service ID: Read Data by Identifier
2	09	ID High byte
3	44	ID Low byte
4	XX	00-A1 01-A

3.6.147 Telematics Event Flag: $22 $09 45

This DID is used to read a bitwise status field where each bit represents a specific event or alert in the TCU. The value of each bit indicates whether that event is active or inactive.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Response ID: Read data by Identifier
2	09	ID High byte
3	45	ID Low byte

Positive Response Message		**Message Length: 7 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Service ID: Read Data by Identifier
2	09	ID High byte
3	44	ID Low byte
4-7	XX	Bit=0→Event inactive Bit = 1 → Event active

3.6.148 TCU Modem Application Version : $22 $09 46

This DID is used read the software version of the modem application currently flashed on the TCU.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Response ID: Read data by Identifier
2	09	ID High byte

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3	46	ID Low byte

Positive Response Message		**Message Length: 6 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Service ID: Read Data by Identifier
2	09	ID High byte
3	46	ID Low byte
4-6	XX	0:Major 1:Minor 2: Micro

3.6.149 GPS Data: $22 $09 47

This DID is used to read a unique identifier used in UDS diagnostics to access specific data items stored inside an TCU.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Response ID: Read data by Identifier
2	09	ID High byte
3	47	ID Low byte

Positive Response Message		**Message Length: 15 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Service ID: Read Data by Identifier
2	09	ID High byte
3	47	ID Low byte
4	XX	Satellite in Use
5	XX	Satellite in View
6	XX	Validity
7	XX	GPS Error Info
8-11	XX	latitude
12-15	XX	Longitude

3.6.150 TCU OTA Status: $22 $09 48

This DID is used to read the 24-byte snapshot of the OTA update progress for all major (TCU, VIC, BMS, VCU, MCU).Each subsystem is represented by a 4-byte block, indicating its individual OTA state along with the overall OTA status

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name

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1	22	Response ID: Read data by Identifier
2	09	ID High byte
3	48	ID Low byte

Positive Response Message		**Message Length: 27 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Service ID: Read Data by Identifier
2	09	ID High byte
3	48	ID Low byte
4-7	XX	Overall OTA status
8-11	XX	TCU OTA status
12-15	XX	VIC OTA status
16-19	XX	BMS OTA status
20-23	XX	VCU OTA status
24-27	XX	MCU OTA status

3.6.151 Mqtt Server Details: $22 $09 49

This DID is used to read a 12-byte information of all the Server public key infrastructure status. (DEV, PROD, UAT)

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Response ID: Read data by Identifier
2	09	ID High byte
3	49	ID Low byte

Positive Response Message		**Message Length: 15 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Service ID: Read Data by Identifier
2	09	ID High byte
3	49	ID Low byte
4-7	XX	DEV PKI status
8-11	XX	UAT PKI status
12-15	XX	PROD PKI status

3.6.152 GSM Data (RSSI, RSRP, RSRQ, Network Band) : $22 $09 4A

This DID is used to read an 8-byte snapshot of key GSM radio metrics used by the TCU, including RSSI, RSRP, RSRQ, and Network Band.

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Acronym	Full Form
GSM	Global System for Mobile Communications
RSSI	Received Signal Strength Indicator
RSRP	Reference Signal Received Power
RSRQ	Reference Signal Received Quality

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Response ID: Read data by Identifier
2	09	ID High byte
3	4A	ID Low byte

Positive Response Message		Message Length: 11 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Service ID: Read Data by Identifier
2	09	ID High byte
3	4A	ID Low byte
4,5	XX	RSSI
6,7	XX	RSRP
8,9	XX	RSRQ
10,11	XX	Network Band

3.6.153 Public Key Infrastructure Status: $22 $09 4B

This DID is used to read a 4-byte snapshot of the TCU's PKI lifecycle, covering key generation, registration, and enrolment processes.

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Response ID: Read data by Identifier
2	09	ID High byte
3	4B	ID Low byte

Positive Response Message		Message Length: 7 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Service ID: Read Data by Identifier

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2	09	ID High byte
3	4B	ID Low byte
4	XX	Key Generation status
5	XX	Registration status
6	XX	Enrolment status
7	XX	Overall PKI status

3.6.154 Key Fob Command Execution Status: $22 $09 4C

This DID is used to read the Key fob command execution status. It gives only the latest command execution status.

The bit value indicates the button state:

1→ Command Execution successful

0→ Command Execution failed

Bit-to-Button Mapping Table

Bit	Button Function	Status Description
b0	Lock command	1 = Successful, 0 = failed
b1	Reserved	—
b2	Unlock command	1 = Successful, 0 = failed
b3	Reserved	—
b4	Find Me command	1 = Successful, 0 = failed
b5	Reserved	—
b6	Seat Unlock command	1 = Successful, 0 = failed
b7	Reserved	—

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Response ID: Read data by Identifier
2	09	ID High byte
3	4C	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Service ID: Read Data by Identifier
2	09	ID High byte
3	4C	ID Low byte
4	XX	Latest Command execution status [HEX]

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3.6.155 NVM and Software version: $22 $F2 04

This DID is used to read the internal config file version .

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Response ID: Read data by Identifier
2	F2	ID High byte
3	04	ID Low byte

Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Service ID: Read Data by Identifier
2	F2	ID High byte
3	04	ID Low byte
4	XX	NVM Version
5	XX	NVM Revision
6	XX	Application Software Block ID
7	XX	Application Software Major version
8	XX	Application Software Minor version
9-10	XX	Boot SW Block ID
11-12	XX	Boot SW Major version
13-14	XX	Boot SW Minor version
15	XX	Calibration SW1 Block ID
16	XX	Calibration SW1 Major Version
17	XX	Calibration SW1 Minor version
18	XX	Calibration SW2 Block ID
19	XX	Calibration SW2 Major Version
20	XX	Calibration SW2 Minor version
21	XX	Calibration SW3 Block ID
22	XX	Calibration SW3 Major Version
23	XX	Calibration SW3 Minor version
24	XX	Calibration SW4 Block ID
25	XX	Calibration SW4 Major Version
26	XX	Calibration SW4 Minor version

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27	XX	Calibration SW5 Block ID
28	XX	Calibration SW5 Major Version
29	XX	Calibration SW5 Minor version
30	XX	Calibration SW6 Block ID
31	XX	Calibration SW6 Major Version
32	XX	Calibration SW6 Minor version
33	XX	Calibration SW7 Block ID
34	XX	Calibration SW7 Major Version
34	XX	Calibration SW7 Minor version
36	XX	Calibration SW8 Block ID
37	XX	Calibration SW8 Major Version
38	XX	Calibration SW8 Minor version
39	XX	ESP SW Block ID
40	XX	ESP SW Major version
41	XX	ESP SW Minor Version

3.6.156 Reprograming Status Flag: $22 $F2 0A

This DID is used to read the status of the Reprogramming the flag.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Response ID: Read data by Identifier
2	F2	ID High byte
3	0A	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Service ID: Read Data by Identifier
2	F2	ID High byte
3	0A	ID Low byte
4	XX	Status Flag

3.6.157 Integrated Circuit Card Identifier1: $22 $F2 0B

This DID is used to read Integrated Circuit Card Identifier (ICCI) in Global System for Mobile Communication (GSM)is a unique identification number assigned to a SIM card.

Request Message	**Message Length: 3 Bytes**

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Byte number	Value (Hex)	Parameter Name
1	22	Response ID: Read data by Identifier
2	F2	ID High byte
3	0B	ID Low byte

Positive Response Message	**Message Length: 23 Bytes**

Byte number	Value (Hex)	Parameter Name
1	62	Service ID: Read Data by Identifier
2	F2	ID High byte
3	0B	ID Low byte
4-23	XX	Integrated Circuit Card Identifier

3.6.158 Integrated Circuit Card Identifier2: $22 $F2 0C

This DID is used to read Integrated Circuit Card Identifier (ICCI) in Global System for Mobile Communication (GSM)is a unique identification number assigned to a SIM card.

Request Message	**Message Length: 3 Bytes**

Byte number	Value (Hex)	Parameter Name
1	22	Response ID: Read data by Identifier
2	F2	ID High byte
3	0C	ID Low byte

Positive Response Message	**Message Length: 23 Bytes**

Byte number	Value (Hex)	Parameter Name
1	62	Service ID: Read Data by Identifier
2	F2	ID High byte
3	0C	ID Low byte
4-23	XX	Integrated Circuit Card Identifier2

3.6.159 Secondary Key fob Mac ID :$22 $F2 0D

The purpose of this DID is to write the MAC_ID of keyfob(s) to be configured.

Request Message	**Message Length: 3 Bytes**

Byte number	Value (Hex)	Parameter Name
1	22	Response ID: Read data by Identifier
2	F2	ID High byte
3	0D	ID Low byte

Positive Response Message	**Message Length: Bytes**

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Byte number	Value (Hex)	Parameter Name
1	62	Service ID: Read Data by Identifier
2	F2	ID High byte
3	0D	ID Low byte
4-9	XX	Secondary KeyFob ID

3.6.160 E-SIM Activation Status :$22 $09 4D

This DID is used to read the status of the device for the current active eSIM.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Response ID: Read data by Identifier
2	09	ID High byte
3	4D	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Service ID: Read Data by Identifier
2	09	ID High byte
3	4D	ID Low byte
4	XX	b0 - eSIM 1 Activation Status (0 - Inactive ; 1 - Active) b1 - eSIM 2 Activation Status (0 - Inactive ; 1 - Active) b2 to b7 - Reserved.

3.6.161 Cruise Control Feature: $22 $09 4E

This DID is used to enable / disable the Cruise control Feature

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Response ID: Read data by Identifier
2	09	ID High byte
3	4E	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Service ID: Read Data by Identifier

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2	09	ID High byte
3	4E	ID Low byte
4	XX	0 - Enable, 1 - disable

3.6.162 Heated Grip Feature : $22 $09 4F

This DID is used to enable / disable the Heated Grip Feature.

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Response ID: Read data by Identifier
2	09	ID High byte
3	4F	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Service ID: Read Data by Identifier
2	09	ID High byte
3	4F	ID Low byte
4	XX	0 - Enable, 1 - disable

3.6.163 Hazard Lamp Cancellation:$22 $09 50

This DID is used to configure the Hazard Lamp auto cancellation time

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Response ID: Read data by Identifier
2	09	ID High byte
3	50	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Service ID: Read Data by Identifier
2	09	ID High byte
3	50	ID Low byte
4	XX	(0 to 120)mins Resolution : not applicable, hence 0x01 - 1mins , 0x78 - 120 mins, 0x3C - 60min

3.6.164 Fuel sensor mean :$22 $09 51

This DID is used to read the fuel sensor value provided by the Dash Board.

Request Message	**Message Length: 3 Bytes**

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Byte number	Value (Hex)	Parameter Name
1	22	Response ID: Read data by Identifier
2	09	ID High byte
3	51	ID Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Service ID: Read Data by Identifier
2	09	ID High byte
3	51	ID Low byte
4	XX	scaling factor of the data: 0.1

3.6.165 Live Parameter:

ECU	VCU	BMS	Cluster	MCU	ABS	iVCU	Charger	LCM
Live Parameter DID range	0100~01FF	0200~02FF	0300~03FF	0400~04FF	0500~05FF	0600~06FF	0700~07FF	0800~08FF

3.6.165.1 VCU

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier: Live Parameter
2	01	ID High byte: Read Data by Identifier High byte
3	00 – FF	ID Low byte: Read Data by Identifier Low byte

Positive Response Message		**Message Length: n Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	01	ID High byte: Read Data by Identifier High byte
3	XX	ID Low byte: Read Data by Identifier Low byte
4	XX	Data byte 1
…	…	…
n	XX	Data byte m

3.6.165.2 BMS

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier: Live Parameter
2	02	ID High byte: Read Data by Identifier High byte

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3	00 – FF	ID Low byte: Read Data by Identifier Low byte

Positive Response Message		**Message Length: n Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	02	ID High byte: Read Data by Identifier High byte
3	XX	ID Low byte: Read Data by Identifier Low byte
4	XX	Data byte 1
…..	……	…….
n	XX	Data byte m

3.6.165.3 Cluster

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier: Live Parameter
2	03	ID High byte: Read Data by Identifier High byte
3	00 – FF	ID Low byte: Read Data by Identifier Low byte

Positive Response Message		**Message Length: n Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	03	ID High byte: Read Data by Identifier High byte
3	XX	ID Low byte: Read Data by Identifier Low byte
4	XX	Data byte 1
…..	……	…….
n	XX	Data byte m

3.6.165.4 MCU

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier: Live Parameter
2	04	ID High byte: Read Data by Identifier High byte

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3	00 – FF	ID Low byte: Read Data by Identifier Low byte

Positive Response Message		**Message Length: n Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	04	ID High byte: Read Data by Identifier High byte
3	XX	ID Low byte: Read Data by Identifier Low byte
4	XX	Data byte 1
…..	……	…….
n	XX	Data byte m

3.6.165.5 ABS

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier: Live Parameter
2	05	ID High byte: Read Data by Identifier High byte
3	00 – FF	ID Low byte: Read Data by Identifier Low byte

Positive Response Message		**Message Length: n Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	05	ID High byte: Read Data by Identifier High byte
3	XX	ID Low byte: Read Data by Identifier Low byte
4	XX	Data byte 1
…..	……	…….
n	XX	Data byte m

3.6.165.5.1 ECU Input signal (1)

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier: Live Parameter
2	05	ID High byte: Read Data by Identifier High byte
3	02	ID Low byte: Read Data by Identifier Low byte

Positive Response Message		**Message Length: 7 Bytes**
Byte number	Value (Hex)	Parameter Name

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1	62	Response ID: Read data by Identifier
2	05	ID High byte: Read Data by Identifier High byte
3	02	ID Low byte: Read Data by Identifier Low byte
4	XX	Front Wheel Speed (1 LSB = 1 Km/h)
5	XX	Read Wheel Speed (1 LSB = 1 Km/h)
6	XX	Power Supply Voltage (1 LSB = 0.078 V)
7	XX	Valve status * -Bit0: Valve relay status, 1=ON -Bit1: Pump motor status, 1=ON -Bit2: Front Inlet Valve status, 1=ON -Bit3: Front Outlet Valve status, 1=ON -Bit4: Reserved -Bit5: Reserved -Bit6: Rear Inlet Valve status, 1=ON * -Bit7: Rear Outlet Valve status, 1=ON*

Note

1. After the execution of Actuator test, the status of the Valves and pump can be read using this service.
2. Maximum speed shown as 0xFF, even if speed greater than 255 kmph.

3.6.165.5.2 ECU Input signal (2)

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier: Live Parameter
2	05	ID High byte: Read Data by Identifier High byte
3	03	ID Low byte: Read Data by Identifier Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	05	ID High byte: Read Data by Identifier High byte
3	03	ID Low byte: Read Data by Identifier Low byte
4	XX	Digital input -Bit0: Reserved -Bit1: Reserved -Bit2: Reserved -Bit3: Reserved -Bit4: Reserved -Bit5: Reserved -Bit6: ABS Off switch status (0: Pushed, 1: Released) -Bit7: Reserved

Note: *If ABSOFF feature is not available, the ECU answers with a Negative Response [7F 22 12].*

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3.6.165.5.3 ABSOFF History Data

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier: Live Parameter
2	05	ID High byte: Read Data by Identifier High byte
3	F8	ID Low byte: Read Data by Identifier Low byte

Positive Response Message		Message Length: 19 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	05	ID High byte: Read Data by Identifier High byte
3	F8	ID Low byte: Read Data by Identifier Low byte
4	XX	History Data Set 1 Byte 0
5	XX	History Data Set 1 Byte 1
6	XX	History Data Set 1 Byte 2
7	XX	History Data Set 1 Byte 3
8	XX	History Data Set 1 Byte 4
9	XX	History Data Set 1 Byte 5
10	XX	History Data Set 1 Byte 6
11	XX	History Data Set 1 Byte 7
12	XX	History Data Set 2 Byte 0
13	XX	History Data Set 2 Byte 1
14	XX	History Data Set 2 Byte 2
Byte number	Value (Hex)	Parameter Name
15	XX	History Data Set 2 Byte 3
16	XX	History Data Set 2 Byte 4
17	XX	History Data Set 2 Byte 5
18	XX	History Data Set 2 Byte 6
19	XX	History Data Set 2 Byte 7

ABS OFF History Data:

Byte	0	1	2	3	4	5	6	7
Contents	A		B		C	D	E	

A : Sequence Counter (will be incremented by one for every successful Enable/Disable of ABS)
B : IgnitionCounter
C : ABS Disabled (0x00 : No and 0xFF:Yes)
D : Reserved
E : VehicleSpeed 1bit=0.015625m/s (1/64m/s)

3.6.165.5.4 Flash History Data

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier: Live Parameter

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2	05	ID High byte: Read Data by Identifier High byte
3	F9	ID Low byte: Read Data by Identifier Low byte

Positive Response Message		**Message Length: 16 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	05	ID High byte: Read Data by Identifier High byte
3	F9	ID Low byte: Read Data by Identifier Low byte
4	XX-BCD	Programming Date Year (YY) MSB
5	XX-BCD	Programming Date Month (YY)
6	XX-BCD	Programming Date Day (YY) LSB
7	XX-ASCII	Part Number Position #1 (High Byte)
8	XX-ASCII	Part Number Position # 2
9	XX-ASCII	Part Number Position # 3
10	XX-ASCII	Part Number Position # 4
11	XX-ASCII	Part Number Position # 5
12	XX-ASCII	Part Number Position # 6
13	XX-ASCII	Part Number Position # 7
14	XX-ASCII	Part Number Position # 8
15	XX-ASCII	Part Number Position # 9
16	XX-ASCII	Part Number Position # 10(LOW Byte)

3.6.165.5.5 Process byte

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier: Live Parameter
2	05	ID High byte: Read Data by Identifier High byte
3	FA	ID Low byte: Read Data by Identifier Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	05	ID High byte: Read Data by Identifier High byte
3	FA	ID Low byte: Read Data by Identifier Low byte
4	XX	Process Byte

3.6.165.5.6 FSW/ASW Validity Check

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier: Live Parameter
2	05	ID High byte: Read Data by Identifier High byte
3	FD	ID Low byte: Read Data by Identifier Low byte

Positive Response Message	**Message Length: 4 Bytes**

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Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	05	ID High byte: Read Data by Identifier High byte
3	FD	ID Low byte: Read Data by Identifier Low byte
4	XX	00-FSW valid, 01-FSW not valid

3.6.165.5.7 Pressure Sensor Signals

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier: Live Parameter
2	05	ID High byte: Read Data by Identifier High byte
3	09	ID Low byte: Read Data by Identifier Low byte

Positive Response Message		**Message Length: 11 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	05	ID High byte: Read Data by Identifier High byte
3	09	ID Low byte: Read Data by Identifier Low byte
4	XX	Front WC Pressure value Low Byte (*1)
5	XX	Front WC Pressure value High Byte (*1)
6	XX	Rear WC Pressure value Low Byte (*)
7	XX	Rear WC Pressure value High Byte (*)
8	XX	Front Wheel MC Pressure value Low Byte (*)
9	XX	Front Wheel MC Pressure value High Byte (*)
10	XX	Rear Wheel MC Pressure value Low Byte (*)
11	XX	Rear Wheel MC Pressure value High Byte (*)

*1: The service is applicable only for “MP” system,
Pressure value: Resolution: 0.0153bar | bit signed
Value range: -42.5 … 425 bars (F527 H … 6C81 H)

3.6.165.5.8 Variant

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier: Live Parameter
2	05	ID High byte: Read Data by Identifier High byte
3	FF	ID Low byte: Read Data by Identifier Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	05	ID High byte: Read Data by Identifier High byte
3	FF	ID Low byte: Read Data by Identifier Low byte
4	XX	Variant

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3.6.165.5.9 IMU Sensor Data

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier: Live Parameter
2	05	ID High byte: Read Data by Identifier High byte
3	07	ID Low byte: Read Data by Identifier Low byte

Positive Response Message		Message Length: 6 Bytes
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	05	ID High byte: Read Data by Identifier High byte
3	07	ID Low byte: Read Data by Identifier Low byte
4	XX	Lean Angle Low Byte
5	XX	Lean Angle High Byte
6	XX	MM5.10 Sensor Availability Sensor Status Bit0: Lean Angle Sensor Availability, 1=ON Bit1: Reserved Bit2: Reserved Bit3: Reserved Bit4: Reserved Bit5: Reserved Bit6: Reserved Bit7: Reserve

Resolution: 1 Degree per bit

Note: Angle has to be read in degrees in signed word format.

ReadDataByLocalIdentifier			
SL No	Angle	HB	LB
1	Positive Values	XX	XX
2	Negative Values	FX	XX
3	0	00	00
4	180	00	B4
5	-180	FF	4C

3.6.165.5.10EOL Lean Angle Data

Request Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier: Live Parameter

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2	05	ID High byte: Read Data by Identifier High byte
3	08	ID Low byte: Read Data by Identifier Low byte

Positive Response Message		**Message Length: 5 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	05	ID High byte: Read Data by Identifier High byte
3	08	ID Low byte: Read Data by Identifier Low byte
4	XX	Lean Angle Low Byte
5	XX	Lean Angle High Byte

3.6.165.5.11EOL Mode

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier: Live Parameter
2	05	ID High byte: Read Data by Identifier High byte
3	F1	ID Low byte: Read Data by Identifier Low byte

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	05	ID High byte: Read Data by Identifier High byte
3	F1	ID Low byte: Read Data by Identifier Low byte
4	XX	00-Successful Completion of EOL 01-Phase-1 of EOL Entered 02-EOL has entered the lock mode

3.6.165.6 iVCU

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier: Live Parameter
2	06	ID High byte: Read Data by Identifier High byte
3	00 – FF	ID Low byte: Read Data by Identifier Low byte

Positive Response Message		**Message Length: n Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	06	ID High byte: Read Data by Identifier High byte
3	XX	ID Low byte: Read Data by Identifier Low byte
4	XX	Data byte 1
…..	……	…….

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n	XX	Data byte m

3.6.165.7 Charger

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier: Live Parameter
2	07	ID High byte: Read Data by Identifier High byte
3	00 – FF	ID Low byte: Read Data by Identifier Low byte

Positive Response Message		**Message Length: n Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	07	ID High byte: Read Data by Identifier High byte
3	XX	ID Low byte: Read Data by Identifier Low byte
4	XX	Data byte 1
…..	……	…….
n	XX	Data byte m

3.6.165.8 LCM

Request Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	22	Service ID: Read Data by Identifier: Live Parameter
2	08	ID High byte: Read Data by Identifier High byte
3	00 – FF	ID Low byte: Read Data by Identifier Low byte

Positive Response Message		**Message Length: n Bytes**
Byte number	Value (Hex)	Parameter Name
1	62	Response ID: Read data by Identifier
2	08	ID High byte: Read Data by Identifier High byte
3	XX	ID Low byte: Read Data by Identifier Low byte

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4	XX	Data byte 1
…..	……	…….
n	XX	Data byte m

3.7 Write Data by Identifier: $2E

The Write Data by Identifier service allows the client to write information into the server at an internal location specified by the provided data identifier. The Write Data by Identifier service is used by the client to write a data Record to a server.

Supported NRCs

Negative Response	
Server Response **(Hex)**	Response Description
13	Incorrect message length or Invalid Format.
22	Operating conditions of the server are not met to perform the required action.
31	The requested DID is either is not supported or has not been assigned.
33	If the DID is secured and the server is not in unlocked state.
72	Server detects an error while writing to a memory location.
7F	Service not supported in active session.

3.7.1 Test stamp: $2E $10 FF

Request Message		**Message Length: 6 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	10	ID High byte: Test Stamp High Byte
3	FF	ID Low byte: Test Stamp Low Byte
4	XX	Test-stamp Data byte 1 (MSB)
5	XX	Test-stamp Data byte 2
6	XX	Test-stamp Data byte 3 (LSB)

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name

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1	6E	Response ID: Write data by Identifier
2	10	ID High byte: Test Stamp High byte
3	FF	ID Low byte: Test Stamp Low byte

The Test-stamp Data bytes are to be mutually decided between Diagnostic Tool provider and TVSM.

3.7.2 VIN WRITE: $2E $F1 90

Request Message		**Message Length: 20 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F1	ID High byte: VIN Write ID High byte
3	90	ID Low byte: VIN Write ID Low byte
4	XX	VIN Data byte 1 (MSB) [ASCII]
5	XX	VIN Data byte 2[ASCII]
6	XX	VIN Data byte 3[ASCII]
….	…..	……
20	XX	VIN Data byte 17 (LSB) [ASCII]

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	F1	ID High byte: VIN Write ID High byte
3	90	ID Low byte: VIN Write ID Low byte

3.7.3 SKU WRITE: $2E $F1 91

This DID is used to write the vehicle variant information

Request Message		**Message Length: 19 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F1	ID High byte: SKU Write ID High byte
3	91	ID Low byte: SKU Write ID Low byte
4-19	XX	SKU (ASCII)

Positive Response Message	**Message Length: 3 Bytes**

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Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	F1	ID High byte: SKU Write ID High byte
3	91	ID Low byte: SKU Write ID Low byte

Note: Unused bytes of the SKU shall be filled with 0x2D.

3.7.4 ODO WRITE: $2E $F1 0A

Request Message		**Message Length: 7 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F1	ID High byte: ODO Write ID High byte
3	0A	ID Low byte: ODO Write ID Low byte
4 - 7	XX	Odometer

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	F1	ID High byte: ODO Write ID High byte
3	0A	ID Low byte: ODO Write ID Low byte

3.7.5 Service Reminder Reset: $2E $F1 80

Request Message		**Message Length: 5 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F1	ID High byte: Service Reminder Write ID High byte
3	80	ID Low byte: Service Reminder Write ID Low byte
4	32	Service Reminder Data byte 1 (MSB)
5	01	Service Reminder Data byte 2 (LSB)

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier

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2	F1	ID High byte: Service Reminder Write ID High byte
3	80	ID Low byte: Service Reminder Write ID Low byte

3.7.6 Pack Variant: $2E $ F1 01

Request Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write data by Identifier
2	F1	ID High byte: pack variant write ID high byte
3	01	ID Low byte: pack variant write ID low byte
4	XX	Variant: No of Battery packs[Hex]

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	F1	ID High byte: pack variant write ID high byte
3	01	ID Low byte: pack variant write ID low byte

3.7.7 Process Data: $2E $ FA 01

Request Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write data by Identifier
2	FA	ID High byte: process data write ID high byte
3	01	ID Low byte: process data write ID low byte
4	XX	Process data

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	FA	ID High byte: process data write ID high byte
3	01	ID Low byte: process data write ID low byte

3.7.8 Checksum Calculated from tester: $2E $ F1 A1

This checksum will be calculated and sent by the tester (during reprogramming) over CAN.

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Request Message		Message Length: 5 Bytes
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write data by Identifier
2	F1	ID High byte: Checksum Calculated from tester write ID high byte
3	A1	ID Low byte: Checksum Calculated from tester write ID low byte
4	XX	Checksum Hex Value (MSB)
5	XX	Checksum Hex Value (LSB)

Positive Response Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	F1	ID High byte: Checksum Calculated from tester write ID high byte
3	A1	ID Low byte: Checksum Calculated from tester write ID low byte

3.7.9 ECU Serial Number: $2E $ F1 8A

Request Message		Message Length: 9 Bytes
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Read Data by Identifier
2	F1	ID High byte: ECU serial number write ID high byte
3	8A	ID Low byte: ECU serial number write ID low byte
4	XX	Hex Value (MSB)
5	XX	Hex Value
6	XX	Hex Value
7	XX	Hex Value
8	XX	Hex Value
9	XX	Hex Value (LSB)

Positive Response Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Read data by Identifier

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2	F1	ID High byte: ECU serial number write ID high byte
3	8A	ID Low byte: ECU serial number write ID low byte

Note: ECU serial number is written once per ECU.

3.7.10 Vehicle Software Version: $2E $ F1 89

Request Message		**Message Length: 12 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Read Data by Identifier
2	F1	ID High byte: vehicle software version write ID high byte
3	89	ID Low byte: vehicle software version write ID low byte
4	XX	ASCII Value (MSB)
5	XX	ASCII Value
6	XX	ASCII Value
7	XX	ASCII Value
8	XX	ASCII Value
9	XX	ASCII Value
10	XX	ASCII Value
11	XX	ASCII Value
12	XX	ASCII Value (LSB)

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Read data by Identifier
2	F1	ID High byte: vehicle software version write ID high byte
3	89	ID Low byte: vehicle software version write ID low byte

3.7.11 GA Number/Software BOM ID: $2E $F1 88

This service is to write the General Assembly/Software assembly number (8 Bytes).

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Request Message		Message Length: 11 Bytes
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F1	ID High byte: GA Number Write ID High byte
3	88	ID Low byte: GA Number Write ID Low byte
4	XX	GA Number byte 1 (MSB)
5	XX	GA Number byte 2
6	XX	GA Number byte 3
….	…..	……
11	XX	GA Number byte 8 (LSB)
Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	F1	ID High byte: GA Number High byte
3	88	ID Low byte: GA Number Low byte

3.7.12 Application Presence Pattern: $2E $F1 2D

This service is to write the Application presence pattern (4 Bytes).

Request Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F1	ID High byte: App presence pattern Write ID High byte
3	2D	ID Low byte: App presence pattern Write ID Low byte
4	01	App presence pattern write status

Note:

Sending the Application presence pattern write status as 01 in the App presence pattern write request will trigger the write function which will write the hard-coded values of the Application presence pattern.

Positive Response Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	F1	ID High byte: App presence pattern Write ID High byte
3	2D	ID Low byte: App presence pattern Write ID Low byte

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3.7.13 Fingerprint/Tester serial Number: $2E $F1 84

This service is to write Fingerprint (2 Bytes).

Request Message		**Message Length: 5 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F1	ID High byte: Fingerprint request Write ID High byte
3	84	ID Low byte: Fingerprint request Write ID Low byte
4	XX	Index (MSB)
5	XX	Index (LSB)

Value Table

Serial Number	Index (Hex)	Client/Tester Name
1	00 01	Infotainment System
2	00 02	Telematics Control unit
3	00 03	MAHLE tool
4	00 04	CAPL script
5	00 05	R&D SW CoC UDS Programmer
6	00 06 – 00 FF	Reserved for future

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	F1	ID High byte: Fingerprint request Write ID High byte
3	84	ID Low byte: Fingerprint request Write ID Low byte

3.7.14 Battery UID or BIN: $2E $F1 1E

The purpose of this service is to write the Battery unique Identification Number [BIN] (12 Bytes).

There are 2 modes for the usage of Battery UID.

Mode1:

This is the default mode where the UIDs will be locked in a vehicle.

Mode 2:

In this mode the Vehicle will be made open to any UID.

Note:

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In BMS (battery pack) the Battery UID should be restricted to writing one-time.
To do the same, the Battery UID should be locked automatically to avoid rewriting/ modification.

However, if the Battery UID written on the battery pack production line goes wrong. There is a provision to unlock the Battery UID writing process. This unlocking of the ECU can be done using a routine control service ($31) with subfunction ($01, $02, $03).

To use the service to unlock the Battery UID writing, Cryptographic security must be passed (A seed-key authentication must be done).

The BIN writing, locking, unlocking flow specification can be provided based on request.

Request Message		**Message Length: 15 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F1	ID High byte: Data ID High byte
3	1E	ID Low byte: Data ID Low byte
4	XX	Data Byte 1 (ASCII): Make (TVS marked as “T”)
5	XX	Data Byte 2 (ASCII): Battery Type (MSB)
6	XX	Data Byte 3 (ASCII): Battery Type (LSB)
7	XX	Data Byte 4 (ASCII): Month Code
8	XX	Data Byte 5 (ASCII): Year Code
9	XX	Data Byte 6 (ASCII): Plant Code
10	XX	Data Byte 7 (ASCII): Check Digit
11	XX	Data Byte 8 (HEX): Running Serial Number (MSB)
12	XX	Data Byte 9 (HEX): Running Serial Number
13	XX	Data Byte 10 (HEX): Running Serial Number
14	XX	Data Byte 11 (HEX): Running Serial Number
15	XX	Data Byte 12 (HEX): Running Serial Number (LSB)

For more Byte-wise details, please refer to the reference document “KE210130_A3”.

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier

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2	F1	ID High byte: Data ID High byte
3	1E	ID Low byte: Data ID Low byte

3.7.15 Theta: $2E $F1 30

This service is to write the motor's Initial angle of sync (Theta). The range of theta is from 0 to 4095 counts.

Request Message		**Message Length: 13 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F1	ID High byte
3	30	ID Low byte
4	XX	Theta Init (MSB)
5	XX	Theta Init (LSB)
6	XX	Sine Offset [0 – 255]
7	XX	Cosine Offset [0 – 255]
8	XX	Correction gain (MSB)
9	XX	Correction gain (LSB)
10	XX	Phase lag [0 – 255]
11	XX	Theta Calib status [0 – 3]
12	XX	Theta Init Offset (MSB)
13	XX	Theta Init Offset (LSB)

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	F1	ID High byte
3	30	ID Low byte

3.7.16 TPMS Pairing Data: $2E $ F1 32

The purpose of this service is to write the MAC ID of the Tire Pressure Measurement System (TPMS). This will be written at the Production's End of Line (EOL) or Service Centre for which the vehicle speed should be zero.

Request Message		**Message Length: 10 Bytes**
Byte number	Value (Hex)	Parameter Name

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1	2E	Service ID: Write Data by Identifier
2	F1	ID High byte
3	32	ID Low byte
4	X'X''	Position of the tire Upper Nibble (X'): 1 – Front, 2 – Rear Lower Nibble (X''): 0 – Centre (Single), 1 – Left, 2 – Right
5	XX	Data Byte 1 (HEX): MAC ID (MSB)
6	XX	Data Byte 2 (HEX): MAC ID
…	…	…
10	XX	Data Byte 6 (HEX): MAC ID (LSB)

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	F1	ID High byte
3	32	ID Low byte

3.7.17 Tire Pressure Threshold: $2E $ F1 33

This service is to write the Tire pressure threshold values. The range is from 0 to 100.

Request Message		**Message Length: 7 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F1	ID High byte
3	33	ID Low byte
4	XX	Tire pressure Threshold: Very low pressure
5	XX	Tire pressure Threshold: Low pressure
6	XX	Tire pressure Threshold: High pressure
7	XX	Tire pressure Threshold: Very High pressure

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	F1	ID High byte
3	33	ID Low byte

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3.7.18 Deleted:

3.7.19 Deleted

3.7.20 Immobilization Status: $2E $ F1 36

The purpose of this DID is to enable or disable the Immobilization feature.

Value description:

01: Enable

00: Disable

Request Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F1	ID High byte: Immobilization Status High byte
3	36	ID Low byte: Immobilization Status Low byte
4	XX	Immobilization Status

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	F1	ID High byte: Immobilization Status High byte
3	36	ID Low byte: Immobilization Status Low byte

3.7.21 Hill Hold Feature: $2E $ F1 37

The purpose of this DID is to write the pitch angle for a vehicle.

Request Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F1	ID High byte: Hill Hold High byte
3	37	ID Low byte: Hill Hold Low byte
4	XX	HEX: Pitch angle (degrees)

Positive response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	F1	ID High byte: Hill Hold High byte

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3	37	ID Low byte: Hill Hold Low byte

3.7.22 MAC_ID Bluetooth chip: $2E $ B1 D5

The purpose of this DID is to write the MAC_ID of Bluetooth chip.

Request Message		**Message Length: 9 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	B1	ID High byte: MAC_ID High byte
3	D5	ID Low byte: MAC_ID Low byte
4	XX	Byte1[Hex][MSB]
5	XX	Byte2[Hex]
6	XX	Byte3[Hex]
7	XX	Byte4[Hex]
8	XX	Byte5[Hex]
9	XX	Byte6[Hex][LSB]

Positive Response		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write Data by Identifier
2	B1	ID High byte: MAC_ID High byte
3	D5	ID Low byte: MAC_ID Low byte

3.7.23 Secure Key: $2E $ F1 39

The purpose of this DID is to write the secure key.

Request Message		**Message Length: 19 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F1	ID High byte: secure key High byte
3	39	ID Low byte: secure key Low byte
4-19	XX	Byte1 – Byte16 [Hex]

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write Data by Identifier

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2	F1	ID High byte: secure key High byte
3	39	ID Low byte: secure key Low byte

3.7.24 IMEI Number: $2E $ F1 43

The purpose of this DID is to write the IMEI number.

Request Message		**Message Length: 18 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F1	ID High byte: IMEI number High byte
3	43	ID Low byte IMEI number Low byte
4-18	XX	Byte1 – Byte15[ASCII][MSB]

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write Data by Identifier
2	F1	ID High byte: IMEI number High byte
3	43	ID Low byte IMEI number Low byte

3.7.25 Distance unit: $2E $ E1 17

This DID represent the unit of the distance.
If the byte value is 0, unit is kilometers (default)
If the byte value is 1, unit is miles

Request Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	E1	ID High byte: Distance unit High byte
3	17	ID Low byte: Distance unit Low byte
4	XX	Byte1[Hex]

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	E1	ID High byte: Distance unit High byte
3	17	ID Low byte: Distance unit Low byte

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3.7.26 Vehicle Manufacturing Time and Date: $2E $ E1 2B

Time and date are read in this DID
The time and date format will be HHmmDDMMYYYY (24Hr Hex Format)
Example:
If the date is April 3rd, 2019, and time is 06:45Pm then the response byte value shall be 0x12 0x2D 0x03 0x04 0x14 0x13

Request Message		**Message Length: 9 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	E1	ID High byte: Time and Date High byte
3	2B	ID Low byte: Time and Date Low byte
4	XX	Hour [Hex]
5	XX	Minute [Hex]
6	XX	Day [Hex]
7	XX	Month [Hex]
8	XX	Year [MSB][Hex]
9	XX	Year [LSB][Hex]

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	E1	ID High byte: Time and Date High byte
3	2B	ID Low byte: Time and Date Low byte

3.7.27 ABS Variant: $2E $ F1 97

2-byte NVM is allocated for ABS Variant.

- 0XE1 -> U489C-flared
- 0xE2 -> U489c-naked
- 0xE3 -> N597

Example: If the variant is U489c Flared then the value is 0x00E1

Request Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F1	ID High byte: ABS variant High byte
3	97	ID Low byte: ABS variant Low byte

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4	XX	Byte1[Hex]
5	XX	Byte2[Hex]

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	F1	ID High byte: ABS variant High byte
3	97	ID Low byte: ABS variant Low byte

3.7.28 Deleted

3.7.29 Deleted

3.7.30 Tire pressure threshold for rear and front tire: $2E $ F1 9A

This DID shall be used when the thresholds are different for front and rear tire.
Threshold for Front and Rear Tire for low or high pressure shall be provided in Hex format

- 0th position -> Front very Low tire pressure
- 1st position -> Front Low tire pressure
- 2nd byte position -> Front high tire pressure
- 3rd byte position -> Front very high tire pressure
- 4th position -> Rear very low tire pressure
- 5th position -> Rear low tire pressure
- 6th position -> Rear high tire pressure
- 7th position -> Rear very high tire pressure

01 indicates both rear and front tire low pressure value and 02 indicates both rear and front tire high pressure value.

Example:

- Front very low tire pressure = 13 psi (0x0D)
- Front low tire pressure = 19 psi(0x13)
- Front high tire pressure = 40 psi (0x28)
- Front very high tire pressure = 44(0x2C)
- Rear very low tire pressure = 15(0x0F)
- Rear low tire pressure = 21 psi(0x15)
- Rear high tire pressure = 42 psi (0x2A)
- Rear very high tire pressure = 46(0x2E)

Request Message		**Message Length: 11 Bytes**
Byte number	Value (Hex)	Parameter Name

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1	2E	Service ID: Write Data by Identifier
2	F1	ID High byte: Tire Pressure threshold for front and rear High byte
3	9A	ID Low byte: Tire Pressure threshold for front and rear Low byte
4	XX	Byte1[Hex][MSB]
5	XX	Byte2[Hex]
6	XX	Byte3[Hex]
7	XX	Byte4[Hex]
8	XX	Byte5[Hex]
9	XX	Byte6[Hex]
10	XX	Byte7[Hex]
11	XX	Byte8[Hex][LSB]

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	FF	ID High byte: Configuring low or high threshold for front and rear High byte
3	9A	ID Low byte Configuring low or high threshold for front and rear Low byte

3.7.31 Head lamp control: $2E $ FF 55

The purpose of this DID is to read the Head lamp control enable/disable status.
Adaptive headlamp control shall be enable/disable by setting 'Byte1'. For Enabling, it is '1' and disabling it is '0'(Default is 1).
The reserved bytes shall be kept as 0.

Request Message		**Message Length: 10 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	FF	ID High byte: Head lamp control High byte
3	55	ID Low byte: Head lamp control Low byte
4	XX	Byte1[Hex][MSB]: Head lamp control enable/disable status
5	XX	Byte2[Hex] Reserved
6	XX	Byte3[Hex]: Reserved

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7	XX	Byte4[Hex]: Reserved
8	XX	Byte5[Hex]: Reserved
9	XX	Byte6[Hex]: Reserved
10	XX	Byte7[Hex] [LSB]: Reserved

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	FF	ID High byte: Head lamp control High byte
3	55	ID Low byte: Head lamp control Low byte

3.7.32 MAC ID for Go pro pairing: $2E $ F2 00

GO PRO ID Config shall be provided in Hex format (6 bytes).

Request Message		**Message Length: 9 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F2	ID High byte: Go pro High byte
3	00	ID Low byte: Go pro Low byte
4	XX	Byte1[Hex][MSB]
5	XX	Byte2[Hex]
6	XX	Byte3[Hex]
7	XX	Byte4[Hex]
8	XX	Byte5[Hex]
9	XX	Byte6[Hex][LSB]

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	F2	ID High byte: Go pro High byte
3	00	ID Low byte: Go pro Low byte

3.7.33 Climate controlled seat: $2E $ F2 05

Climate controlled seat feature will be enabled by writing data '01' and disabled by writing data '00'.

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Request Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F2	ID High byte: Climate controlled seat High byte
3	05	ID Low byte: Climate controlled seat Low byte
4	XX	Byte1[Hex]

Positive Response Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	F2	ID High byte: Climate controlled seat High byte
3	05	ID Low byte: Climate controlled seat Low byte

3.7.34 BTO animation: $2E $ F2 06

The purpose of this DID is to enable/disable the BTO animation.

Value description:

Enabled-1

Disabled-0

Request Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F2	ID High byte: BTO animation High byte
3	06	ID Low byte: BTO animation Low byte
4	XX	Byte1[Hex]

Positive Response Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	F2	ID High byte: BTO animation High byte
3	06	ID Low byte: BTO animation Low byte

3.7.35 Tacho swipe: $2E $ F2 07

Request Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name

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1	2E	Service ID: Write Data by Identifier
2	F2	ID High byte: Tacho swipe High byte
3	07	ID Low byte: Tacho swipe Low byte
4	XX	Byte1[Hex]

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	F2	ID High byte: Tacho swipe High byte
3	07	ID Low byte: Tacho swipe Low byte

3.7.36 Celebration alerts: $2E $ F2 08

The purpose of this DID is to read the celebration alerts . It shall be provided in 2 bytes

Value mapped to Byte1:

- Enable – 1(default)
- Disable – 0

Value mapped to Byte2:

- 00 -> Default Alerts (Birthday, etc.)
- 01 -> Days of National Importance (only for IN)
- 02 -> Vehicle related Alerts (First Month, Anniversary, etc.)
- FF -> Enabled for all Celebration Alerts

Request Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F2	ID High byte: Celebration alerts High byte
3	08	ID Low byte: Celebration alerts Low byte
4	XX	Byte1[Hex]

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	F2	ID High byte: Celebration alerts High byte
3	08	ID Low byte: Celebration alerts Low byte

3.7.37 Navigation for TPMS/Low fuel: $2E $ F1 A6

To enable/disable rerouting when fuel level is Low or the tire pressure is low/critical.

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Request Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F1	ID High byte: Navigation for TPMS/Low fuel High byte
3	A6	ID Low byte: Navigation for TPMS/Low fuel Low byte
4	XX	Value Description: Enable: 1 Disable: 0

Positive Response Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	F1	ID High byte: Navigation for TPMS/Low fuel High byte
3	A6	ID Low byte: Navigation for TPMS/Low fuel Low byte

3.7.38 Display Language DID: $2E $ F1 A5

Display language shall be interpreted as mentioned below.

Value description:

EN-0(default), FR-1, DE-2, ES-3, JP-4

Request Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F1	ID High byte: Display Language High byte
3	A5	ID Low byte: Display Language Low byte
4	XX	Byte1[Hex]

Positive Response Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	F1	ID High byte: Display Language High byte
3	A5	ID Low byte: Display Language Low byte

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3.7.39 Celebration settings: $2E $ F2 09

Celebration dates can be read through tester. It shall be provided in 3 bytes
Happy Birthday Celebration, Vehicle Anniversary Celebration shall be written in this DID Celebration End Date .

Value Mapped to Byte1:

- 00 - do not consider
- 01 - Vehicle Anniversary
- 02 - Happy Birthday

Value Mapped to Byte2:

- 00 - Default

Value Mapped to Byte3:

- 00 - Default

Request Message		**Message Length: 6 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F2	ID High byte: Celebration settings High byte
3	09	ID Low byte: Celebration settings Low byte
4	XX	Byte1[Hex][MSB]
5	XX	Byte2[Hex]
6	XX	Byte3[Hex][LSB]

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	F2	ID High byte: Celebration settings High byte
3	09	ID Low byte: Celebration settings Low byte

3.7.40 TPMS Configuration Status: $2E $ B1 DB

Status of TPMS shall be as mentioned below.

Value Description:

0-Not Configured, 1 – both configured, 2- Front Configured, 3- Rear Configured

Request Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	B1	ID High byte: TPMS Configuration Status High byte
3	DB	ID Low byte: TPMS Configuration Status Low byte

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4	XX	Byte1[Hex]

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	B1	ID High byte: TPMS Configuration Status High byte
3	DB	ID Low byte: TPMS Configuration Status Low byte

3.7.41 Country DID: $2E $ F1 A7

Vehicle country code can be written through this DID.

Value Description:

IN-0(default), EU-1, LATAM-2, AU-3, JP-4, AF - 5

Request Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F1	ID High byte: Country High byte
3	A7	ID Low byte: Country Low byte
4	XX	Byte1[Hex]

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	F1	ID High byte: Country High byte
3	A7	ID Low byte: Country Low byte

3.7.42 Fog lamp: $2E $ FF 56

FOG Lamp enable status shall be considered as mentioned below

Value Description:

Enable - 1 (default)/Disable - 0

Request Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	FF	ID High byte: Fog lamp High byte
3	56	ID Low byte: Fog lamp Low byte
4	XX	Byte1[Hex][MSB]

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Positive Response Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	FF	ID High byte: Fog lamp High byte
3	56	ID Low byte: Fog lamp Low byte

3.7.43 Odometer reset: $2E $ E1 1A

The purpose of this DID is to reset the odometer.

Value Description:

To reset the odometer - 1

Request Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	E1	ID High byte: Odometer reset High byte
3	1A	ID Low byte: Odometer reset Low byte
4	XX	Byte1[Boolean]

Positive Response Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	E1	ID High byte: Odometer reset High byte
3	1A	ID Low byte: Odometer reset Low byte

3.7.44 Debug frame: $2E $ E0 63

Debug Write through Tester to enable debug message over CAN

Request Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	E0	ID High byte: Debug frame High byte
3	63	ID Low byte: Debug frame Low byte
4	XX	Byte1[Boolean]

Positive Response Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	E0	ID High byte: Debug frame High byte

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3	63	ID Low byte: Debug frame Low byte

3.7.45 Dealer ID: $2E $ F1 00

Dealer ID can be written through this DID.

Request Message		**Message Length: 8 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F1	ID High byte: Dealer ID High byte
3	00	ID Low byte: Dealer ID Low byte
4	XX	Dealer ID[MSB]
5	XX	Dealer ID
6	XX	Dealer ID
7	XX	Dealer ID
8	XX	Dealer ID[LSB]

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	F1	ID High byte: Dealer ID High byte
3	00	ID Low byte: Dealer ID Low byte

3.7.46 Project code: $2E $ F1 03

Project code can be written through this DID.

Request Message		**Message Length: 7 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	F1	ID High byte: Project code High byte
3	03	ID Low byte: Project code Low byte
4	XX	Project code [ASCII] (MSB)
5	XX	Project code [ASCII]
6	XX	Project code [ASCI] (LSB)

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier

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2	F1	ID High byte: Project code High byte
3	03	ID Low byte: Project code Low byte

3.7.47 Feature enable: $2E $ 09 19

Request Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	09	ID High byte: Feature enable High byte
3	19	ID Low byte: Feature enable Low byte
4	XX	Byte1[Hex]

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	09	ID High byte: Feature enable High byte
3	19	ID Low byte: Feature enable Low byte

3.7.48 Timing parameter for SIA: $2E $ 09 1A

Request Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write Data by Identifier
2	09	ID High byte: Timing parameter for SIA High byte
3	1A	ID Low byte: Timing parameter for SIA Low byte
4	XX	Byte1[Hex]

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	09	ID High byte: Timing parameter for SIA High byte
3	1A	ID Low byte: Timing parameter for SIA Low byte

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3.7.49 Variant control: $2E $ 01 00

Request Message		Message Length:4 Bytes
Byte number	Value (Hex)	Parameter Name
1	2E	Response ID: Write data by Identifier
2	01	ID High byte: Variant control High byte
3	00	ID Low byte: Variant control Low byte
4	XX	Variant control [Hex]

Positive Response Message		Message Length:4 Bytes
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write data by Identifier
2	01	ID High byte: Variant control High byte
3	00	ID Low byte: Variant control Low byte

3.7.50 MAC ID for key fob pairing: $2E $ F1 38

The purpose of this DID is to write the MAC_ID of keyfob(s) to be configured.

Request Message		Message Length:10 Bytes (minimum)
Byte number	Value (Hex)	Parameter Name
1	2E	Response ID: Write data by Identifier
2	F1	ID High byte: MAC_ID High byte
3	38	ID Low byte: MAC_ID Low byte
4	XX	Number of keyfobs to be paired
5-10	XX	MAC Id of Keyfob1[Hex]
11-16	XX	MAC Id of Keyfob2[Hex]

Note: The DID 0xF138 is defined with the assumption that maximum 2 keyfobs are paired per vehicle. If there are more than 2 keyfobs to be paired, it can be extended.

Positive Response Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	6E	Service ID: Write data by Identifier
2	F1	ID High byte: MAC_ID High byte
3	38	ID Low byte: MAC_ID Low byte

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3.7.51 Voice Assist Feature : $2E $A1 2B

To enable/disable Voice Assist feature.

Request Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID : Write data by Identifier
2	A1	ID High byte: Voice Assist Feature High byte
3	2B	ID Low byte: Voice Assist Feature Low byte
4	XX	Value Description: Enable -1 Disable - 0
Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write Data by Identifier
2	A1	ID High byte: Voice Assist Feature High byte.
3	2B	ID Low byte: Voice Assist Feature Low byte.

3.7.52 Number of key fobs to be Paired: $2E $44 3C

This DID is used to write total number of keyfobs to be learnt with keyless Ecu

Request Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write data by Identifier
2	44	ID High byte: Number of key fobs Paired High byte
3	3C	ID Low byte: Number of key fobs Paired Low byte
4	XX	Number of key fobs Paired Byte1[Hex]
Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write Data by Identifier
2	44	ID High byte: Number of key fobs Paired High byte
3	3C	ID Low byte: Number of key fobs Paired Low byte

3.7.53 Service Due Date : $2E $E1 2C

This DID is used to write the next service due date.
The service date format shall be (DDMMYYYY 24Hr BCD format)

Request Message		**Message Length: 7 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write data by Identifier

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2	E1	ID High byte: Service Date High byte
3	2C	ID Low byte: Service Date Low byte
4	XX	DAY
5	XX	MONTH
6,7	XX	YEAR

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Response ID: Write Data by Identifier
2	E1	ID High byte: Service Date High byte
3	2C	ID Low byte: Service Date Low byte

3.7.54 Service Distance: $2E $E1 2D

This DID records the distance value for the next scheduled service.

Request Message		**Message Length: 5 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Service ID: Write data by Identifier
2	E1	ID High byte: Service Distance High byte
3	2D	ID Low byte: Service Distance Low byte
4	XX	MSB[Hex]
5	XX	LSB [Hex]

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Service ID: Write Data by Identifier
2	E1	ID High byte: Service Distance High byte
3	2D	ID Low byte: Service Distance Low byte

3.7.55 Product Variant ID: $2E $F1 B1

It consists of a lookup table used to identify a separate project Variant from a product platform . Ex: Such as iQube 09 from iQube platform etc.

Request Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Response ID: Read data by Identifier
2	F1	ID High byte: Product Variant ID High byte
3	B1	ID Low byte: Product Variant ID Low byte
4	XX	Product Variant ID [HEX]

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Positive Response Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	6E	Service ID: Read Data by Identifier
2	F1	ID High byte: Product Variant ID High byte
3	B1	ID Low byte: Product Variant ID Low byte

Note: Allow secure write in programming session (To be written along with VIN and SKU).

3.7.56 Battery Type_0: $2E $F1 B2

It contains information of each BMS pack type that the vehicle is initially configured to.

Request Message		Message Length: 6 Bytes
Byte number	Value (Hex)	Parameter Name
1	2E	Response ID: Read data by Identifier
2	F1	ID High byte: Battery Type_0 ID High byte
3	B2	ID Low byte: Battery Type_0 ID Low byte
4-6	XX	Battery Type_0 ID [HEX]

Positive Response Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	6E	Service ID: Read Data by Identifier
2	F1	ID High byte: Battery Type_0 ID High byte
3	B2	ID Low byte: Battery Type_0 ID Low byte

Note:Allow secure write in programming session (To be written along with VIN and SKU).

3.7.57 Erase Battery Type: $2E $09 2B

This DID is used to erase the Battery type value which is stored in NVM.

Request Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	2E	Response ID: Read data by Identifier
2	09	ID High byte
3	2B	ID Low byte
4	01	Erase Battery Type ID [HEX]

Positive Response Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	6E	Service ID: Read Data by Identifier
2	09	ID High byte
3	2B	ID Low byte

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3.7.58 Battery Type Feature Check Enable: $2E $09 2C

This DID is used to enable or disable the battery type in a project.

Request Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	2E	Response ID: Read data by Identifier
2	09	ID High byte
3	2C	ID Low byte
4	XX	Battery type Check Enable [HEX] 01- Enable,00- Disable

Positive Response Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	6E	Service ID: Read Data by Identifier
2	09	ID High byte
3	2C	ID Low byte

3.7.59 Launch Control Counter: $2E $09 21

This DID is used to limit the maximum number of launches possible.

Request Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	2E	Response ID: Read data by Identifier
2	09	ID High byte
3	21	ID Low byte
4	XX	Launch Threshold Max[Hex]

Positive Response Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	6E	Service ID: Read Data by Identifier
2	09	ID High byte
3	21	ID Low byte

3.7.60 Enable or Disable Emergency Brake Lamp: :$2E $09 2D

The DID is used To Enable or Disable Emergency Brake Lamp functionality.

Request Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name

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1	2E	Response ID: Read data by Identifier
2	09	ID High byte
3	2D	ID Low byte
4	XX	Emergency Brake Lamp Enable [HEX] 01- Enable,00- Disable

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Service ID: Read Data by Identifier
2	09	ID High byte
3	2D	ID Low byte

3.7.61 Vehicle variant and Region Parameters: $2E $09 2E

The DID is used to configure the vehicle settings before delivering the vehicle to the customer.

Request Message		**Message Length:8 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Response ID: Read data by Identifier
2	09	ID High byte
3	2E	ID Low byte
4	00-0F	Vehicle Type [HEX]
5	00-0F	Vehicle Trim Level[HEX]
6	00-0F	Model Year[HEX]
7	00-0F	Region of Sale[HEX]
8	00-0F	Language

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Service ID: Read Data by Identifier
2	09	ID High byte
3	2E	ID Low byte

3.7.62 Variant Configuration Method: $2E $09 43

The DID is Method of variant configuration in cluster. If Value is set as 00, Cluster shall allow variant configuration via CAN bus message. If Value is set as 01, Cluster shall allow variant configuration via DID.

Request Message	**Message Length: 4 Bytes**

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Byte number	Value (Hex)	Parameter Name
1	2E	Response ID: Read data by Identifier
2	09	ID High byte
3	43	ID Low byte
4	XX	00-CAN 01-DID

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Service ID: Read Data by Identifier
2	09	ID High byte
3	43	ID Low byte

3.7.63 A/A2 License configuration: $2E $09 44

Cluster shall support both A and A2 license configurations within the same software package, Cluster shall support switching b/w A and A2 license configurations via UDS using DID. Based on A/A2 license configuration enabled, the following CAN signal shall be updated.

Request Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Response ID: Read data by Identifier
2	09	ID High byte
3	44	ID Low byte
4	XX	00-CAN 01-DID

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Service ID: Read Data by Identifier
2	09	ID High byte
3	44	ID Low byte

3.7.64 Reprogramming Status Flag: $2E $F2 0A

This DID is used to write the Flag for Reprogramming status.

Request Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Response ID: Read data by Identifier
2	09	ID High byte

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3	F2	ID Low byte
4	0A	Reprogramming Status Flag

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Service ID: Read Data by Identifier
2	F2	ID High byte
3	0A	ID Low byte

3.7.65 E-SIM Activation switching $22 $09 4D

This DID shall be used to request eSIM switching via a write operation.

Request Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Response ID: Read data by Identifier
2	09	ID High byte
3	4D	ID Low byte
4	XX	Valid Request combinations ▪ b0 - 1 & b1 – 0, b0 - 0 & b1 - 1 Invalid Request combinations • b0 - 0 & b1 – 0 , b0 - 1 & b1 - 1

Positive Response Message		**Message Length: 3 Bytes**
Byte number	Value (Hex)	Parameter Name
1	6E	Service ID: Read Data by Identifier
2	09	ID High byte
3	4D	ID Low byte

3.7.66 Cruise Control Feature $22 $09 4E

This DID is used to enable / disable the Cruise control Feature

Request Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	2E	Response ID: Read data by Identifier
2	09	ID High byte
3	4E	ID Low byte
4	XX	0 - Enable, 1 - disable

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Positive Response Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	6E	Service ID: Read Data by Identifier
2	09	ID High byte
3	4D	ID Low byte

3.7.67 Heated Grip Feature:$22 $09 4F

This DID is used to enable / disable the Heated Grip Feature

Request Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	2E	Response ID: Read data by Identifier
2	09	ID High byte
3	4F	ID Low byte
4	XX	0 - Enable, 1 - disable

Positive Response Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	6E	Service ID: Read Data by Identifier
2	09	ID High byte
3	4F	ID Low byte

3.7.68 Hazard Lamp cancellation:$22 $09 50

This DID is used to enable / disable the Hazard lamp cancellation.

Request Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	2E	Response ID: Read data by Identifier
2	09	ID High byte
3	50	ID Low byte
4	XX	0 to 120)mins Resolution : not applicable, hence 0x01 - 1mins , 0x78 - 120 mins, 0x3C - 60min

Positive Response Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	6E	Service ID: Read Data by Identifier
2	09	ID High byte
3	50	ID Low byte

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3.8 Tester Present: $3E

It is used by ECU to continue communication with Tester. Testers need to send the request within S3 Server-timeout to keep the server in the requested non-default session.

Request Message		**Message Length: 2 Bytes**
Byte number	Value (Hex)	Parameter Name
1	3E	Service ID: Tester Present
2	00/80*	Zero Sub-function

Positive Response Message		**Message Length: 2 Bytes**
Byte number	Value (Hex)	Parameter Name
1	7E	Response ID: Tester Present
2	00	Zero Sub-function

(*) Positive response suppression

Negative Response	
Server Response **(Hex)**	Response Description
12	Sub-function is not supported
13	Incorrect Message Length or Invalid Format

3.9 Security Access: $27

The purpose of this service is to provide a means to access data and/or diagnostic services, which have restricted access for security, emissions, or safety reasons. Diagnostic services for downloading/uploading routines or data into a server and reading specific memory locations from a server are situations where security access may be required. Improper routines or data downloaded into a server could potentially damage the electronics or other vehicle components or risk the vehicle's compliance to emission, safety, or security standards. The security concept uses a seed and key relationship.

A typical example of the use of this service is as follows:
– client requests the "Seed",
– server sends the "Seed",
– client sends the "Key" (appropriate for the Seed received),
– server responds that the "Key" was valid and that it will unlock itself.

Tester Requesting for Seed Value for Security Access
Security Access Algorithm is based **on AES 128 ECB Mode**
Security Access Algorithm detailed in Attached Document Reference (d).

3.9.1 Seed request: $27 01

Request Message	**Message Length: 2 Bytes**

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Byte number	Value (Hex)	Parameter Name
1	27	Service ID: Security Access
2	01	Request Seed Sub function

Server Responding with Random seed Value with a Positive Response

Positive Response Message		**Message Length: 10 Bytes**
Byte number	Value (Hex)	Parameter Name
1	67	Response ID: Security Access
2	01	Zero Sub-function
3	xx	Seed Value Byte 1 (MSB)
4	xx	Seed Value Byte 2
---	---	----
10	xx	Seed Value Byte 8 (LSB)

3.9.2 Security Access – Authentication Challenge: $27 02

Tester Sends the Calculated Key for Authenticating with Server
Security Access Algorithm detailed in Appendix – A

Request Message		**Message Length: 10 Bytes**
Byte number	Value (Hex)	Parameter Name
1	27	Service ID: Security Access
2	02	Send Key Sub function
3	xx	Key Value Byte 1 (MSB)
4	xx	Key Value Byte 2
---	---	---- (LSB)
10	xx	Key Value Byte 8

Server Responding with a Positive Response if Authenticated successfully.

Positive Response Message		**Message Length: 2 Bytes**
Byte number	Value (Hex)	Parameter Name
1	67	Response ID: Security Access
2	02	Send Key Sub Function Response

Server Responding with Negative Response due to the reasons below.

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Negative Response	
Server Response **(Hex)**	Response Description
12	Sub-function is not supported
13	Incorrect Message Length or Invalid Format
22	Criteria for the request Security Access is not met.
24	The 'Send Key' sub-function is received without first receiving a 'Request Seed' request message.
35	The value of the key does not match the server's internally stored/ calculated key.
36	Exceeded the maximum number [3rd time] of allowed false access attempts.
7F	Service Not supported in the active session.

3.10 Routine Control: $31

The Routine Control service is used by the client to execute a defined sequence of steps and obtain any relevant results. There is a lot of flexibility with this service, but typical usage may include functionality such as erasing memory, resetting or learning adaptive data, running a self-test, overriding the normal server control strategy, and controlling a server value to change over time including predefined sequences (e.g., close convertible roof) to name a few. In general, when used to control outputs this service is used for more complex type control whereas input Output Control by Identifier is used for relatively simple (e.g., static) output control.

Request Message		Message Length: N Bytes
Byte number	Value (Hex)	Parameter Name
1	31	Service ID: Routine Control
2	xx	Sub-function: Routine Control Type
3	xx	EOL Routine Identifier
4	xx	RoutineEntryOption#1
5	xx	--
6	xx	Routine Entry Option #m

Server Responding with a Positive Response if accepted successfully.

Positive Response Message		Message Length: 3 Bytes
Byte number	Value (Hex)	Parameter Name
1	71	Response ID: Routine Control

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2	xx	Sub-function: Routine Control Type
3	xx	EOL Identifier

Supported NRC's:

Negative Response	
Server Response **(Hex)**	Response Description
12	Sub-function is not supported
13	Incorrect Message Length or Invalid Format
22	Criteria for the request Routine Control are not met.
31	The server does not support the requested Routine Identifier.
33	The server's security feature is currently active. (not unlocked)
7F	Service Not supported in the active session.

3.10.1 Erase Flash Request: $31 01 FF 00

This value shall be used to start the server's memory erase routine. The Control option and status record format shall be ECU specific and defined by the vehicle manufacturer.

Tester Sends the request for Erasing Flash after Successful Security Access to Server

Request Message		**Message Length: 13 Bytes**
Byte number	Value (Hex)	Parameter Name
1	31	Service ID: Routine Control
2	01	Routine Control Type – Start Routine
3,4	FF 00	Erase memory
5	44	4 – Start Address length, 4 – Length in Bytes of Data Size to Erase
6	XX	MSB of Start Address Derived from Hex file
7	XX	Lower MSB of Start Address
8	XX	Higher LSB of Start Address
9	XX	LSB of Start Address
10	YY	MSB of Data Size to be Erased Derived from Hex file
11	YY	Lower MSB of Data Size
12	YY	Higher LSB of Data Size
13	YY	LSB of Data Size

Server Responding with a Positive Response if Authenticated successfully.

Positive Response Message	**Message Length: 5 Bytes**

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Byte number	Value (Hex)	Parameter Name
1	71	Response ID: Routine Control
2	01	Routine Control Type – Start Routine Response
3,4	FF 00	Erase memory
5	01 00	Successful Unsuccessful

3.10.2 EOL Routine:

These EOL Routine are taken from Bosch standard (For ABS).

Definition of routineControlType:

No	Sub Function description	SubFunctionValue
1	Start Routine	01 H/ 81 H
2	Stop Routine	02 H/ 82 H
3	RequestRoutineResults	03 H/ 83 H

Definition of 'routinIdentifiers':

No	Routine Identifier description	routineIdentifier Value
1	Evacuation and Fill	B001 H
2	Repair Bleed	B002 H
3	Actuator test	B003 H
4	Wheel Speed Sensor Test	B004 H
5	ABS SILA Blinking Test	B005 H
6	Sensor Mounting Validation Test	B006 H

3.10.3 Check Precondition: $31 01 02 01

Tester Requests Server to verify if precondition to enter/move server to reprograming session.
The list of error codes needs to be decided by the ECU owners.

Request Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	31	Service ID: Routine Control
2	01	Routine Control Type – Start Routine
3, 4	02 01	RID for pre-condition check

Server Responding with a Positive Response and respective error code.

Positive Response Message	**Message Length: 6 Bytes**

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Byte number	Value (Hex)	Parameter Name
1	71	Response ID: Routine Control
2	01	Routine Control Type – Start Routine
3, 4	02 01	RID for pre-condition check
5, 6	XX XX	Error Code (Byte 1, 2) 00 00: Pre-conditions are OK The Bit-wise details for the errors can be found in section 5.

3.10.4 Check Memory [Checksum]: $31 01 02 02

Tester Requests Server to verify if the Checksum communicated by Tester to Server matches the Checksum calculated by Server after successful Transfer Exit

Request Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	31	Service ID: Routine Control
2	01	Routine Control Type – Start Routine
3, 4	02 02	RID for Check Memory

Server Responding with a Positive Response if Authenticated successfully.

Positive Response Message		**Message Length: 7 Bytes**
Byte number	Value (Hex)	Parameter Name
1	71	Response ID: Routine Control
2	01	Routine Control Type – Start Routine Response
3,4	02 02	RID for Check Memory
5	01 00	Successful Verification Not successful
6	XX	Checksum byte1 [Hex] (MSB)
7	XX	Checksum byte2 [Hex] (LSB)

Note: The Checksum Calculation Algorithm can be found in the Annexure (e).

3.10.5 Unlock Battery UID writing: $31 01 02 03

Battery Unique ID writing (done at Battery EOL) should be restricted to one-time. However, if the Battery UID written on the battery pack goes wrong. There is a provision to unlock the Battery UID writing process. This unlocking of the ECU for Writing Battery UID can be done using this routine control service.

Note: Security Access is required to use this service.

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Request Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	31	Service ID: Routine Control
2	01	Routine Control Type – Start Routine
3, 4	02 03	To Unlock the Battery UID writing

Server Responding with a Positive Response if Authenticated successfully.

Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	71	Response ID: Routine Control
2	01	Routine Control Type – Start Routine
3,4	02 03	To Unlock the Battery UID writing

3.10.6 Battery Discharge Port: $31 01 02 04/05

At the time of flashing an ECU at the Production line and during the Over the air update (re-flashing) of any ECU the battery needs to be in the Discharge mode. This routine control service can be used to make sure that the ECU is powered by Battery throughout the flashing process.

Request Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	31	Service ID: Routine Control
2	01	Routine Control Type – Start Routine
3, 4	02 04 02 05	Battery Discharge port ON Battery Discharge port OFF

Server Responding with a Positive Response if Authenticated successfully.

Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	71	Response ID: Routine Control
2	01	Routine Control Type – Start Routine
3,4	02 04 02 05	Battery Discharge port ON Battery Discharge port OFF

*Note: Security Access is **NOT** required to use this service.*

3.10.7 Keyless sensor learning: $31 01 02 06

The routine service starts the learning of the Keyless Hall based sensor, where it learns its minimum and maximum range for handlebar rotation from one extreme to the other.

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Request Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	31	Service ID: Routine Control
2	01	Routine Control Type – Start Routine
3, 4	02 06	RID

Server Responding with a Positive Response if Authenticated successfully.

Positive Response Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	71	Response ID: Routine Control
2	01	Routine Control Type – Start Routine
3,4	02 06	RID

3.10.8 Erase all paired Key fobs routine: $31 01 4F 17

The routine service initiates the erase of all the paired key fobs.

Request Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	31	Service ID: Routine Control
2	01	Routine Control Type – Start Routine
3, 4	4F 17	RID

Server Responding with a Positive Response if Authenticated successfully.

Positive Response Message		Message Length: 5 Bytes
Byte number	Value (Hex)	Parameter Name
1	71	Response ID: Routine Control
2	01	Routine Control Type – Start Routine
3,4	4F 17	RID
5	XX	Erase successful – 0 Erase failed - 1

3.10.9 Key fob learning routine: $31 01 02 DD

The routine service initiates the pairing of Key fob(s).

Request Message		Message Length: 4 Bytes
Byte number	Value (Hex)	Parameter Name
1	31	Service ID: Routine Control

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2	01	Routine Control Type – Start Routine
3, 4	02 DD	RID

Server Responding with a Positive Response if Authenticated successfully.

Positive Response Message		**Message Length: 5 Bytes**
Byte number	Value (Hex)	Parameter Name
1	71	Response ID: Routine Control
2	01	Routine Control Type – Start Routine
3,4	02 DD	RID
5	XX	Pairing failed – 0 Pairing successful - 1

3.10.10 Fast Boot Mode: $31 01 02 07

The routine service is used for Fast boot mode purpose, to send cluster to fast boot mode for MPU flashing

Request Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	31	Service ID: Routine Control
2	01	Routine Control Type – Start Routine
3, 4	02 07	RID

Server Responding with a Positive Response if Authenticated successfully.

Positive Response Message		**Message Length: 5 Bytes**
Byte number	Value (Hex)	Parameter Name
1	71	Response ID: Routine Control
2	01	Routine Control Type – Start Routine
3,4	02 07	RID
5	XX	Pairing failed – 0 Pairing successful - 1

3.11 Control DTC Setting: $85

Session: Extended (Check the note)

The Control DTC Setting service shall be used by a client to stop or resume the updating of DTC status bits in the server(s). DTC status bits are reported in the Status of DTC parameter of the positive response to certain subfunctions of Read DTC Information.

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This service enables or disables the DTC settings.

Request Message		**Message Length: 2 Bytes**
Byte number	Value (Hex)	Parameter Name
1	85	Service ID: Control DTC setting
2	xx	01/81*: Enable the DTC settings 02/82*: Disable the DTC settings

(*) Positive response suppression

01/81: Enable the DTC settings

The server(s) shall resume the updating of diagnostic trouble code status bits according to normal operating conditions.

02/82: Disable the DTC settings

The server(s) shall stop the updating of diagnostic trouble code status bits.

Server Responding with a Positive Response if Download Request accepted successfully.

Positive Response Message		**Message Length: 2 Bytes**
Byte number	Value (Hex)	Parameter Name
1	C5	Response ID: Control DTC setting
2	xx	Requested Lid

Server Responding with Negative Response due to below reasons.

Negative Response	
Server Response **(Hex)**	Response Description
12	Sub-function is not supported
13	Incorrect Message Length or Invalid Format
22	The server is in a critical normal mode activity and therefore cannot perform the requested DTC control functionality.
31	There is an error in the DTC Setting Control Option Record.
7F	Service Not supported in active session

Note:

If the server session transitions to the default, then all the DTCs operations will start again.

3.12 Communication Control: $28

Session: Extended (Check the note)

The purpose of this service is to switch on/off the transmission and/or the reception of certain messages of (a) server(s) (e.g. application communication messages).

This service enables or Disables the CAN transmission.

Request Message	**Message Length: 3 Bytes**

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Byte number	Value (Hex)	Parameter Name
1	28	Service ID: Communication Control
2	xx	Sub-function 01/81*: Disable the CAN Transmission 00/80*: Enable the CAN Transmission
3	xx	Message type 01: Normal Communication Messages (used now) 02: Network Management communication messages 03: Both type of messages

(*) Positive response suppression

Server will respond with a Positive Response if the Communication control Request is accepted successfully.

Positive Response Message	**Message Length: 3 Bytes**

Byte number	Value (Hex)	Parameter Name
1	68	Response ID: Communication Control
2	xx	Requested Sub-function
3	xx	Message type requested

Server may respond with a Negative Response due to the reasons below.

Negative Response	
Server Response **(Hex)**	Response Description
12	Sub-function is not supported.
13	Incorrect Message Length or Invalid Format.
22	Server is in critical normal mode activity.
31	Detected an error in the communication type.
7F	Service Not supported in active session

Note:
If the server session transitions to the default, then all communication will start again.

3.13 Request Download: $34

This service is used by the client to initiate a data transfer from the client to the server. (download).

After the server has received this request message the server shall take all necessary actions to receive data before it sends a positive response message.

Request Message	**Message Length: 11 Bytes**

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Byte number	Value (Hex)	Parameter Name
1	34	Request Download
2	00	Data format Identifier
3	44	4 – Start Address length, 4 – Length in Bytes of Data Size to Download
4	XX	MSB of Start Address Derived from Hex file
5	XX	Lower MSB of Start Address
6	XX	Higher LSB of Start Address
7	XX	LSB of Start Address
8	UU	MSB of Data Size to be Flashed Derived from Hex file
9	UU	Lower MSB of Data Size to be Flashed
10	UU	Higher LSB of Data Size to be Flashed
11	UU	LSB of Data Size to be Flashed

Server Responding with a Positive Response if Download Request accepted successfully

Positive Response Message		**Message Length: 4 Bytes**
Byte number	Value (Hex)	Parameter Name
1	74	Response ID: Routine Control
2	20	– bit 7 - 4: Length (number of bytes) of the Max Number of Block Length parameter – bit 3 - 0: reserved by document, to be set to '0'
3	xx	Block Length MSB
4	xx	Block Length LSB

Note: The length (in the above positive response) reflects the complete message length, including the service identifier and the data-parameters present in the TransferData request message.

Server Responding with Negative Response due to the reasons below.

Negative Response	
Server Response **(Hex)**	Response Description
13	Incorrect message length or Invalid format.
22	Conditions not correct.
31	Request out of range.
33	Security is secure.

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3.14 Read memory by address: $23

This service allows the client to request memory data from the server via provided starting address and size of memory to be read.

The Read Memory by Address request message is used to request memory data from the server identified by the parameter Memory Address and Memory Size. The number of bytes used for the Memory Address and Memory Size parameter is defined by Address and Length Format Identifier (low and high nibble).

Request Message		**Message Length: n Bytes**
Byte number	Value (Hex)	Parameter Name
1	23	Request Service ID
2	xx	Address and length format identifier bit 7-4: Length (**k** bytes) of the Memory Size parameter bit 3-0: Length (**m** bytes) of the Memory Address parameter
#3 **:** **#(m-1) + 3**	Byte #1(MSB) : Byte #m	The memory address of **m** bytes.
#n-(k-1) **:** **#n**	byte#1(MSB) : Byte #k	The memory size of **k** bytes.

Server Responding with a Positive Response if the Request is accepted successfully.

Positive response message		**Message Length: n' Bytes**
Byte number	Value (Hex)	Parameter Name
1	63	Response Service ID
#2 **:** **#n'**	Byte #1(MSB) : Byte #m	Data record values

Server Responding with Negative Response due to the reasons below.

Negative Response	
Server Response **(Hex)**	Response Description
13	Incorrect message length or Invalid format

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22	Criteria for the request are not met.
31	Request out of range
33	Security access denied
7F	Service Not supported in the active session.

3.15 Transfer Data: $36

The Transfer Data service is used by the client to transfer data either from the client to the server (download) or from the server to the client (upload). The data transfer direction is defined by the preceding Request Download or Request Upload service. If the client initiated a Request Download the data to be downloaded is included in the parameter(s) Transfer Request Parameter in the Transfer Data request message(s). If the client initiated a Request Upload the data to be uploaded is included in the parameter(s) Transfer Response Parameter in the Transfer Data response message(s).

Tester Sends the Request to Start Transfer of Data to Server after Successful Positive response from Server for Request Download Service

Request Message		**Message Length: n Bytes**
Byte number	Value (Hex)	Parameter Name
1	36	Request SID
2	m	Sequence Counter (0x00 – 0xFF) depending on total bytes to transfer
3	xx	Data Byte 1
4	xx	Data Byte 2
--	---	----
n	xx	Data Byte n-2

Server Responding with a Positive Response if the Request is accepted successfully.

Positive Response Message		**Message Length: 2 Bytes**
Byte number	Value (Hex)	Parameter Name
1	76	Response SID
2	m	Sequence Counter

Server Responding with Negative Response due to the reasons below.

Negative Response	
Server Response **(Hex)**	Response Description

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13	Incorrect Message Length or Invalid Format
24	Service is not active when a request for this service is received.
31	Parameters records are not consistent with the maximum block length or server's memory alignment constraints.
71	Length does not meet the requirements of the memory size parameter.
72	Error detected when programming a memory location in the Flash memory.
73	Error in the sequence of the Block sequence Counter.
7F	Service Not supported in the active session.

3.16 Transfer Data Exit: $37

Once Tester completes the Transfer of all the Flash Data to Server, Tester tries to Exit from the Transfer Data by sending Transfer Data Exit service request.

Request Message		**Message Length: 1 Bytes**
Byte number	Value (Hex)	Parameter Name
1	37	Request SID

Server Responding with a Positive Response if the Request is accepted successfully.

Positive Response Message		**Message Length: 1 Bytes**
Byte number	Value (Hex)	Parameter Name
1	77	Response SID

Server Responding with Negative Response due to the reasons below.

Negative Response	
Server Response **(Hex)**	Response Description
13	Incorrect Message Length or Invalid Format
24	Either the Request Download service is not active, or Programming process is not complete.
31	Transfer request parameter record contains invalid data.
72	Error detected when finalizing the data transfer.
7F	Service Not supported in the active session.

3.17 Access timing parameter: $83

The Access Timing Parameter service is used to read and change the default timing parameters of a communication link for the duration this communication link is active.

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Only one extended timing parameter set will be supported per diagnostic session. It is recommended to use this service only with physical addressing, because of the different sets of extended timing parameters supported by the servers.

Request Message		**Message Length: n Bytes**
Byte number	Value (Hex)	Parameter Name
1	83	Request Service ID
2	xx	Sub-function (Timing Parameter Access Type) 00: ISO SAE Reserved 01: Read Extended Timing Parameter Set 02: Set Timing Parameters to Default Values 03: Read Currently Active Timing Parameters 04: Set Timing Parameters to Given Values
#3 **:** **#n**	Byte #1(MSB) : Byte #m	Timing Parameter Request Record of **m** bytes.

Server Responding with a Positive Response if the Request is accepted successfully.

Positive response message		**Message Length: n' Bytes**
Byte number	Value (Hex)	Parameter Name
1	C3	Response Service ID
2	xx	Timing Parameter Access Type 6-0 bits of the sub-function parameter from the request message.
#3 **:** **#n'**	Byte #1(MSB) : Byte #m	Timing Parameter Response Record (**m** bytes)

Server Responding with Negative Response due to the reasons below.

Negative Response	
Server Response **(Hex)**	Response Description
12	Sub-function not supported.
13	Incorrect message length or Invalid format.
22	Criteria for the request are not met.
31	Timing Parameter request record contains invalid timing parameter values.

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3.18 Response on Event: $86

The Response on Event service requests a server to start or stop transmission of responses on a specified event.

This service provides the possibility to automatically execute a diagnostic service in case a specified event occurs on the server. The client specifies the event (including optional event parameters) and the service (including service parameters) to be executed in case the event occurs.

Request Message		**Message Length: n Bytes**
Byte number	Value (Hex)	Parameter Name
1	86	Service ID
2	xx	Sub-function (event Type) Bit 6 value; 0: Do Not Store Event 1: Store Event Bit 5-0 value; 00: Stop Response on Event 01: on DTC Status Change 02: on Timer Interrupt 03: on Change of Data Identifier 04: report Activated Events 05: start Response on Event 06: clear Response on Event 07: on Comparison of Values
3	xx	Event Window Time
#4 **:** **#(m-1)+4**	Byte #1(MSB) : Byte #m	Event Type Record: **m** bytes.
#n-(r-1)-1 **:** **#n**	Byte #1(MSB) : Byte #r	Service to Respond to Record: **r** bytes.

Server Responding with a Positive Response if the Request is accepted successfully.

Positive response message		**Message Length: n' Bytes**
Byte number	Value (Hex)	Parameter Name
1	C6	Service ID response
2	xx	Event Type
3	xx	Number of Identified Events
4	xx	Event Window Time
#3 **:** **#(m-1)+5**	Byte #1(MSB) : Byte #m	Event Type Record (**m** bytes)
#n'-(r-1)-1 **:**	Byte #1(MSB) :	Service to Respond to Record (**r** bytes)

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#n'	Byte #r	

Server Responding with Negative Response due to the reasons below.

Negative Response	
Server Response **(Hex)**	Response Description
12	Sub-function not supported.
13	Incorrect message length or Invalid format.
22	Criteria for the request are not met.
31	Request parameters invalid or DID not supported.

3.19 Link Control: $87

The Link Control service is used to control the communication between the client and the server(s) in order to gain bus bandwidth for diagnostic purposes (e.g., programming). This service optionally applies to those data link layers, which provides the capability to reconfigure its communication parameter (e.g. change the baud-rate on CAN or reconfigure a Flex-Ray cycle design) during a non-default diagnostic session.

Request Message		**Message Length: 2 Bytes**
Byte number	Value (Hex)	Parameter Name
1	87	Request Service ID
2	03	Sub-function (Link Control Type = Transition Mode)

Server Responding with a Positive Response if the Request is accepted successfully.

Positive response message		**Message Length: 2 Bytes**
Byte number	Value (Hex)	Parameter Name
1	C7	Response Service ID
2	03	Link Control Type = Transition Mode

Server Responding with Negative Response due to the reasons below.

Negative Response	
Server Response **(Hex)**	Response Description
12	Sub-function not supported.
13	Incorrect message length or Invalid format.
22	Criteria for the request are not met.
24	The client requests the transition of the mode of operation without a preceding verification step.

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31	Requested Identifier or parameter is invalid.

3.20 Control Vehicle Mobilization: $0C

The purpose of this service is to provide a means to mobilize or immobilize the vehicle. For safety reasons access to this service has been restricted. So, to use this service first the ECU needs to be unlocked first.

Mobility Control request.

Request Message		**Message Length: 2 Bytes**
Byte number	Value(Hex)	Parameter Name
1	0C	Service ID: Mobility control
2	xx	Sub-function; 00: Mobilize the vehicle 01: Immobilize the vehicle 02: Force immobilization

Server Responding with a Positive Response.

Positive Response Message		**Message Length: 10 Bytes**
Byte number	Value(Hex)	Parameter Name
1	4C	Response ID: Mobility control
2	xx	Requested sub-function

Negative Response	
Server Response **(Hex)**	Response Description
12	Sub-function is not supported.
13	Incorrect Message Length or Invalid Format.
22	Criteria for this request are not met.
33	The server's security feature is currently active. (not unlocked)
7F	Service Not supported in the active session.

3.21 Input Output Control by Identifier: $2F

This service is used by the client to substitute a value for an input signal, and internal server function, and force control to a value for an output of an electronic system. Generally, this service is used for input substitution and output control.

The request message contains the data Identifier to reference the input signal, internal server function, and force control of the output of the server. The vehicle manufacturer may require that the request message contains a control Enable Mask if the data identifier to be controlled references more than one parameter. If the vehicle manufacturer chooses to support the Enable Mask concept, then the control Enable Mask parameter is mandatory on all types of Input Output Control by Identifier requests for this service.

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If input Output Control by Identifier is requested on a data Identifier that references a measured output value or feedback value, for the standard server control strategy to attempt to reach the desired state from the client request message, the server must substitute the proper target value within the server control strategy. There is an optional byte. So, the tester can request the server through optional bytes. There are 4 types of control parameters that are used in the optional byte.

Supported Control parameters:

Control parameter	Parameter Name	Description
0	Return Control to ECU	This Return control to the ECU parameter tells the server that the client has no longer access and it gets back control of the mentioned signal.
1	Reset to default	This Reset to Default parameter tells the server to set the default value to the mentioned input and output signal and internal parameters.
2	Freeze Current State	This Freeze current state parameter tells the server to freeze mentioned input, output signal, and internal parameters to their current value.
3	Short term adjustment	This Short-term adjustment parameter tells the server to set its input, output signal and internal parameters to the provided value.

Request Message		**Message Length: N Bytes**
Byte number	Value (Hex)	Parameter Name
1	2F	Service ID: Input Output Control by Identifier Request SID
2	XX	ID High byte
3	XX	ID Low byte
4	XX	Control parameter
…	…	…
N	XX	Control parameter

Positive Response Message		**Message Length: N Bytes**
Byte number	Value (Hex)	Parameter Name
1	6F	Response ID: Read data by Identifier
2	XX	ID High byte
3	XX	ID Low byte
…N	XX	Control parameter

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Supported NRCs

Negative Response	
Server Response **(Hex)**	Response Description
13	Incorrect message length or Invalid Format.
22	Operating conditions of the server are not met to perform the required action.
31	This NRC shall be sent if: • The requested data Identifier value is not supported by the device. • The value contained in the input Output Control Parameter is invalid (see definition of input Output Control Parameter). • one or multiple of the applicable control State values of the control option record are invalid.
33	This NRC shall be returned if a client sends a request with a valid secure data Identifier and the server's security feature is currently active.
7F	Service not supported in active session.

Short Term Adjustment Control parameter: $03

S.No	Parameter name	Request	Response
1	**Heated grip - LH**	$2F $F5 A1 $03 $Adjustment value	$6F $F5 A1 $03 $Adjustment value
2	**Heated grip - RH**	$2F $F5 A2 $03 $Adjustment value	$6F $F5 A2 $03 $Adjustment value
3	**12V socket**	$2F $F5 A3 $03 $Adjustment value	$6F $F5 A3 $03 $Adjustment value
4	**Fuel tank actuator**	$2F $F5 A4 $03 $Adjustment value	$6F $F5 A4 $03 $Adjustment value
5	**Seat actuator**	$2F $F5 A5 $03 $Adjustment value	$6F $F5 A5 $03 $Adjustment value
6	**Heated seat-Rider**	$2F $F5 A6 $03 $Adjustment value	$6F $F5 A6 $03 $Adjustment value
7	**Heated seat-Pillion**	$2F $F5 A7 $03 $Adjustment value	$6F $F5 A7 $03 $Adjustment value

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8	**Keyless actuator**	$2F $F5 A8 $03 $Adjustment value	$6F $F5 A8 $03 $Adjustment value
9	**Haptic grip-LH**	$2F $F5 A9 $03 $Adjustment value	$6F $F5 A9 $03 $Adjustment value
10	**Haptic grip-RH**	$2F $F5 B1 $03 $Adjustment value	$6F $F5 B1 $03 $Adjustment value
11	**Horn**	$2F $F5 B2 $03 $Adjustment value	$6F $F5 B2 $03 $Adjustment value
12	**USB charger**	$2F $F5 B3 $03 $Adjustment value	$6F $F5 B3 $03 $Adjustment value
13	**Fuel volume**	$2F $F5 $E0 $64 $Adjustment value	$6F $F5 $E0 $64 $Adjustment value
14	**Fuel level sensor resistance**	$2F $F5 $E0 $65 $Adjustment value	$6F $F5 $E0 $65 $Adjustment value
15	**Gear position**	$2F $F5 $E1 $27 $Adjustment value	$6F $F5 $E1 $27 $Adjustment value
16	**Warning lights control**	$2F $F5 $E1 $BD $Adjustment value	$6F $F5 $E1 $BD $Adjustment value
17	**Indicator status**	$2F $F5 $E1 $B3 $Adjustment value	$6F $F5 $E1 $B3 $Adjustment value
18	**Mode connection status**	$2F $F5 $E1 $D0 $Adjustment value	$6F $F5 $E1 $D0 $Adjustment value
19	**Front position lamp status**	$2F $F5 $01 $14 $Adjustment value	$6F $F5 $01 $14 $Adjustment value

4. Vehicle Safe State

S. No.	Bit position in the Error Code	Safe State Checks
1	0	Battery SOC >=15%
2	1	Speed = 0 KMPH
3	2	Charger not connected?

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4	3	OTA campaign DL ready?
5	4	Configured packs = Communicating Packs
6	5	Aux Battery Voltage >12.0 V
7	6	Kill Switch Enable
8	7	Vehicle State = ON
9	8	Side Stand = Present (Down)
10	9	Battery Current < Safe state threshold value
11	10	Reserved for future use case
12	11	Reserved for future use case
13	12	Reserved for future use case
14	13	Reserved for future use case
15	14	Reserved for future use case
16	15	Reserved for future use case

Note: *The Error code of the pre-condition check (Section 3.10.3) is 16 bits.*
The corresponding bit being "0" denotes the passing of Safe State Check.

5. References

The following referenced documents are indispensable for the application of this document.

a) **ISO 14229-1**, Second edition (2013-03-15), Road vehicles — Unified diagnostic services (UDS) — **Part 1**: Specification and requirements.
b) **ISO 15765-3**, First edition (2004-10-15), Road vehicles — Diagnostics on Controller Area Networks (CAN) — **Part 3**: Implementation of unified diagnostic services (UDS on CAN).
c) **SAE J2012,** Diagnostic Trouble Code Definitions (April 2002).
d) **KE210130_A3**: Diagram for the Battery UID.

6. Annexure

The following attached documents can be referred to for the detailed lists.

a) **TVSM_Diagnostics_Info.xlsx**

b) **TVSM_ECU_MetaData_Management.xlsx**

c) **TVSM_Value_Tables_for_Diagnostics.xlsx**

d) **TVSM_SecurityAccess_Specification.pdf**

e) **TVSM_Checksum_Specification.pdf**

f) **TVSM_UDS_FlashingSequence.pdf**

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7. Revision History

Rev No	Remarks	Date	Author
V1	Base: UDS_Bootloader Flashing Sequence and Diagnostic specification U347 Initial Revision	16/08/2021	NC Giresha
V2	Section 2.5.6 Checksum: $22 $ F1 F0- Added Section 1.3 [Diag-mode, function table added] Section 1.4 : P2 min is set to 0msec Section 1.4.1 : Added Section 1.4.2 : Added Section 2.2.2 : Added Section 4.0: Cluster DTC added	17/09/2021	NC Giresha
V3	1) Added 1.4.3 CAN node Addresses 2) Added 1.7Overview of Functional addressing support 3) section 2.2.3 [19 0A] , Number of DTC added in the response section 2.2.3 [19 0A] , Request increased to 3bytes [19 0A FF] 4) section 5.0 EEPROM added 5) section 2.5 New services are added 6) section 2.6 New services are added	20/12/2021	NC Giresha
V4	1) EV-3W added 2) Section 4.2 added 3) Section 4.3 added 4) Section 4.4 added 5) Section 4.5 created for MCU 6) Section 2.5.13 added	05/01/2022	NC Giresha
V5	1) Section 1.7 Added 2) Section 2.2.2 modified [UDS adapted] 3) Section 2.5.13.2,2.5.13.3,2.5.13.4,2.5.13.5 added 4) Section 2.10.2 Added 5) Section 2.11 added 6) Section 2.6.9 added 7) Section 2.6.10 added 8) Section 2.6.3 SKU number added 9) Section 2.5.3 SKU number added 10) Section 2.5.20 increased to three bytes	19/01/2022	NC Giresha
V6	1) Section 5.0 updated Vehicle software version added ECU serial Number added 2) Section 2.5.14 added 3) Section 2.5.15 added 4) Section 2.6.12 added 5) Section 2.6.13 added	21/01/2022	NC Giresha

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V7	Section:2.11 Control DTC $85 modified Section:7.0 added new [ECU status on CAN] Section: 2.1.2 precondition added [failure added]	13/07/2022	NC Giresha
V8	2.5.16: Battery pack count Read 2.6.16: Battery pack count Write	16/12/2022	NC Giresha
V9	1.4.3: Added more CAN node (ECU) addresses	20/03/2023	Arunesh Singh
V10	Section 1.4.3: Modified the CAN node address table Section 1.6: Modified as per the implementation Section 2.6.17: Bootloader SW version read [$F1 85] added Section 2.6.18: GA Number read [$F1 88] added Section 2.6.19: App presence pattern read [$F1 82] added Section 2.6.20: Fingerprint read [$F1 84] added Section 2.6.21: SW BOM ID read [$F1 83] added Section 2.7.14: GA Number write [$F1 88] added Section 2.7.15: App Presence pattern write [$F1 2D] added Section 2.7.16: Fingerprint write [$F1 84] added Section 2.7.17: SW BOM ID write [$F1 83] added Section 2.6.22.1: Live Parameters Added to VCU Section 2.10.2: Added the Erase Flash request Section 2.10.3: Added the Check memory request Section 2.12: Updated the number of bytes in the $28 service Section 2.13: Added the request download service $34 Section 2.14: Added the read memory by address service $23 Section 2.15: Added the transfer data service $36 Section 2.16: Added the transfer exit service $37 Section 2.17: Added the access timing parameter service $83 Section 2.18: Added the response on event service $86 Section 2.19: Added the link control service $87 Section 5.0: Edited EEPROM services table	04/05/2023	Arunesh Singh
V10.1	The following sections has been added Section 2.0: Flashing Procedure Section 2.0.1: Flashing Sequence [UDS 1.1] Section 2.0.2: Flashing Flow Diagram Section 3.6.22: Vehicle's Current Mobility Status: $22 $ F1 0C Section 3.6.23: Partition Verify: $22 $ F1 1B Section 3.6.24: Key Verify: $22 $ F1 1C Section 6.0: Vehicle Safe State Section 1.4.3: CAN node addresses corrected Section 3.10.2: Data size bytes in the request made variable Section 3.13: Positive response size corrected to 4 bytes Section 3.15: Positive response corrected to 2 bytes	12/06/2023	Arunesh Singh
V10.2	Section 1.4.3: Irrelevant CAN addresses removed Section 4.1: DTCs list in the current VCU implementation added	26-06-2023	Arunesh Singh

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	Section 1.9: Currently implemented NRCs added		
V10.3	Section 2.0.1: Bootloader information reading moved to step 4 in the flashing sequence. Section 3.6.1: Added the Tester stamp DID for reading. Section 3.6.5: Corrected the last 3 bytes format to YY/MM/DD. Section 3.6.16: Corrected the last 3 bytes format to YY/MM/DD Section 3.6.11: Made Reprogramming times as Read-only. Section 3.6.12: Made Crash counter as Read-only. Section 3.6.19: Added value table for the fingerprint. Section 3.6.22: Added one more byte to identify the key. Section 3.7.5: Edited the data bytes. Section 3.7.8: Updated the EOL mode DID. Section 3.7.13: Added value table for the fingerprint.	26-07-2023	Arunesh Singh
V10.4	Section 1.4.3: Included LCM CAN node address. Section 1.7: Modified the LIDs and included LCM. Section 2.0.1: Added the flashing sequence flow for the segmented memory. Section 3.6.3: Included Byte-wise details for DIDs. Section 3.6.4: Included Byte-wise details for DIDs. Section 3.6.5: Included Byte-wise details for DIDs. Section 3.6.14: Included Byte-wise details for DIDs. Section 3.6.17: Included Byte-wise details for DIDs. Section 3.6.19: Included Byte-wise details for DIDs. Section 3.6.23: Added reading of EMS Unique Code Section 3.6.24: Added reading of Date of Manufacturing. Section 3.6.25: Live parameters section added for LCM and iVCU. Section 4.0: DTC section added for LCM and iVCU. Edited the DTCs with SAE standard codes. Section 5.0: Modified the DIDs applicable. Updated the following attached files, the details can be seen in their revision history. “OBD-II_Diagnostics_Mahle_APK_dev_Ver3” “TVSM_AEG_AutoMation_VersionManagement_Ver3” “Value_Table_for_UDS_Ver2”	08-09-2023	Arunesh Singh
V10.5	Section 1.4.3: Added CAN node addresses for TACO BMS and ARAS ECU. Section 1.7: Updated the table. Added the BIN, Theta reading and writing. Added Routine Control services 04 and 06. Section 3.5: Added the details of Reset subfunctions. Section 3.6.2: Added the 5th byte in the response message. Section 3.6.25: Added DID for Battery UID reading. Section 3.6.26: Added DID for Theta reading. Section 3.6.27: Moved the list of all the live parameters into the attachment. Section 3.7.2: Added Security for writing the VIN.	09-02-2024	Arunesh Singh

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	Section 3.7.14: Added DID for Battery UID writing. Section 3.7.15: Added DID for Theta writing. Section 3.9: Attached Security Access Algorithm document for Security Access service. Section 3.10.5: Added a Routine control subfunction for Unlocking the Battery UID writing. Section 3.10.6: Added a Routine control subfunction for Controlling the Battery discharge port. Section 4.0: Modified according to the services and security required. Moved the List of DTCs into the attachments. Section 7.0: Updated the attachments. Updated the following attached files, the details can be seen in their revision history. "OBD-II_Diagnostics_Mahle_APK_dev_Ver4" "TVSM_AEG_AutoMation_VersionManagement_Ver4" The following attached. "UDS-SecurityAcess-Specification_Ver5.0"		
V10.6	Section 1.3: Updated the P2 and P2* Timing Parameters' selection method. Section 1.3.1: Updated the Timeout handling diagrams Section 1.6: Removed the Communication Control service and Control DTC settings services support from the Bootloader SW. Section 2.1: Corrected typing errors. Updated the comments with recovery mechanism. Modified the Tester Present request frequency to 2 Seconds to remain in the non-default sessions. Section 3.6.6: Modified the Hardware Version Read Byte details. Section 3.6.26: Updated the Theta Read DID details (13 Bytes). Section 3.7.15: Updated the Theta Write DID details (13 Bytes). Section 3.10.5: Corrected the Number of response Bytes. Section 3.10.6: Corrected the Number of response Bytes. Updated the following attached files, the details can be seen in their revision history. "OBD-II_Diagnostics_Mahle_APK_dev_Ver5" "Value_Table_for_UDS_Ver3"	16-05-2024	Arunesh Singh
V11.0	Section 1.3: Incorporated the P2 and P2* Timing min, max and recommended values. Section 1.6: Added the Communication Control support for the Bootloader software. Removed the ECU reset support from the Default diagnostic session. Added notes. Section 1.7: The mismatch of IDs with other sections have been corrected. Added positive response suppression Sub-IDs.	18-06-2024	Vinod Pendela, Arunesh Singh

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	Column added to identify the service as secured or not. Section 2.1: Organized the Flashing sequence into pre-programming, programming and post-programming stages. Modifications for sending of the Communication Control, DTC setting control and Tester Present on functional ID with the positive response suppression sub-function ID. Section 3.1.1: Specified the resolution of P2 and P2* in the response. Section 3.1.2: Specified the resolution of P2 and P2* in the response. Section 3.1.3: Specified the resolution of P2 and P2* in the response. Section 3.1.4: Specified the resolution of P2 and P2* in the response. Section 3.1.5: Added the section to list the supported Negative response codes for $10 service. Section 3.1.6: Modified the flow diagram. Section 3.3: Modified DTCs to 2 Bytes as per the SAE Standards. Section 3.3.1: Listed the supported Negative response codes. Corrected the Status Mask Byte to “2F”. Section 3.3.2: Listed the supported Negative response codes. Section 3.3.3: Listed the supported Negative response codes. Section 3.4: Modified DTCs to 2 Bytes as per the SAE Standards. Modified the supported Negative response codes. Section 3.4.1: Modified the supported Negative response codes. Section 3.5: Modified the supported Negative response codes. Section 3.6: Listed the supported Negative response codes. Section 3.7: Listed the supported Negative response codes. Section 3.8: Added the option of positive suppression in sub-function. Modified the supported Negative response codes. Section 3.9: Modified the supported Negative response codes. Section 3.10: Re-ordered the Sub-sections of Routine control. Section 3.10.1: Modified the supported Negative response codes. Section 3.10.3: The Pre-condition check RID has been corrected to $31 01 02 01 and the Error code Byte added. This routine control service has been mandated for all the ECUs. Modified the supported Negative response codes Section 3.10.4: Modified the supported Negative response codes. Section 3.10.5: RID has been modified to comply with ISO standard. Listed the supported Negative response codes. Section 3.10.6: RID has been modified to comply with ISO standard. Modified the supported Negative response codes. Section 3.11: Added the option of positive suppression in subfunction. Modified the supported Negative response codes. Removed the tester present description from the notes and added in section 3.8.		

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	Section 3.12: Added the option of positive suppression in subfunction. Modified the supported Negative response codes. Removed the tester present description from the notes and added in section 3.8. Section 3.15: Modified the supported Negative response codes. Section 3.16: Modified the supported Negative response codes. Section 3.20: Listed the supported Negative response codes. Section 5: Vehicle Safe State Parameters have been modified. Updated the following attached files, the details can be seen in their revision history. "TVSM_Diagnostics_Info_Ver6" "TVSM_ECU_MetaData_Management_Ver5" "TVSM_Value_Tables_for Diagnostics_Ver4"		
V11.1	Section 1.6: Service table has been updated for Client Session. A Note related to services implementation has been added. Section 1.7: Added the DID F1 31 in the table. Updated the applicability of Security Access. Section 2.1: Added a Periodic tester present request in the programming sequence. Section 2.2: The flashing flow diagram has been removed as it was redundant. Section 3.1.4: Description has been updated for Client session Section 3.3: Message format has been corrected for the subfunctions: 0x02, 0x04 and 0x0A Number of bytes for DTC has been changed from 2 to 3 to align with the ISO 14229-1. Section 3.3.3: To align with the ISO standard 14229-1, removed the Byte 4 (Number of DTCs) from the response Bytes. Section 3.4: Service request format for DTC group has been updated Section 3.4.1: Service request and response format for a single DTC has been updated Section 3.6.28: Read DID $F1 $31 added for TPMS pairing data Section 3.7.16: Write DID $F1 $32 added for TPMS pairing data Updated the following attached files, the details can be seen in their revision history. "TVSM_Diagnostics_Info_Ver7"	30-09-2024	Vinod Pendela, Arunesh Singh
V11.2	Section 1.1.2.4: Note has been updated for Flow Control frame Section 1.1.3: Note has been added that Tester Present service shall support Functional addressing mode. Section 1.4: BMS description in the CAN Node address table has been updated Section 1.6, 1.7: Security Access condition has been removed for	04-12-2024	Vinod Pendela, Arunesh Singh, Bharath Gullipilli

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	the Services 0x11, 0x28 and 0x85 to be in sync with the Flashing sequence. Section 2.1: Service ID has been corrected for the step "Write Fingerprint" Section 3.4.1: Clearing of an individual DTC by Service 0x14 is optional. Section 3.6.4: Message description has been corrected Section 3.6.6: Message description has been corrected Section 3.6.21: Note has been added for DID 0xF1 1B Section 3.6.24: Note has been corrected. Section 3.6.30: Read DID $F1 $33 added for Tire pressure Threshold Section 3.7.2: Removed the note. Section 3.7.4: Removed the note. Section 3.7.8: Removed the note. Section 3.7.9: Note added for one time writing of serial number. Section 3.7.13: Updated the table headings. Section 3.7.14: Updated the note. Section 3.7.15: Removed the note. Section 3.7.17: Write DID $F1 $33 added for Tire Pressure Threshold. Section 3.9.2: Positive Response has been updated for Seed-Key. Removed NRC 31. Section 3.10: Corrected the request and response format. Corrected the NRC response. Section 3.10.3: Modified the Pre-condition Error Code to 2 Bytes. Section 3.10.4: Note added for the attachment. Section 3.10.7: NRC description has been updated for better clarity. Section 3.11: NRC number has been corrected Section 3.12: NRC number has been corrected Section 3.13: Note has been added for Request Download service. Section 3.15: Positive Response message format has been updated. Section 4: Removed the security column since it is redundant. Section 5: Modified the list of Pre-condition check. Added bit-wise column to interpret the error code in the pre-condition check. Added note. Section 7: Attached the following file, "TVSM_Checksum_Specification_Ver1"		
V11.2.1	Section 1.7: Updated the overview of project specific supported services. Section 3.1.6: Corrected the session transition diagram. Section 3.6.3: Corrected the data format of VIN number. Section 3.6.23: Corrected the data format of EMS unique code number and data length. Section 3.6.30: Added ECU Hardware part Number read DID Section 3.6.31: Added ECU Software part number read DID Section 3.6.32: Added Immobilization Status read DID. Section 3.6.33: Added Hill Hold Feature read DID.		Arunesh Singh, Bharath Gullipilli

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	Section 3.6.34: Added MAC_ID of Bluetooth read DID. Section 3.6.35: Added IMEI number read DID. Section 3.6.36: Added Battery voltage read DID. Section 3.6.37: Added photosensor read DID. Section 3.6.38: Added Ignition status number read DID. Section 3.6.39: Added Current Fuel volume number read DID. Section 3.6.40: Added Fuel level sensor resistance read DID. Section 3.6.41: Added Gear position read DID. Section 3.6.42: Added High beam input switch read DID. Section 3.6.43: Added Indicator switch Activation Status read DID. Section 3.6.44: Added Distance unit read DID. Section 3.6.45: Added Indicator Activation status read DID. Section 3.6.46: Added Bluetooth connection status read DID. Section 3.6.47: Added Mode connection status read DID. Section 3.6.48: Added Hazard switch status read DID. Section 3.6.49: Added Speedometer software version number read DID. Section 3.6.50: Added Bluetooth firmware version number read DID. Section 3.6.51: Added ABS variant read DID. Section 3.6.52: Added MAC ID for front tire read DID. Section 3.6.53: Added MAC ID for rear tire read DID. Section 3.6.54: Added battery level for front tire read DID. Section 3.6.55: Added battery level for rear tire read DID. Section 3.6.56: Added Front tire pressure read DID Section 3.6.57: Added Rear tire pressure read DID Section 3.6.58: Added Tire pressure threshold for front and rear read DID. Section 3.6.59: Added cruising range read DID. Section 3.6.60: Added head lamp control read DID. Section 3.6.61: Added go pro read DID. Section 3.6.62: Added climate-controlled seat read DID. Section 3.6.63: Added cruise set status read DID. Section 3.6.64: Added cruising reset status read DID. Section 3.6.65: Added cruise on/off status read DID. Section 3.6.66: Added cruise functionality status read DID. Section 3.6.67: Added QS functionality status read DID. Section 3.6.68: Added brake switch status read DID. Section 3.6.69: Added brake lamp control status read DID. Section 3.6.70: Added BTO animation enable read DID. Section 3.6.71: Added Tacho swipe read DID. Section 3.6.72: Added celebration alerts read DID. Section 3.6.73: Added navigation for TPMS/Low fuel read DID. Section 3.6.74: Added Language read DID. Section 3.6.75: Added celebration settings read DID. Section 3.6.76: Added enable/disable TPMS read DID. Section 3.6.77: Added country read DID. Section 3.6.78: Added fog lamp read DID.		

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	Section 3.6.79: Added cluster housing temperature read DID. Section 3.6.80: Added ambient light sensor value read DID. Section 3.6.81: Added time and date read DID. Section 3.6.82: Added TC functionality status read DID. Section 3.7.18: Added Hardware part number write DID. Section 3.7.19: Added software part number write DID. Section 3.7.20: Added Immobilization Status write DID. Section 3.7.21: Added Hill Hold Feature write DID Section 3.6.22: Added MAC_ID for Bluetooth chip write DID. Section 3.7.23: Added Secure key write DID. Section 3.7.24: Added IMEI number write DID. Section 3.7.25: Added Distance unit write DID. Section 3.7.26: Added time and date write DID. Section 3.7.27: Added ABS variant write DID. Section 3.7.28: Added MAC ID for front tire write DID. Section 3.7.29: Added MAC ID for rear tire write DID. Section 3.7.30: Added Tire Pressure threshold for rear and front write DID. Section 3.7.31: Added head lamp control write DID. Section 3.7.32: Added Go pro write DID. Section 3.7.33: Added climate-controlled seat write DID. Section 3.7.34: Added BTO animation write DID. Section 3.7.35: Added Tacho swipe seat write DID. Section 3.7.36: Added celebration alerts write DID. Section 3.7.37: Added navigation for TPMS/Low fuel write DID. Section 3.7.38: Added language write DID. Section 3.7.39: Added service reminder reset write DID. Section 3.7.40: Added celebration settings write DID. Section 3.7.41: Added enable/disable TPMS write DID. Section 3.7.42: Added country write DID. Section 3.7.43: Added fog lamp write DID. Section 3.7.44: Added odometer reset write DID. Section 3.7.45: Added debug frame write DID. Section 3.10.7: Added Keyless sensor learning RID. Section 4: Corrected Application presence pattern write DID in the table		
V11.2.2	Section 3.6.82: Added Quick shifter cluster switch test read DID. Section 3.6.83: Added TC functionality status read DID. Section 3.6.84: Added speedometer VCU authentication status read DID. Section 3.6.85: Added VCU EMS authentication status read DID. Section 3.6.86: Added Keyless actuator status read DID. Section 3.6.87: Added Keyless sensor status read DID.	02-05-2025	Arunesh Singh, Bharath Gullipilli

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	Section 3.6.88: Added Haptic grip status read DID. Section 3.6.89: Added Fuel tank actuator status read DID. Section 3.6.90: Added Seat actuator status read DID. Section 3.6.91: Added Seat lock sense read DID. Section 3.6.92: Added Fuel tank sense read DID. Section 3.6.93: Added Variant Control read DID. Section 3.6.94: Added Dealer ID read DID. Section 3.6.95: Added Project code read DID. Section 3.6.96: Added Date of last programming read DID. Section 3.6.97: Added Heated grip thermistor input-LH read DID. Section 3.6.98: Added Heated grip thermistor input-RH read DID. Section 3.6.99: Added Heated grip heat level-RH read DID. Section 3.6.100: Added Heated grip heat level-LH read DID. Section 3.6.101: Added Heated seat heat level-rider read DID. Section 3.6.102: Added Heated seat heat level-pillion read DID. Section 3.6.103: Added Horn status read DID. Section 3.6.104: Added USB charger status read DID. Section 3.6.105: Added MFR/Projector lamp ID – read DID. Section 3.6.106: Added MFR/Projector lamp ID - DRL read DID. Section 3.6.107: Added Dynamic Head lamp read DID. Section 3.6.108: Added Cornering/Bend Lamp read DID. Section 3.6.109: Added Turn Signal Lamp read DID. Section 3.6.110: Added Parking lamp read DID. Section 3.6.111: Added Live Fuel Economy Dashboard read DID. Section 3.6.112: Added Telematics Status read DID. Section 3.6.113: Added Carpet lamp read DID. Section 3.6.114: Added Drive mode read DID. Section 3.6.115: Updated Live Data Parameters section . Section 3.7.8: Request message length has been corrected. Section 3.7.46: Added Dealer code write DID. Section 3.7.47: Added Project code write DID. Section 3.7.48: Added Feature enable write DID. Section 3.7.49: Added Timing parameter for SIA write DID. Section 3.7.50: Added Variant control write DID. Updated the following attached files, the details can be seen in their revision history. “TVSM_Diagnostics_Info_Ver9”		
V11.2.3	Section 3.6.10: Added description for IGN Counter read DID. Section 3.6.11: Added description for Reprograming times read DID. Section 3.6.61: Updated name for Go pro read DID. Section 3.6.63: Added description for Cruise set status. Section 3.6.64: Added description for Cruise reset status. Section 3.6.65: Added description for Cruise ON or OFF status. Section 3.6.66: Added description for Cruise functionality status.	21-05-2025	Bharath Gullipilli, Pujitha Yalamanchili

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	Section 3.6.67: Updated the name for QS cluster switch DID. Section 3.6.71: Added description for Tacho Swipe read DID. Section 3.6.79: Changed the DID number for Cluster housing temperature. Section 3.6.83: Updated the name and added the description of the for the TC functionality status read DID. Section 3.6.105: Added description for MFR/Projector lamp ID - Headlamp read DID. Section 3.6.106: Added description for MFR/Projector lamp ID - DRL read DID. Section 3.6.107: Changed the name of Dynamic Head lamp to Head Lamp Intensity and added description for this DID. Section 3.6.108: Added description for Cornering/Bend Lamp read DID. Section 3.6.109: Added description for Turn Signal Lamp read DID. Section 3.6.110: Added description for Parking lamp read DID. Section 3.6.111: Added description for Live Fuel Economy Dashboard read DID. Section 3.6.112: Added description for Telematics Status read DID. Section 3.6.113: Added description for Carpet lamp read DID. Section 3.6.114: removed Drive mode values of Drive mode read DID. Section 3.6.115: Added MAC ID for Key fob pairing read DID. Section 3.7.20: Updated the Immobilization Feature status DID name and description. Section 3.7.32: Updated name for Go pro read DID. Section 3.7.39: Removed the Service reminder reset DID ($ E1 2E). Section 3.7.50: Added MAC ID for key fob pairing write DID. Section 4: Removed EEPROM section since storage location of data is not in the scope of diagnostic specification.		
V11.3	Section 2.1: Flashing sequence has been optimized Section 3.1.5: Added EOL Diagnostic session Section 3.6.109: Updated for checking TSL auto cancellation status Section 3.6.115: Updated for MAC ID of multiple Keyfobs Section 3.6.116: Added Take me bike read DID. Section 3.6.117: Added Dynamic brake light read DID. Section 3.6.118: Added Fuel Filler Flap read DID. Section 3.6.119: Added Voice Assist Feature read DID Section 3.6.120: Added No.of keyfobs Paired read DID Section 3.6.121: Added Service Date read DID Section 3.6.122: Added Service Distance read DID Section 3.6.123: Added Key fob authentication status read DID Section 3.6.124: Added ECU authentication status read DID Section 3.7.51: Added Voice Assist Feature write DID Section 3.7.52: Added no.of keyfob's paired write DID Section 3.7.53: Added Service Date write DID Section 3.7.54: Added Service Distance Write DID Section 3.10.8: Added erase all paired Keyfob Routine ID	30-06-2025	Bharath Gullipilli, Pujitha Yalamanchili

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	Section 3.10.9: Added Keyfob learning routine Routine ID DID Numbers have been changed for Variant Control, Traction Control Functionality, Heated grip thermistor-LH, Heated grip thermistor input-RH , Heated grip heat level-RH , Heated grip heat level- LH , Heated seat heat level-Rider , Heated seat heat level-Pillion ,Horn Status , USB charger status ,MFR/Projector lamp ID-Head Lamp , MFR/Projector Lamp ID- DRL, Head lamp Intensity Level , Cornering / Blend Lamp , Turn Signal Lamp , Parking Lamp , Live Fuel Economy , Telematics status. Feature enable, Timing parameter for SIA		
V11.3.1	Section 1.7 : Reference to 443D has been removed Section 2.1: Note has been added for ECU specific requirements handling after the flashing sequence has been completed. Section 3.6.121: DID short name and description have been corrected based on the use case. Section 3.7.53: DID short name and description have been corrected based on the use case. 3.6.107: Added Map Mirroring Enable Status read DID 3.6.108: Added Traffic Screen Enable Status read DID 3.6.109: Added Music Enable Status read DID 3.6.110: Added SMS notification Enable read DID 3.6.112: Added Product Variant ID read DID 3.6.113: Added Battery Type_0 read DID 3.7.55: Added Product variant ID write DID 3.7.56: Added Battery Type_0 write DID The DIDs below have been deleted from here and moved to Live parameters category: 0908- Heated grip thermistor input-LH 090A- Heated grip thermistor input-RH 090B- Heated grip heat level-RH 090C-Heated grip heat level-LH 090D-Heated seat heat level-rider 090E-Heated seat heat level-pillion F1B1-Speedometer VCU authentication status F1B2-Keyless actuator status F1B3-VCU-EMS AUTHENTICATION STATUS F1B4-Keyless Sensor status 0900-Carpet Lamp (puddle lamp) F1B5-Haptic grip status F1B8-Fuel tank actuator status 0910-USB charger status F1B9-Seat actuator status F1C1-Seat lock sense F1C2-Fuel tank sense 090F-Horn Status	31-07-2025	Pujitha Yalamanchili
V11.3.2	Section 1.4: CAN Nodes address table has been deleted and moved to TVSM_Diagnostics_Info.xlsx.	02-09-2025	Pujitha Yalamanchili

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	Section 1.6: Link Control Service usage has been updated Section 1.7 : Overview supported services has been deleted. Added in TVSM_Diagnostics_Info.Xlsx in new format -DID_List Section 3.1.7: Clarification has been added in a note for session transition possibilities. Section 3.6.95: BIT Description is updated Section 3.6.73: Modified the description based on the use case Section 3.6.113: Added RPM Threshold- a for Gear shifting read DID Section 3.6.114: Added RPM Threshold- b for Gear shifting read DID Section 3.6.115: Added RPM Threshold- c for Gear shifting read DID Section 3.6.116: Added RPM Threshold- d for Gear shifting read DID Section 3.6.117: Added RPM Threshold- e for Gear shifting read DID Section 3.6.118: Added RPM Threshold- f for Gear shifting read DID Section 3.6.119: Added RPM Threshold- g for Gear shifting read DID Section 3.6.120: Added RPM Threshold- h for Gear shifting read DID Section 3.6.121: Added ODO Threshold-x for Gear shifting read DID Section 3.6.122: Added Coolant Temp Threshold-y for Gear shifting read DID Section 3.6.123: Added Indicator Blinking Period read DID Section 3.6.124: Added Engine High Temp Threshold read DID Section 3.6.125: Added Fuel Thermistor Threshold read DID Section 3.6.126: Live Parameter range for the ECUs have been corrected Section 3.7.37: Modified the description based on the use case		
V11.3.3	Section 3.3.1: Read Number of DTC by status mask: $19 01 sub service is added. Section 3.3.2: Request format has been corrected. Section 3.3.4: Read extended data record of DTC: $ 19 06 sub service is added. Section 3.6.4: Modified SKU read DID byte length and Description of bytes Section 3.6.5: Modified byte length for SW part number for Application SW version DID Section3.6.6: Modified byte length for HW part number for Hardware version Read DID Section 3.6.9: Modified data length of ODO Read DID Section 3.6.16: Modified SW part number for Bootloader SW version Read DID Section 3.6.30: Modified HW part number for ECU SW part number Read DID Section 3.6.31: Modified SW part number for Ecu SW part number read DID. Section 3.6.41: Added gear position description for Gear position read DID. Section 3.6.52: Deleted MAC ID for Front tire read DID Section 3.6.53: Deleted MAC ID FOR rear tire read DID	22-11-2025	Pujitha Yalamanchili

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	Section 3.6.112: Modified byte length for Battery Type_0 Read DID Section 3.6.126: Added Suspension setting Active Status read DID. Section 3.6.127: Added Battery Type erase status read DID. Section 3.6.128: Added Battery Feature Enable read DID. Section 3.6.129: Added Enable or Disable Emergency Brake Lamp read DID. Section 3.6.130: Added Vehicle variant and Region Parameters read DID Section 3.7.3: Modified SKU write DID byte length and Description of bytes Section 3.7.4: Modified data length of Odometer DID. Section 3.7.18: Modified HW part number for SAP Hardware part number Write DID. Section 3.7.19: Modified SW part number for SAP SW part number Write DID. Section 3.7.23: Modified byte length for Secure key DID. Section 3.7.28: Deleted MAC ID for Front tire Write DID Section 3.7.29: Deleted MAC ID for Rear tire Write DID Section 3.7.56: Modified byte length for Battery Type_0 write DID. Section 3.7.57: Added Battery Type erase status write DID. Section 3.7.58: Added Battery Type Feature Enable write DID. Section 3.7.59: Added Launch Control counter write DID. Section 3.7.60: Added Enable or Disable Emergency Brake Lamp write DID. Section 3.7.61: Added Vehicle variant and Region Parameters write DID.		
V11.4	Section 2.1 Deleted and moved to separate document Section 3.6.52: Deleted Hardware SAP DID -F134 Section 3.6.53:Deleted Software SAP DID-F135 Section 3.6.84:Added value description and DID description Section 3.6.131:Added Vehicle Steering Status Read DID Section 3.6.132:Added seat and fuel cap solenoid status Read DID Section 3.6.133:Added VCU sleep status Read DID Section 3.6.134:Added Ignition Load Relay Status Read DID Section 3.6.135: Added Find Me TSL Status Read DID. Section 3.6.136: Added PWM Signal Overrun Read DID Section 3.6.137:Added Watchdog Reset Count Read DID. Section 3.6.138:Added HSD diagnostics pin status Read DID. Section 3.6.139:Added HSD overcurrent value Read DID. Section 3.6.140:Added Authentication Fail Count Read DID. Section 3.6.141:Added EEPROM Operation Fail Count Read DID. Section 3.6.142:Added Hardware security Module Init Status Read DID. Section 3.6.143:Added Hot Seat Feature Read DID. Section 3.6.144:Added Cool Seat Feature Read DID. Section 3.6.145:Added Variant Configuration Method Read DID	26-03-2026	Pujitha Yalamanchili & Shaik Nafeesa Khaisar

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	Section 3.6.146:Added A/A2 License Configuration Read DID Section 3.6.147:Added Telematics Event Flag Read DID. Section 3.6.148:Added TCU Modem Application Version Read DID Section 3.6.149:Added GPS Data Read DID Section 3.6.150:Added TCU OTA Status Read DID Section 3.6.151:Added MQTT Server Details Read DID. Section 3.6.152:Added GSM Data Read DID Section 3.6.153:PKI Status Read DID Section 3.6.154:KeyFob Button Status Read DID Section 3.6.155:Added NVM and Software Version Read DID. Section 3.6.156:Added Reprogramming Status Flag Read DID. Section 3.6.157:Added Integrated Circuit Card Identifier Read DID. Section 3.7.18:Deleted SAP Hardware part number Write DID. Section 3.7.19:Deleted SAP Software Part Number Write DID. Section 3.7.62:Added Variant Configuration Method Write DID. section 3.7.63:Added A/A2 License Configuration Write DID. Section 3.7.64:Added Reprogramming Status Flag write DID.		
11.4.1	Section 3.6.97:Renamed the DID and Changed the Byte length Section 3.6.157:Renamed the DID Section 3.6.158:Added the Integrated Circuit Card Identifier2 read DID. Section 3.6.159:Added the Secondary Key Fob MAC ID read DID. Section 3.6.160: Added the E-SIM Activation Status read DID. Section 3.6.161:Added the Cruise control feature read DID. Section 3.6.162:Added the Heated Grip Feature read DID. Section 3.6.163:Added the Hazard Lamp Feature read DID. Section 3.6.164:Added the fuel sensor mean read DID. Section 3.7.65:Addec the E-SIM Activation Switching Write DID Section 3.7.66:Added the Cruise Control Feature Write DID. Section 3.7.67:Added the Heated Grip Feature Write DID. Section 3.7.68:Added the Hazard Lamp Cancellation write DID.	29-04-26	Shaik Nafeesa Khaisr & Pujitha Yalamanchili